Topics in Mathematics Science VII
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Introduction to Koszul Duality

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Convention

Throughout the course, $\Bbbk$ is a field. All rings are unital and associative. All modules are right modules unless otherwise specified. When we discuss dimensions, we mean dimensions as $\Bbbk$-vector spaces.

There are a lot of contents not covered in lecture:

> This is supplementary material; often for 'unnecessarily long pedantic check of definitions' or content of similar nature – things that are only there for rigour

> This is advanced material; things that I personally would like to add, from personal experience in the theory, but definitely not necessary for lecture purpose.

Neither of these are rigourously checked since they are mostly produced by AI, so the reader should read these contents with extra cares themselves. All other contents are drafted by AI and QA'ed (heavily edited) by myself.

1 Graded algebras, modules, and duals

1.1 Algebras and modules

Definition 1.1. *A $\Bbbk$-algebra is a $\Bbbk$-vector space with an associative unital multiplication that is bilinear, meaning linear in each variable.*

A right A-module is understood to be a $\Bbbk$-vector space M with a bilinear action satisfying $(ma)b = m(ab)$ and $m1_A = m$ for $m \in M$ and $a, b \in A$. A module homomorphism $f : M \to N$ is linear and

satisfies $f(ma) = f(m)a$. A homomorphism of $\Bbbk$-algebras is a linear map preserving multiplication and the identity.

Example 1.2. *(1) The matrix ring* $\mathrm{Mat}_n(\Bbbk)$ *is a* $\Bbbk$*-algebra. The space of* $1 \times n$ *row vectors is a right* $\mathrm{Mat}_n(\Bbbk)$*-module by matrix multiplication.*

(2) Let $\Bbbk A_n$ *be the upper triangular matrix ring, which is also a* $\Bbbk$*-algebra.*

$$\Bbbk A_n = \begin{pmatrix} \Bbbk & \Bbbk & \cdots & \Bbbk \\ 0 & \Bbbk & \cdots & \Bbbk \\ 0 & 0 & \ddots & \vdots \\ 0 & \cdots & 0 & \Bbbk \end{pmatrix}$$

For those familiar with path algebras, $\Bbbk A_n$ *is the path algebra of a linearly oriented* A_n *quiver, which explains the notation. We use matrix calculations, so no knowledge of quivers is required. Write* e_{ij} *for the matrix with* 1 *in entry* (i, j) *and zero elsewhere. Our convention composes paths from left to right:* e_{ij} *represents* $i \to j$*, and* $e_{ij}e_{j\ell} = e_{i\ell}$*.*

1.2 Categories and functors

We organise modules together with their homomorphisms. A category records both: specifying the objects alone does not specify a category.

Definition 1.3. *A category* $\mathcal{C}$ *consists of objects, a set* $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ *of morphisms* $f : X \to Y$ *for each pair of objects, and composition sending* $f : X \to Y$ *and* $g : Y \to Z$ *to* $g \circ f : X \to Z$*. Each object* X *has an identity* $\mathrm{id}_X : X \to X$*, and*

$$h \circ (g \circ f) = (h \circ g) \circ f, \qquad \mathrm{id}_Y \circ f = f = f \circ \mathrm{id}_X \quad \text{for all composable } f, g, h.$$

A morphism $f : X \to Y$ is an *isomorphism* if some $g : Y \to X$ satisfies $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$; we then write $X \cong Y$. A *full subcategory* of $\mathcal{C}$ consists of chosen objects of $\mathcal{C}$, with all morphisms between them and the same composition.

Example 1.4. *For a* $\Bbbk$*-algebra* A*, we use the notation*

$$\begin{aligned} \mathrm{Mod}\, A &:= \textit{category of right } A\textit{-modules and } A\textit{-linear maps}, \\ \mathrm{mod}\, A &:= \textit{full subcategory of finitely generated right } A\textit{-modules}. \end{aligned}$$

In either category, $\mathrm{Hom}_A(M, N)$ *is the space of* A*-linear maps* $M \to N$*; composition is the usual composition of maps. For* $A = \Bbbk$*, these are the categories of all and of finite-dimensional vector spaces, respectively.*

Note that, if $\dim_{\Bbbk} A < \infty$*, then* $M \in \mathrm{mod}\, A \Leftrightarrow M$ *is finite-dimensional (*$\because$ *finite generation implies that* M *is a quotient of* $A^{\oplus n}$ *some* n*, and a finite* $\Bbbk$*-basis of* M *is clearly a finite generating set.)*

Example 1.5. *For* $M \in \mathrm{mod}\, A$*, let* $\mathrm{add}\, M := \mathrm{add}_{\mathrm{mod}\, A}(M)$ *be the full subcategory with*

$$N \in \mathrm{add}\, M \quad \Longleftrightarrow \quad N \oplus N' \cong M^{\oplus r} \quad \text{for some } N' \in \mathrm{mod}\, A \text{ and integer } r \geq 0.$$

Thus $\mathrm{add}\, M$ *consists of direct summands of finite direct sums of* M*.*

In modern representation theory, remembering A and $\mathrm{add}\, A$ holds the same categorical information is crucial in utilising cateogircal technique in representation theory; this allows one to push many results to a more abstract framework.

Definition 1.6. *A functor $F : \mathcal{C} \to \mathcal{D}$ assigns an object $F(X)$ to each object X and a morphism $F(f) : F(X) \to F(Y)$ to each $f : X \to Y$, such that*

$$F(\mathrm{id}_X) = \mathrm{id}_{F(X)}, \qquad F(g \circ f) = F(g) \circ F(f) \quad \text{for all composable } f, g.$$

For example, forgetting the A-action gives a functor $\operatorname{Mod} A \to \operatorname{Mod} \Bbbk$: it keeps the underlying vector spaces and linear maps. The *identity functor* $\mathrm{id}_{\mathcal{C}}$ leaves every object and morphism unchanged; functors compose by applying one after the other. Every functor preserves isomorphisms, since $F(f^{-1})$ is inverse to $F(f)$.

The *opposite category* $\mathcal{C}^{\mathrm{op}}$ has the same objects as $\mathcal{C}$, with $\operatorname{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, Y) = \operatorname{Hom}_{\mathcal{C}}(Y, X)$ and composition reversed. A *contravariant functor* from $\mathcal{C}$ to $\mathcal{D}$ means a functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Thus it sends $f : X \to Y$ in $\mathcal{C}$ to $F(f) : F(Y) \to F(X)$ and satisfies $F(g \circ f) = F(f) \circ F(g)$. Duals will provide our main examples.

Definition 1.7. *For functors $F, G : \mathcal{C} \to \mathcal{D}$, a natural transformation $\eta : F \to G$ is a family of morphisms $\eta_X : F(X) \to G(X)$ such that*

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow[\eta_Y]{} & G(Y) \end{array} \qquad G(f) \circ \eta_X = \eta_Y \circ F(f) \quad \text{for every } f : X \to Y. \tag{1.1}$$

If every η_X is an isomorphism, η is a natural isomorphism, written $F \cong G$.

The square in (1.1) is said to *commute*: its two routes from $F(X)$ to $G(Y)$ give the same morphism. We often say a family of morphisms (resp. isomorphisms) in a category $\mathcal{C}$ is *natural in $M \in \mathcal{C}$* to mean that it defines a natural transformation (resp. isomorphism).

For example, for fixed $c \in \Bbbk$, the maps $c\,\mathrm{id}_M : M \to M$ define a natural transformation $\mathrm{id}_{\operatorname{Mod} A} \to \mathrm{id}_{\operatorname{Mod} A}$ because $f(cm) = cf(m)$ for every module homomorphism $f : M \to N$. It is a natural isomorphism when $c \neq 0$.

A functor $F : \mathcal{C} \to \mathcal{D}$ is an *equivalence* if there is a functor $G : \mathcal{D} \to \mathcal{C}$ with natural isomorphisms $GF \cong \mathrm{id}_{\mathcal{C}}$ and $FG \cong \mathrm{id}_{\mathcal{D}}$. A *duality* between $\mathcal{C}$ and $\mathcal{D}$ is an equivalence $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Thus an equivalence recovers objects and maps up to natural isomorphism; a duality also reverses the direction of maps.

Definition 1.8. *Let A be a $\Bbbk$-algebra, and let $L, M, N \in \operatorname{Mod} A$, with $i : L \to M$ and $p : M \to N$ module homomorphisms. The sequence $L \xrightarrow{i} M \xrightarrow{p} N$ is exact at M if $\operatorname{Im} i = \operatorname{Ker} p$. A sequence of module homomorphisms is exact if it is exact at every term with an incoming and an outgoing map. A short exact sequence is an exact sequence*

$$0 \longrightarrow L \xrightarrow{i} M \xrightarrow{p} N \longrightarrow 0, \qquad \operatorname{Ker} i = 0, \quad \operatorname{Im} i = \operatorname{Ker} p, \quad \operatorname{Im} p = N. \tag{1.2}$$

Definition 1.9. *Let A, B be $\Bbbk$-algebras. A functor $F : \operatorname{Mod} A \to \operatorname{Mod} B$ is exact if $F(0) = 0$ and every short exact sequence (1.2) gives a short exact sequence*

$$0 \longrightarrow F(L) \xrightarrow{F(i)} F(M) \xrightarrow{F(p)} F(N) \longrightarrow 0.$$

An exact functor preserves injective and surjective maps: apply it to $0 \to L \xrightarrow{i} M \to M/i(L) \to 0$ and $0 \to \operatorname{Ker} p \to M \xrightarrow{p} N \to 0$, respectively. For a functor $F : (\operatorname{Mod} A)^{\mathrm{op}} \to \operatorname{Mod} B$, exactness instead means that (1.2) gives

$$0 \longrightarrow F(N) \xrightarrow{F(p)} F(M) \xrightarrow{F(i)} F(L) \longrightarrow 0.$$

Thus an exact contravariant functor exchanges injective and surjective maps.

Example 1.10 (𝕜-linear duality). *For a vector space V, define its 𝕜-linear dual by $D(V) := \mathrm{Hom}_{\Bbbk}(V, \Bbbk)$. For a linear map $u : V \to W$, set*

$$D(u) : D(W) \longrightarrow D(V), \qquad \lambda \longmapsto \lambda \circ u.$$

The identities $D(\mathrm{id}_V) = \mathrm{id}_{D(V)}$ and $D(v \circ u) = D(u) \circ D(v)$ make $D : (\operatorname{Mod} \Bbbk)^{\mathrm{op}} \to \operatorname{Mod} \Bbbk$ a functor. The functor D is exact.

> *For a short exact sequence (1.2) of vector spaces, $D(p)$ is injective because p is surjective, and $D(i)$ is surjective because a functional on a subspace extends to the whole vector space. If $\lambda \in D(M)$ satisfies $\lambda \circ i = 0$, then*
>
> $$\mu : N \longrightarrow \Bbbk, \qquad p(m) \longmapsto \lambda(m)$$
>
> *is well-defined since $\operatorname{Ker} p = i(L)$, and $\lambda = \mu \circ p = D(p)(\mu)$. Conversely, $p \circ i = 0$ gives $D(i) \circ D(p) = 0$, so $\operatorname{Ker} D(i) = \operatorname{Im} D(p)$. Thus D exchanges injective and surjective linear maps.*

For every vector space V, evaluation defines a natural injective map

$$\mathrm{ev}_V : V \longrightarrow D(D(V)), \qquad v \longmapsto (\lambda \mapsto \lambda(v)). \tag{1.3}$$

If V is finite-dimensional, choose a basis $v_1, \dots, v_n$ and its dual basis $v_1^*, \dots, v_n^*$, defined by $v_i^*(v_j) = \delta_{ij}$. Then

$$\mathrm{ev}_V^{-1} : D(D(V)) \longrightarrow V, \qquad \Phi \longmapsto \sum_{i=1}^{n} \Phi(v_i^*) v_i.$$

Consequently, $D : (\operatorname{mod} \Bbbk)^{\mathrm{op}} \to \operatorname{mod} \Bbbk$ is a duality.

> *If $v \neq 0$, choose $\lambda \in D(V)$ with $\lambda(v) \neq 0$; hence ev_V is injective. For $u : V \to W$, $v \in V$, and $\lambda \in D(W)$,*
>
> $$D(D(u))(\mathrm{ev}_V(v))(\lambda) = \lambda(u(v)) = \mathrm{ev}_W(u(v))(\lambda),$$
>
> *which proves naturality.*
>
> *In finite dimension, writing $\lambda = \sum_{i=1}^n \lambda(v_i) v_i^*$ gives $\Phi(\lambda) = \lambda(\sum_{i=1}^n \Phi(v_i^*) v_i)$. Thus the displayed formula is the inverse of (1.3), and is independent of the chosen basis.*

1.3 Semisimple algebras

Definition 1.11. *A module L is simple if $L \neq 0$ and its only submodules are 0 and L. A module is semisimple if it is a direct sum of simple modules. An algebra B is semisimple if it is semisimple as a right module B_B over itself. The Jacobson radical $\operatorname{rad} B$ is the intersection of all maximal right ideals of B. For a right B-module M, the radical $\operatorname{rad} M$ is the intersection of all maximal submodules of M, with $\operatorname{rad} M = M$ if there are none.*

We use the following standard ring-theoretic facts without proof. For a finite-dimensional 𝕜-algebra B and any right B-module M,

$$\operatorname{rad} M = M \operatorname{rad} B.$$

Recall that a *division 𝕜-algebra* is a non-zero 𝕜-algebra whose non-zero elements are invertible.

Theorem 1.12 (Artin–Wedderburn). *For a non-zero finite-dimensional 𝕜-algebra B,*

$$B \text{ is semisimple} \iff \operatorname{rad} B = 0 \iff B \cong \prod_{i=1}^{n} \mathrm{Mat}_{n_i}(D_i),$$

where $n, n_i \geq 1$ and each D_i is a finite-dimensional division $\Bbbk$-algebra. If $\Bbbk$ is algebraically closed, then $D_i = \Bbbk$ for every i. For semisimple B, every left or right B-module is semisimple, and every submodule has a complement. In particular, the definition is independent of the chosen side.

Corollary 1.13. *Every finite-dimensional left or right module over a finite-diemnsional semisimple algebra S is a direct summand of $S^{\oplus n}$ for some n, i.e. $M \in \operatorname{add}(S)$ for all $M \in \operatorname{mod} S$.*

Proof Choose a surjection $p : S^{\oplus n} \twoheadrightarrow M$. By Theorem 1.12, there is a submodule L such that $S^{\oplus n} = \operatorname{Ker} p \oplus L$. The restriction $p|_L : L \to M$ has kernel $L \cap \operatorname{Ker} p = 0$ and image $p(L) = p(S^{\oplus n}) = M$. Hence $L \cong M$, so $S^{\oplus n} \cong \operatorname{Ker} p \oplus M$. □

Example 1.14. *The algebras $\Bbbk^{\oplus n}$ and $\operatorname{Mat}_n(\Bbbk)$ are semisimple by Theorem 1.12. Upper triangular matrices need not form a semisimple algebra: for $A = \Bbbk A_2$,*

$$\operatorname{rad} A = \Bbbk e_{12} \neq 0, \qquad A/\operatorname{rad} A \cong \Bbbk^{\oplus 2}.$$

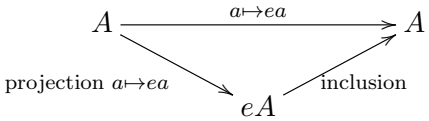
> *Put $J = \Bbbk e_{12}$, so $J^2 = 0$. For a simple right module L, simplicity gives $LJ = 0$ or $LJ = L$; the second alternative implies $L = LJ^2 = 0$, a contradiction. Hence J is contained in every maximal right ideal. The quotient $A/J \cong \Bbbk^{\oplus 2}$ has exactly two maximal right ideals, whose inverse images have intersection*
>
> $$(\Bbbk e_{11} \oplus \Bbbk e_{12}) \cap (\Bbbk e_{12} \oplus \Bbbk e_{22}) = J.$$

Definition 1.15. *An element $e \in A$ is idempotent if $e^2 = e$. Idempotents e, f are orthogonal if $ef = fe = 0$. A non-zero idempotent is primitive if it is not the sum of two non-zero orthogonal idempotents. A finite family $e_1, \ldots, e_r$ of pairwise orthogonal idempotents is complete if $\sum_{i=1}^r e_i = 1_A$.*

Remark 1.16. For orthogonal idempotents $e, f \in A$,

$$(e + f)A = eA \oplus fA.$$

In particular, we always have $A \cong eA \oplus (1 - e)A$. This strengthen the viewpoint of idempotent-as-local-identity: left-multiplying e defines a canonical projection onto the direct summand eA of A, and yields a commutative diagram



So multiplying e means we work inside the direct summand eA. The corner subalgebra $eAe = \{eae \mid a \in A\}$ has identity e (note that this is not a unital subalgera; but rather a 'subcategory'). For $A = \prod_{i=1}^r A_i$, the coordinate idempotents $e_i = (0, \ldots, 1_{A_i}, \ldots, 0)$ are a complete orthogonal family and are the identities of the individual factors. In $A = \Bbbk A_n$, the diagonal matrix units $e_{11}, \ldots, e_{nn}$ are primitive, since $e_{ii} A e_{ii} = \Bbbk e_{ii}$. For $n \geq 2$, the sum $e = e_{11} + \cdots + e_{n-1,n-1}$ gives the corner

$$eAe = \left\{ \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} \;\middle|\; B \in \Bbbk A_{n-1} \right\} \cong \Bbbk A_{n-1}.$$

Idempotents also describe maps out of the summands of A_A.

Lemma 1.17 (Yoneda lemma). *For an idempotent $e \in A$ and a right A-module M, evaluation at e gives a natural (in M) vector space isomorphism*

$$\operatorname{Hom}_A(eA, M) \xrightarrow{\sim} Me, \qquad f \longmapsto f(e), \quad \text{with inverse} \quad m \longmapsto (ea \mapsto ma).$$

Proof We have $f(e) = f(e)e$ and $f(ea) = f(e)a$. If $m = me$, the map $ea \mapsto ma$ is well-defined because $ea = ea'$ implies $ma = mea = mea' = ma'$. These formulas prove that the maps are inverse; postcomposition by a module map $M \to N$ proves naturality. □

1.4 Graded algebras

Definition 1.18. *Let G be a group with identity e. A G-graded $\Bbbk$-algebra is a $\Bbbk$-algebra A with $1_A \in A_e$ and a decomposition*

$$A = \bigoplus_{g \in G} A_g \quad \text{such that} \quad A_g A_h \subset A_{gh} \quad \text{for all } g, h \in G.$$

An element of A_g is homogeneous of degree g.

Convention. *Unless stated otherwise, a graded algebra is $\mathbb{Z}$-graded and concentrated in non-negative degrees.*

Note that the decomposition is into a direct sum, so every element of a graded algebra A can always be written as a *finite* sum of homogeneous elements.

Example 1.19. *For the upper triangular matrix ring $\Bbbk A_n$ of Example 1.2, the decomposition*

$$(\Bbbk A_n)_d = \operatorname{span}_{\Bbbk}\{e_{ij} \mid j - i = d\}$$

makes $\Bbbk A_n$ a graded algebra.

Example 1.20. *The algebra $\Bbbk A_3$ also has a $\mathbb{Z}^{\oplus 2}$-grading with*

$$\deg e_{ii} = (0,0), \qquad \deg e_{12} = (1,0), \qquad \deg e_{23} = (0,1), \qquad \deg e_{13} = (1,1).$$

The homomorphism $\mathbb{Z}^{\oplus 2} \to \mathbb{Z}$, $(p,q) \mapsto p+q$, gives the grading of Example 1.19. Alternatively, setting $\deg e_{ij} = 2(j-i)$ gives a $\mathbb{Z}$-grading with $A_1 = 0$ but $A_2 \neq 0$.

Definition 1.21. *A homomorphism of graded algebras $f : A \to B$ is a $\Bbbk$-algebra homomorphism with $f(A_i) \subset B_i$ for all $i \in \mathbb{Z}$.*

Definition 1.22. *A graded algebra A is locally finite-dimensional (lfd) if $\dim_{\Bbbk} A_i < \infty$ for every $i \in \mathbb{Z}$.*

1.5 Graded modules

Definition 1.23. *Let $A = \bigoplus_{g \in G} A_g$ be a G-graded algebra. A G-graded right A-module is a right A-module M with a direct sum decomposition*

$$M = \bigoplus_{g \in G} M_g \quad \text{such that} \quad M_g A_h \subset M_{gh} \quad \text{for all } g, h \in G.$$

An element of M_g is called homogeneous of degree g.

A G-graded A-module $M = \bigoplus_{g \in G} M_g$ is locally finite-dimensional (lfd) if $\dim_{\Bbbk} M_g < \infty$ for all $g \in G$.

Unless specified otherwise, graded modules are $\mathbb{Z}$-graded; their degrees need not be non-negative.

For an algebra A, the opposite algebra A^{op} has the same vector space as A, with multiplication $a^{\mathrm{op}} b^{\mathrm{op}} = (ba)^{\mathrm{op}}$. A left A-module is a right A^{op}-module by $m \cdot a^{\mathrm{op}} = am$. Thus $\operatorname{Mod}(A^{\mathrm{op}})$ is the category of left A-modules and left A-linear maps; not to be confused with the opposite category $(\operatorname{Mod} A)^{\mathrm{op}}$ though. For a $\mathbb{Z}$-graded algebra $A = \bigoplus_{i \geq 0} A_i$,

$$\text{the opposite algebra is also graded with } (A^{\mathrm{op}})_i = A_i.$$

Definition 1.24. *Let M be a graded A-module and let $d \in \mathbb{Z}$. The degree shift, or grading shift, $M(d)$ is the graded A-module whose underlying A-module is M, but whose grading is*

$$M(d)_j = M_{j+d}.$$

By Definition 1.24, $1 \in A_0 = A(-d)_d$, so $A(-d)$ has a generator in degree d. If M is concentrated in degree 0, then $M(d)$ is concentrated in degree $-d$.

Example 1.25. *For $A = \Bbbk A_3$ with the grading of Example 1.19, the module $P_1 := e_{11}A$ consists of matrices supported on the first row. Its basis e_{11}, e_{12}, e_{13} has degrees $0, 1, 2$, respectively; in $P_1(-2)$, the same basis has degrees $2, 3, 4$.*

Definition 1.26. *Let M, N be graded A-modules. A homomorphism $f : M \to N$ is homogeneous of degree d if $f(M_j) \subset N_{j+d}$ for all $j \in \mathbb{Z}$. A homomorphism of graded modules means a homogeneous A-module homomorphism of degree 0, also said to preserve degrees.*

Example 1.27. *For $A = \Bbbk A_3$, left multiplication by e_{12} yields a map*

$$\begin{aligned} e_{22}A &\longrightarrow e_{11}A, \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & u & v \\ 0 & 0 & 0 \end{pmatrix} &\longmapsto \begin{pmatrix} 0 & u & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad (u, v \in \Bbbk). \end{aligned}$$

The u and v entries have degrees $0, 1$ in the source and $1, 2$ in the target, and so the map has degree 1.

Notation. *For graded right A-modules M, N, the space of homomorphisms preserving degrees is*

$$\mathrm{Hom}_A^{\mathbb{Z}}(M, N) = \{f \in \mathrm{Hom}_A(M, N) \mid f(M_j) \subset N_j \text{ for all } j \in \mathbb{Z}\}. \tag{1.4}$$

Thus $\mathrm{Hom}_A^{\mathbb{Z}}(M, N(d))$ consists of homomorphisms $M \to N$ of degree d, since $N(d)_j = N_{j+d}$. Note that, like the grading shift functor, different literature has different notation for this (there does not seem to be any widely agreed notation).

We write

$$\begin{aligned} \mathrm{GrMod}\, A &:= \text{category of graded right } A\text{-modules with morphism spaces } \mathrm{Hom}_A^{\mathbb{Z}}(M, N), \\ \mathrm{grmod}\, A &:= \text{full subcategory of } \mathrm{GrMod}\, A \text{ consisting of locally finite-dimensional modules.} \end{aligned}$$

Thus $\mathrm{GrMod}(A^{\mathrm{op}})$ is the category of graded left A-modules and degree-preserving maps. Forgetting the grading gives a functor $\mathrm{GrMod}\, A \to \mathrm{Mod}\, A$.

Remark 1.28. $\mathrm{grmod}\, A$ is somewhat analogous to $\mathrm{mod}\, A$ but not really the same: If A is locally finite-dimensional, every finitely generated graded A-module M is locally finite-dimensional. ($\because$ a finite homogeneous generating set of a graded A-module M gives a surjection from a finite direct sum of shifts of A to M, so every M_i has to be finite-dimensional.) However, a locally finite-dimensional M need not be finitely generated.

For $d \in \mathbb{Z}$, the shift of Definition 1.24 is a functor

$$(d) : \mathrm{GrMod}\, A \to \mathrm{GrMod}\, A.$$

It sends $f : M \to N$ to the same underlying map $f(d) : M(d) \to N(d)$. Since $M(d)(-d) = M$ and $f(d)(-d) = f$, the shift is an equivalence with inverse $(-d)$; it restricts to an equivalence $\mathrm{grmod}\, A \to \mathrm{grmod}\, A$.

Remark 1.29. The notion of exactness makes sense for graded module categories – a sequence in $\mathrm{GrMod}\, A$ is exact if the underlying sequence in $\mathrm{Mod}\, A$ is exact. In particular, we talk about functors invovle such categories being exact.

For $L, M, N \in \operatorname{GrMod} A$ with maps $i : L \to M$ and $p : M \to N$, the sequence $0 \to L \xrightarrow{i} M \xrightarrow{p} N \to 0$ is exact if and only if

$$0 \longrightarrow L_j \xrightarrow{i|_{L_j}} M_j \xrightarrow{p|_{M_j}} N_j \longrightarrow 0 \quad \text{is exact for every } j \in \mathbb{Z}.$$

Indeed, i and p preserve the direct sum decompositions, so their kernels and images decompose into the kernels and images of the restrictions.

Example 1.30. *Let $A = \Bbbk A_2$. Consider $P_1 := e_{11}A$ which has $\operatorname{rad} P_1 = \Bbbk e_{12}$ and $S_1 := P_1/\operatorname{rad} P_1$. Then we have a short exact sequence in* $\operatorname{GrMod} A$*:*

$$0 \longrightarrow \operatorname{rad} P_1 \xrightarrow{i} P_1 \xrightarrow{p} P_1/\operatorname{rad} P_1 \longrightarrow 0, \qquad i(ce_{12}) = ce_{12}, \quad p(m) = \overline{m}.$$

Note that $\deg \operatorname{rad} P_1 = 1$ and $\deg S_1 = 0$. Let $S_2 := \Bbbk e_{22} = e_{22}A$, which we treat as a graded A-module concentrated in degree 0. Note that $S_2 \cong \operatorname{rad} P_1$ as ungraded modules but we only have $S_2 \cong (\operatorname{rad} P_1)(1)$, which yields the following short exact sequence in $\operatorname{GrMod} A$*:*

$$0 \longrightarrow S_2(-1) \to P_1 \to P_1/\operatorname{rad} P_1 \longrightarrow 0,$$

Lemma 1.31. *For graded right A-modules M, N, there is an inclusion*

$$\bigoplus_{d\in\mathbb{Z}} \operatorname{Hom}_A^{\mathbb{Z}}(M, N(d)) \subset \operatorname{Hom}_A(M, N), \text{ which is equality when } M \text{ is finitely generated.}$$

Proof For $f \in \operatorname{Hom}_A(M, N)$, define $f_d(m) = f(m)_{j+d}$ when $m \in M_j$, and extend linearly. For $a \in A_i$, we have $f_d(ma) = f(ma)_{j+i+d} = f(m)_{j+d}a$, so f_d is A-linear of degree d. The sum of the homogeneous Hom spaces is direct: on M_j, distinct degrees have values in distinct summands of N.

If M is finitely generated, the homogeneous components of a finite generating set form a finite homogeneous generating set. Only finitely many f_d are non-zero on these homogeneous generators, hence on M, giving $f = \sum_d f_d$. □

1.6 Homogeneous ideals and quotients

Definition 1.32. *Let $A = \bigoplus_{i\in\mathbb{Z}} A_i$ be a graded algebra. A right ideal $I \subset A$ is homogeneous if*

$$I = \bigoplus_{i\in\mathbb{Z}} (I \cap A_i).$$

Equivalently, whenever $a \in I$ and $a = a_0 + \cdots + a_N$ is its decomposition into homogeneous components, each a_i also belongs to I.

For quotient algebras we need two-sided ideals. If I is a homogeneous two-sided ideal, then

$$A/I = \bigoplus_{i\in\mathbb{Z}} A_i/(I \cap A_i)$$

is again a graded algebra.

Example 1.33. *In $A = \Bbbk A_3$ with the grading of Example 1.19, the two-sided ideal $I = (e_{13}) = \Bbbk e_{13}$ is homogeneous. Writing $\overline{e}_{ij}$ for the image of e_{ij},*

$$(A/I)_0 \cong \Bbbk^{\oplus 3}, \qquad (A/I)_1 = \Bbbk\overline{e}_{12} \oplus \Bbbk\overline{e}_{23}, \qquad (A/I)_d = 0 \quad (d \geq 2).$$

In particular, $\overline{e}_{12}\overline{e}_{23} = 0$.

Example 1.34. *Let $A = \Bbbk[x]$ with $\deg x = 1$, and let $I = (x - x^2)$. The ideal I is two-sided because A is commutative. Every element of I vanishes at $x = 1$, whereas neither x nor x^2 does. Thus $x - x^2 \in I$ but $x, x^2 \notin I$, so I is not homogeneous. Writing $\overline{f} = f + I$, we have $0 \neq \overline{x} = \overline{x^2}$. The images of A_1 and A_2 therefore intersect non-trivially, so A/I does not inherit the quotient grading. As an ungraded algebra, the quotient is semisimple:*

$$\Bbbk[x]/(x - x^2) \xrightarrow{\sim} \Bbbk \oplus \Bbbk, \qquad \overline{f} \longmapsto (f(0), f(1)).$$

The inverse sends (a, b) to $a + (b - a)\overline{x}$.

Definition 1.35. *The positive degree ideal of a graded algebra A is $A_+ := \bigoplus_{i>0} A_i$.*[1]

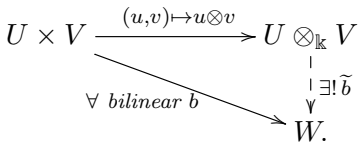
The subspace A_+ is a homogeneous two-sided ideal.

1.7 Tensor products of vector spaces

Definition 1.36. *Choose a basis $\{u_i \mid i \in I\}$ of U and a basis $\{v_j \mid j \in J\}$ of V. The tensor product $U \otimes_{\Bbbk} V$ is the vector space with basis $\{u_i \otimes v_j \mid i \in I,\ j \in J\}$.*

The assignment $(u_i, v_j) \mapsto u_i \otimes v_j$ extends uniquely to a bilinear map $U \times V \to U \otimes_{\Bbbk} V$ sending $(u, v) \mapsto u \otimes v$. Note that this means that elements like $(au_1 + bu_2) \otimes (cv_1 + dv_2)$ in $U \otimes_{\Bbbk} V$ is by definition the linear combination $ac(u_1 \otimes v_1) + ad(u_1 \otimes v_2) + bc(u_2 \otimes v_1) + bd(u_2 \otimes v_2)$.

Proposition 1.37 (Universal property). *Every bilinear map $b : U \times V \to W$ has a unique linear factorisation (commutative diagram)*



Consequently, $U \otimes_{\Bbbk} V$ is independent of the chosen bases up to a unique isomorphism preserving the bilinear map.

Proof The formula $\widetilde{b}(u_i \otimes v_j) = b(u_i, v_j)$ determines a unique linear map $\widetilde{b} : U \otimes_{\Bbbk} V \to W$. Bilinearity gives $\widetilde{b}(u \otimes v) = b(u, v)$ for all $u \in U$ and $v \in V$.

Suppose $t : U \times V \to T$ and $t' : U \times V \to T'$ both satisfy the universal property. They give unique linear maps $a : T \to T'$ and $b : T' \to T$ with $a \circ t = t'$ and $b \circ t' = t$. Both $b \circ a : T \to T$ and $\mathrm{id}_T : T \to T$ factor t through itself, since

$$(b \circ a) \circ t = b \circ t' = t = \mathrm{id}_T \circ t.$$

Uniqueness for (T, t) gives $b \circ a = \mathrm{id}_T$. Likewise, $(a \circ b) \circ t' = a \circ t = t' = \mathrm{id}_{T'} \circ t'$, so uniqueness for (T', t') gives $a \circ b = \mathrm{id}_{T'}$. □

For finite bases with $|I| = r$ and $|J| = s$, we obtain $\dim_{\Bbbk}(U \otimes_{\Bbbk} V) = rs$. We use the canonical identifications

$$(U \otimes_{\Bbbk} V) \otimes_{\Bbbk} W \xrightarrow{\sim} U \otimes_{\Bbbk} (V \otimes_{\Bbbk} W), \quad (u \otimes v) \otimes w \longmapsto u \otimes (v \otimes w),$$
$$\Bbbk \otimes_{\Bbbk} U \xrightarrow{\sim} U, \qquad c \otimes u \longmapsto cu,$$

and set $U^{\otimes_{\Bbbk} 0} := \Bbbk$.

[1] It is also called the augmentation ideal; we avoid that terminology.

More generally, when I, J are infinite, the tensor product can be defined as a pair. The map $(u, v) \mapsto u \otimes v$ can be defined as the linear extension of

$$\Big(\sum_{i\in I} a_i u_i, \sum_{j\in J} b_j v_j\Big) \longmapsto \sum_{(i,j)\in I\times J} a_i b_j (u_i \otimes v_j) \quad \text{for finitely supported families } (a_i)_{i\in I}, (b_j)_{j\in J}.$$

Then universal property still holds for the $U \otimes_{\Bbbk} V$.

1.8 Direct sums and direct products

For an arbitrary family $(M_i)_{i\in I}$ of right A-modules, indexed by a set I, the *direct product* and *direct sum* are

$$\prod_{i\in I} M_i = \{(m_i)_{i\in I} \mid m_i \in M_i \text{ for all } i \in I\},$$
$$\bigoplus_{i\in I} M_i = \{(m_i)_{i\in I} \in \prod_{i\in I} M_i \mid m_i = 0 \text{ for all but finitely many } i\}.$$

Addition and the A-action are componentwise; the same definitions apply to vector spaces by taking $A = \Bbbk$. Thus $\bigoplus_{i\in I} M_i \subset \prod_{i\in I} M_i$, with equality for finite I. For example, $(1, 1, \ldots) \in \prod_{i\geq 0} \Bbbk$ does not belong to $\bigoplus_{i\geq 0} \Bbbk$. In a direct sum, an element is a finite sum of its components; no notion of convergence is involved. For instance, ordinary polynomials use non-negative powers, whereas Laurent polynomials allow integer powers, but both are finite sums:

$$\Bbbk[x] = \bigoplus_{n\geq 0} \Bbbk x^n, \qquad \Bbbk[x, x^{-1}] = \bigoplus_{n\in\mathbb{Z}} \Bbbk x^n.$$

The *formal power series ring* allows arbitrary coefficients:

$$\Bbbk[[x]] = \left\{ \sum_{n\geq 0} a_n x^n \;\middle|\; a_n \in \Bbbk \text{ for every } n \geq 0 \right\} \cong \prod_{n\geq 0} \Bbbk x^n \quad \text{as vector spaces.}$$

Addition is coefficientwise, and multiplication is

$$\Big(\sum_{n\geq 0} a_n x^n\Big)\Big(\sum_{n\geq 0} b_n x^n\Big) = \sum_{n\geq 0} \Big(\sum_{j=0}^{n} a_j b_{n-j}\Big) x^n.$$

Tensor products distribute over direct sums in either variable. For vector spaces U, V and arbitrary families $(U_i)_{i\in I}$, $(V_j)_{j\in J}$ indexed by sets I, J, there are natural linear isomorphisms

$$\begin{aligned} \Big(\bigoplus_{i\in I} U_i\Big) \otimes_{\Bbbk} V &\xrightarrow{\sim} \bigoplus_{i\in I} (U_i \otimes_{\Bbbk} V), \quad (u_i)_{i\in I} \otimes v \longmapsto (u_i \otimes v)_{i\in I}, \\ U \otimes_{\Bbbk} \Big(\bigoplus_{j\in J} V_j\Big) &\xrightarrow{\sim} \bigoplus_{j\in J} (U \otimes_{\Bbbk} V_j), \quad u \otimes (v_j)_{j\in J} \longmapsto (u \otimes v_j)_{j\in J}. \end{aligned} \tag{1.5}$$

The displayed bilinear assignments induce these linear maps by Proposition 1.37. On the ith summand, the inverse of the first map sends $u_i \otimes v$ to the element with u_i in coordinate i, tensored with v. On the jth summand, the inverse of the second sends $u \otimes v_j$ to u tensored with the element having v_j in coordinate j. Extending over the direct sums gives the two inverses.

For direct products, the corresponding map need not be an isomorphism.

Lemma 1.38. *For an arbitrary family $(U_i)_{i\in I}$ of vector spaces, indexed by a set I, and a vector space V, the coordinate projections define a linear map*

$$\Big(\prod_{i\in I} U_i\Big) \otimes_{\Bbbk} V \longrightarrow \prod_{i\in I}(U_i \otimes_{\Bbbk} V), \qquad (u_i)_i \otimes v \longmapsto (u_i \otimes v)_i. \tag{1.6}$$

This map is injective, and is an isomorphism if I is finite or V is finite-dimensional. The same statements hold with the two tensor factors reversed.

Proof The bilinear map $\big(\prod_{i\in I} U_i\big) \times V \to \prod_{i\in I}(U_i \otimes_{\Bbbk} V)$, $((u_i)_i, v) \mapsto (u_i \otimes v)_i$, induces (1.6) by Proposition 1.37. Every element can be written $\sum_{\ell=1}^{r} u^{(\ell)} \otimes v_\ell$ with the v_ℓ linearly independent: choose a basis of the span of its finitely many second factors. If its image is zero, then $\sum_{\ell=1}^{r} u_i^{(\ell)} \otimes v_\ell = 0$ for every i. The tensor-product basis construction gives $u_i^{(\ell)} = 0$ for all i, ℓ, proving injectivity. If $v_1, \dots, v_r$ is a basis of V, the inverse is

$$\prod_{i\in I}(U_i \otimes_{\Bbbk} V) \longrightarrow \Big(\prod_{i\in I} U_i\Big) \otimes_{\Bbbk} V, \qquad \Big(\sum_{\ell=1}^{r} u_i^{(\ell)} \otimes v_\ell\Big)_i \longmapsto \sum_{\ell=1}^{r} (u_i^{(\ell)})_i \otimes v_\ell.$$

For finite I, products equal sums. Reversing the tensor factors gives the other statements, since $U \otimes_{\Bbbk} V \to V \otimes_{\Bbbk} U$, $u \otimes v \mapsto v \otimes u$, is an isomorphism. □

Example 1.39. *The comparison in* (1.6) *need not be surjective when V is infinite-dimensional. Take $U_i = \Bbbk$ for $i \geq 1$ and $V = \bigoplus_{j\geq 1} \Bbbk v_j$. Under $\Bbbk \otimes_{\Bbbk} V \cong V$, the image of the product comparison in* (1.6) *is*

$$\left\{ (w_i)_i \in \prod_{i\geq 1} V \;\middle|\; \dim_{\Bbbk} \operatorname{span}_{\Bbbk}\{w_i \mid i \geq 1\} < \infty \right\}.$$

Indeed, the image of $\sum_{\ell=1}^{r} (c_i^{(\ell)})_i \otimes w_\ell$ has every entry in $\operatorname{span}_{\Bbbk}\{w_1, \dots, w_r\}$. Conversely, expand all entries in a basis of their common finite-dimensional span. Since the v_i are linearly independent, $(v_i)_i$ is not in the image.

1.9 Bimodules and enveloping algebras

Definition 1.40. *For $\Bbbk$-algebras B, C, a bimodule ${}_BM_C$ is a vector space with $\Bbbk$-bilinear left B- and right C-module actions satisfying $(bm)c = b(mc)$ for all $b \in B$, $c \in C$, and $m \in M$. The subscripts indicate the left and right acting algebras; when $B = C = A$, we call M an A-bimodule. A bimodule homomorphism $f : {}_BM_C \to {}_BN_C$ is a linear map satisfying $f(bmc) = bf(m)c$ for all $b \in B$, $c \in C$, and $m \in M$. For graded B, C, a bimodule M is graded if $M = \bigoplus_{i\in\mathbb{Z}} M_i$ and $B_i M_j C_k \subset M_{i+j+k}$ for all $i, j, k \in \mathbb{Z}$. A graded bimodule homomorphism is a bimodule homomorphism $f : M \to N$ satisfying $f(M_i) \subset N_i$ for all $i \in \mathbb{Z}$.*

Convention.

- *Bilinearity implies that $\Bbbk$ acts centrally: $(c1_B)m = cm = m(c1_C)$ for $c \in \Bbbk$. From the perspective of general module theory over general rings (algebras over non-field commutative rings), we should have said that our bimodules are $\Bbbk$-central bimodule. Since we alawys work over a field, all bimodules we considered are always $\Bbbk$-central and we will not write such adjective every time.*
- *Instead of saying (B, C)-bimodule, we usually just write ${}_BM_C$. It would be natural to write $B\operatorname{Mod} C$ for the category of (B, C)-bimodules, but since $\Bbbk$-centrality comes for free from the*

assumption of $\Bbbk$ *being a field, we can use* $\operatorname{Mod} C \otimes_{\Bbbk} B^{\mathrm{op}}$ *instead.*

Example 1.41.
- *Multiplication makes* A_A *a graded right module and* ${}_AA_A$ *a graded bimodule; each degree component is a bimodule* ${}_{A_0}(A_i)_{A_0}$.
- *For a bimodule* ${}_AM_A$ *and idempotents* $e, f \in A$, *the space* $eMf := \{enf \mid n \in M\}$ *is a bimodule* ${}_{eAe}(eMf)_{fAf}$.

Definition 1.42. *For a* $\Bbbk$*-algebra* A, *its enveloping algebra is* $A^{\mathrm{en}} := A \otimes_{\Bbbk} A^{\mathrm{op}}$, *with multiplication extended bilinearly from*

$$(a \otimes b^{\mathrm{op}})(a' \otimes (b')^{\mathrm{op}}) = aa' \otimes (b'b)^{\mathrm{op}}, \qquad 1_{A^{\mathrm{en}}} = 1_A \otimes 1_A^{\mathrm{op}} \quad (a, b, a', b' \in A).$$

A bimodule ${}_AM_A$ becomes a right A^{en}-module by

$$m \cdot (a \otimes b^{\mathrm{op}}) = bma \quad (a, b \in A,\ m \in M). \tag{1.7}$$

The first tensor factor records the right action, and the second the left action:

$$\big(m \cdot (a \otimes b^{\mathrm{op}})\big) \cdot (a' \otimes (b')^{\mathrm{op}}) = b'bmaa' = m \cdot \big(aa' \otimes (b'b)^{\mathrm{op}}\big).$$

Conversely, for $M \in \operatorname{Mod} A^{\mathrm{en}}$, define $ma = m \cdot (a \otimes 1)$ and $bm = m \cdot (1 \otimes b^{\mathrm{op}})$. The two actions commute because $a \otimes 1$ and $1 \otimes b^{\mathrm{op}}$ commute in A^{en}. For bimodules ${}_AM_A$, ${}_AN_A$, a linear map $u : M \to N$ satisfies

$$u(bma) = bu(m)a \quad (a, b \in A,\ m \in M) \quad \Longleftrightarrow \quad u \in \operatorname{Hom}_{A^{\mathrm{en}}}(M, N).$$

Thus $A \operatorname{Mod} A$ is identified with $\operatorname{Mod} A^{\mathrm{en}}$. More generally, for $\Bbbk$-algebras B, C, the vector space $C \otimes_{\Bbbk} B^{\mathrm{op}}$ is an algebra with

$$(c \otimes b^{\mathrm{op}})(c' \otimes (b')^{\mathrm{op}}) = cc' \otimes (b'b)^{\mathrm{op}}, \qquad 1 = 1_C \otimes 1_B^{\mathrm{op}}.$$

The same construction identifies the categories

$$B \operatorname{Mod} C \simeq \operatorname{Mod}(C \otimes_{\Bbbk} B^{\mathrm{op}}), \qquad m \cdot (c \otimes b^{\mathrm{op}}) = bmc.$$

Grant–Iyama [GI20, §2, p. 2591] identify bimodules with left modules over A^{en}, using $(a \otimes b^{\mathrm{op}})m = amb$. Our right module convention is (1.7): $m \cdot (a \otimes b^{\mathrm{op}}) = bma$. Both describe the same bimodules; the side on which A^{en} acts determines the formula.

Remark 1.43. For a finite-dimensional semisimple $\Bbbk$-algebra S, semisimplicity of S^{en} requires an additional condition. For a field extension $K/\Bbbk$, give $K \otimes_{\Bbbk} S$ the multiplication $(c \otimes s)(d \otimes t) = cd \otimes st$. The algebra S is called *separable over* $\Bbbk$ if $K \otimes_{\Bbbk} S$ is semisimple for every field extension $K/\Bbbk$. By [DK94, Theorem 6.1.2, p. 105],

$$S^{\mathrm{en}} \text{ is semisimple} \quad \Longleftrightarrow \quad S \text{ is separable over } \Bbbk.$$

For a product of matrix algebras, every field extension $K/\Bbbk$ gives

$$K \otimes_{\Bbbk} \prod_{j=1}^{n} \operatorname{Mat}_{n_j}(\Bbbk) \xrightarrow{\sim} \prod_{j=1}^{n} \operatorname{Mat}_{n_j}(K), \qquad c \otimes (a_j)_j \longmapsto (ca_j)_j.$$

The target is semisimple by Theorem 1.12. Thus products of matrix algebras over $\Bbbk$ are separable; in particular, every finite-dimensional semisimple algebra over an algebraically closed field

is separable. Semisimplicity of S alone does not imply semisimplicity of S^{en} over an arbitrary field.

Example 1.44. *Let $\Bbbk = \mathbb{F}_2(t)$, where t is an indeterminate, and let $S = \Bbbk[u]/(u^2 - t)$, also writing u for its residue class. The element t is not a square in $\Bbbk$: an equality $p(t)^2 = tq(t)^2$ for non-zero polynomials would give $2\deg p = 1 + 2\deg q$. Thus $u^2 - t$ is irreducible and S is a field, hence semisimple. However, since S is commutative and has characteristic 2,*

$$S^{\mathrm{en}} = S \otimes_{\Bbbk} S \cong S[v]/(v^2 - t) = S[v]/((v-u)^2) \cong S[\varepsilon]/(\varepsilon^2).$$

Here the first tensor factor supplies the coefficients, v corresponds to $1 \otimes u$, and $\varepsilon = 1 \otimes u - u \otimes 1$. The ideal $S\varepsilon$ is non-zero, has square zero, and is the unique maximal ideal, so $\operatorname{rad}(S^{\mathrm{en}}) = S\varepsilon \neq 0$. Thus S^{en} is not semisimple, and the field extension $S/\Bbbk$ already shows that S is not separable over $\Bbbk$.

1.10 Graded $\Bbbk$-linear duals

A linear functional on a graded vector space can be non-zero on infinitely many degrees. To obtain a graded space, retain only finite sums of homogeneous functionals.

Definition 1.45. *For a graded vector space $V = \bigoplus_{i \in \mathbb{Z}} V_i$, the graded $\Bbbk$-linear dual is the graded vector space*

$$\mathbb{D}(V) = \bigoplus_{i \in \mathbb{Z}} \mathbb{D}(V)_i, \qquad \mathbb{D}(V)_i = D(V_{-i}) = \operatorname{Hom}_{\Bbbk}^{\mathbb{Z}}(V, \Bbbk(i)).$$

Note that here $\Bbbk$ is regarded as a graded module concentrated in degree 0.

Linear duality reverses the side of a module action. For a right A-module M and a left A-module N, set

$$(af)(m) = f(ma), \qquad (ga)(n) = g(an) \quad \text{for } a \in A,\ f \in D(M),\ g \in D(N),\ m \in M,\ n \in N. \tag{1.8}$$

For graded modules these actions restrict to the graded linear duals: if $a \in A_p$ and $f \in \mathbb{D}(M)_q$, then af vanishes outside M_{-p-q}.

Example 1.46. *Let $A = \Bbbk A_3$ and $P_1 = e_{11}A$ as in Example 1.25. The functionals $\varepsilon_{1j}(e_{1\ell}) = \delta_{j\ell}$ for $1 \leq j, \ell \leq 3$ form a basis of $\mathbb{D}(P_1)$, with*

$$\deg \varepsilon_{1j} = 1 - j, \qquad e_{12}\varepsilon_{12} = \varepsilon_{11}, \qquad e_{23}\varepsilon_{13} = \varepsilon_{12}, \qquad e_{jj}\varepsilon_{1j} = \varepsilon_{1j}.$$

The actions follow from (1.8).

For any $u \in \operatorname{Hom}_A(M, N)$, we have a dual map

$$D(u) : D(N) \to D(M), \qquad f \mapsto f \circ u,$$

which is a left A-linear map, i.e. $D(u) \in \operatorname{Hom}_{A^{\mathrm{op}}}(D(N), D(M))$. This gives rise to four exact (contravariant) functors (without any finiteness assumption):

$$\begin{aligned} D &: (\operatorname{Mod} A)^{\mathrm{op}} \longrightarrow \operatorname{Mod}(A^{\mathrm{op}}), & D &: (\operatorname{Mod}(A^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{Mod} A, \\ \mathbb{D} &: (\operatorname{GrMod} A)^{\mathrm{op}} \longrightarrow \operatorname{GrMod}(A^{\mathrm{op}}), & \mathbb{D} &: (\operatorname{GrMod}(A^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{GrMod} A. \end{aligned} \tag{1.9}$$

Exactness follows from Example 1.10, degree by degree for $\mathbb{D}$.

For these to be duality, we need to recover a module from its bidual; this requires some finiteness conditions.

Lemma 1.47. *Let A be a $\Bbbk$-algebra.*

(1) *The functor D defines a duality between finite-dimensional right and left A-modules. In particular, if A is finite-dimensional, then we have a duality on finitely generated modules*

$$D : (\operatorname{mod} A)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod}(A^{\mathrm{op}}), \qquad D : (\operatorname{mod}(A^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod} A.$$

(2) *Suppose that A is graded. For locally finite-dimensional $M \in \operatorname{grmod} A$, evaluation is a natural graded module isomorphism*

$$\mathrm{ev}_M : M \xrightarrow{\sim} \mathbb{D}(\mathbb{D}(M)), \qquad m \longmapsto (f \mapsto f(m)). \tag{1.10}$$

In particular, we have a duality between locally finite-dimensional graded right and left A-modules:

$$\mathbb{D} : (\operatorname{grmod} A)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod}(A^{\mathrm{op}}), \qquad \mathbb{D} : (\operatorname{grmod}(A^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod} A. \tag{1.11}$$

Proof First we show that ev_M is A-linear. Indeed,

$$\mathrm{ev}_M(ma)(f) = f(ma) = \mathrm{ev}_M(m)(af) = (\mathrm{ev}_M(m)a)(f) \quad (m \in M,\ a \in A,\ f \in D(M)).$$

For (1), the claim follows from Example 1.10. For (2), apply (1) degree-wise. □

For a locally finite-dimensional graded vector space V, restriction gives

$$\mathbb{D}(V) = \bigoplus_{i \in \mathbb{Z}} D(V_{-i}) \subset D(V) \cong \prod_{i \in \mathbb{Z}} D(V_i).$$

The inclusion is equality exactly when only finitely many V_i are non-zero. For an arbitrary vector space, evaluation $V \to D(D(V))$ is injective but need not be surjective; hence the functors in (1.9) are not dualities on all modules.

Example 1.48. *Graded linear duality need not preserve finite generation, even for a polynomial ring. Let $A = \Bbbk[x]$ with $\deg x = 1$ and let $\varepsilon_n(x^m) = \delta_{nm}$ for $n, m \geq 0$. Then $\deg \varepsilon_n = -n$, $x\varepsilon_0 = 0$, and $x\varepsilon_n = \varepsilon_{n-1}$ for $n \geq 1$. Any finite set of elements of $\mathbb{D}(A)$ lies in the span of $\varepsilon_0, \ldots, \varepsilon_N$ for some N, and multiplication by x preserves that span. Thus $\mathbb{D}(A)$ is locally finite-dimensional but not finitely generated.*

For vector spaces V, W, a functional $f \in D(V)$ and a vector $w \in W$ define a linear map $V \to W$, $v \mapsto f(v)w$. Finite sums of these maps give the following description of Hom.

Proposition 1.49. *For vector spaces V, W, there is a natural linear map*

$$W \otimes_{\Bbbk} D(V) \longrightarrow \operatorname{Hom}_{\Bbbk}(V, W), \qquad w \otimes f \longmapsto \big(v \mapsto f(v)w\big). \tag{1.12}$$

It is an isomorphism if either V or W is finite-dimensional.

Proof If V has a finite basis $\{v_i \mid 1 \leq i \leq n\}$ with dual basis $\{v_i^* \mid 1 \leq i \leq n\}$, the inverse of (1.12) is

$$\operatorname{Hom}_{\Bbbk}(V, W) \longrightarrow W \otimes_{\Bbbk} D(V), \qquad F \longmapsto \sum_{i=1}^{n} F(v_i) \otimes v_i^*,$$

since $v = \sum_{i=1}^{n} v_i^*(v) v_i$. If instead W has a finite basis $\{w_j \mid 1 \leq j \leq m\}$, write $F(v) = \sum_{j=1}^{m} f_j(v) w_j$; the uniquely determined coordinate functionals $f_j \in D(V)$ give the inverse

$F \mapsto \sum_{j=1}^m w_j \otimes f_j$. The formulas commute with precomposition and postcomposition, so the comparison is natural. □

1.11 Left, right, and two-sided duals over S

Fix a finite-dimensional semisimple $\Bbbk$-algebra S, which is regarded as concentrated in degree 0 when treated as a graded algebra. For a graded algebra A with $A_0 = S$, each degree component is a bimodule ${}_S(A_i)_S$.

Definition 1.50. *For a right S-module M and a left S-module N, define their S-duals by*

$$M^* := \operatorname{Hom}_S(M, S) \in \operatorname{Mod}(S^{\mathrm{op}}), \quad (sf)(m) := sf(m),$$
$${}^*N := \operatorname{Hom}_{S^{\mathrm{op}}}(N, S) \in \operatorname{Mod} S, \qquad (gs)(n) := g(n)s.$$

Here $s \in S$, $m \in M$, $n \in N$, $f \in M^*$, and $g \in {}^*N$.

Similar to $\Bbbk$-lienar duality, this definition extends to exact contravariant functors: for a right module map $u : M \to N$, define $u^* : N^* \to M^*$ by precomposition $u^*(f) = f \circ u$; for a left module map $u : M \to N$, the same recipe defines ${}^*u : {}^*N \to {}^*M$. This yields the desired functors

$$(-)^* : (\operatorname{Mod} S)^{\mathrm{op}} \longrightarrow \operatorname{Mod}(S^{\mathrm{op}}), \qquad {}^*(-) : (\operatorname{Mod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{Mod} S.$$

We also have graded analogue similar to Definition 1.45.

Definition 1.51. *For graded right and left S-modules M, respectively, define their graded S-duals by*

$$M^{(*)} := \bigoplus_{i\in\mathbb{Z}} (M_{-i})^* = \bigoplus_{i\in\mathbb{Z}} \operatorname{Hom}_S^{\mathbb{Z}}(M, S(i)), \qquad {}^{(*)}M := \bigoplus_{i\in\mathbb{Z}} {}^*(M_{-i}) = \bigoplus_{i\in\mathbb{Z}} \operatorname{Hom}_{S^{\mathrm{op}}}^{\mathbb{Z}}(M, S(i)),$$

where each indicated summand has degree i.

This also extends to a functor: for a degree 0 right S-module map $u : M \to N$, define $u^{(*)} : N^{(*)} \to M^{(*)}$ by

$$(u^{(*)})_i : (N_{-i})^* \longrightarrow (M_{-i})^*, \qquad f \longmapsto f \circ u|_{M_{-i}} \quad (i \in \mathbb{Z}).$$

For a degree 0 left module map $u : M \to N$, the same formula defines ${}^{(*)}u : {}^{(*)}N \to {}^{(*)}M$. With the actions from Definition 1.50, we obtain functors

$$(-)^{(*)} : (\operatorname{GrMod} S)^{\mathrm{op}} \longrightarrow \operatorname{GrMod}(S^{\mathrm{op}}), \qquad {}^{(*)}(-) : (\operatorname{GrMod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{GrMod} S. \tag{1.13}$$

Each functor in (1.13) restricts to locally finite-dimensional modules: replace GrMod by grmod in its source and target. Indeed, for $R = S$ or S^{op} and $M \in \operatorname{grmod} R$,

$$\dim_{\Bbbk} \operatorname{Hom}_R(M_{-i}, R) \le (\dim_{\Bbbk} M_{-i})(\dim_{\Bbbk} R) < \infty \quad (i \in \mathbb{Z}). \tag{1.14}$$

Lemma 1.52. *We have dualities*

$$(-)^* : (\operatorname{mod} S)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod}(S^{\mathrm{op}}), \qquad {}^*(-) : (\operatorname{mod}(S^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{mod} S,$$
$$(-)^{(*)} : (\operatorname{grmod} S)^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod}(S^{\mathrm{op}}), \quad {}^{(*)}(-) : (\operatorname{grmod}(S^{\mathrm{op}}))^{\mathrm{op}} \xrightarrow{\sim} \operatorname{grmod} S.$$

Proof We show that evaluation gives natural isomorphisms of functors

$$\mathrm{id}_{\mathrm{mod}\, S} \xrightarrow{\sim} {}^*((-)^*), \qquad \mathrm{id}_{\mathrm{mod}(S^{\mathrm{op}})} \xrightarrow{\sim} ({}^*(-))^*,$$

with components

$$\begin{aligned} \mathrm{ev}_M : M &\xrightarrow{\sim} {}^*(M^*), \quad m \longmapsto (f \mapsto f(m)) \quad (M \in \mathrm{mod}\, S), \\ \mathrm{ev}_N : N &\xrightarrow{\sim} ({}^*N)^*, \quad n \longmapsto (g \mapsto g(n)) \quad (N \in \mathrm{mod}(S^{\mathrm{op}})). \end{aligned}$$

For $u : M \to N$ in $\mathrm{mod}\, S$, $m \in M$, and $f \in N^*$,

$$\big({}^*(u^*)(\mathrm{ev}_M(m))\big)(f) = (u^* f)(m) = f(u(m)) = \mathrm{ev}_N(u(m))(f).$$

Hence ${}^*(u^*) \circ \mathrm{ev}_M = \mathrm{ev}_N \circ u$. For $M = S$, evaluation is an isomorphism: its inverse sends $F \in {}^*(S^*)$ to $F(\mathrm{id}_S)$. Indeed, every $f \in S^*$ satisfies $f(s) = f(1)s$, so $F(f) = f(1)F(\mathrm{id}_S) = f(F(\mathrm{id}_S))$. Restriction to summands identifies the dual of a finite direct sum with the direct sum of the duals; evaluation therefore is an isomorphism for $S^{\oplus n}$. By Corollary 1.13, $M \oplus L \cong S^{\oplus n}$ for some L. Under the direct sum identifications, $\mathrm{ev}_{M\oplus L} = \mathrm{ev}_M \oplus \mathrm{ev}_L$, so ev_M is an isomorphism. Replacing S by S^{op} proves the left module statement.

For the graded cases, apply the ungraded case degree-wise. Note that since S is finite-dimensional, finitely generated S-module is the same as finite-dimensional S-modules. □

For a bimodule ${}_S M_S$, the linear and one-sided duals retain both actions:

$$(s\lambda t)(m) = \lambda(tms), \qquad (sft)(m) = sf(tm), \qquad (sgt)(m) = g(ms)t, \tag{1.15}$$

where $s, t \in S$, $m \in M$, $\lambda \in D(M)$, $f \in \mathrm{Hom}_S(M, S)$, and $g \in \mathrm{Hom}_{S^{\mathrm{op}}}(M, S)$. The functionals still take values in $\Bbbk$ or S. With precomposition on morphisms, we obtain

$$\begin{aligned} D,\ (-)^*,\ {}^*(-) &: (\mathrm{Mod}\, S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \mathrm{Mod}\, S^{\mathrm{en}}, \\ \mathbb{D},\ (-)^{(*)},\ {}^{(*)}(-) &: (\mathrm{GrMod}\, S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \mathrm{GrMod}\, S^{\mathrm{en}}. \end{aligned} \tag{1.16}$$

The restrictions to finite-dimensional modules, and to locally finite-dimensional graded modules, are dualities by Lemmas 1.47 and 1.52.

Evaluation preserves both actions: for $m \in M$, $f \in M^*$, and $s, t \in S$,

$$(s\, \mathrm{ev}_M(m)\, t)(f) = f(sm)t = f(smt) = \mathrm{ev}_M(smt)(f).$$

The corresponding checks for D and the left dual use the same evaluation formula; applying them in each degree gives the graded statements.

Recall from Definition 1.42 and (1.7) that $S^{\mathrm{en}} = S \otimes_{\Bbbk} S^{\mathrm{op}}$ and S-bimodules are the same thing as S^{en}-modules via $m \cdot (s \otimes t^{\mathrm{op}}) = tms$.

Definition 1.53. *The two-sided dual of a bimodule ${}_S M_S$ is*

$$M^{*_{\mathrm{en}}} := \mathrm{Hom}_{S^{\mathrm{en}}}(M, S^{\mathrm{en}}), \qquad (sFt)(m) := (s \otimes t^{\mathrm{op}})F(m) \quad (s, t \in S,\ F \in M^{*_{\mathrm{en}}},\ m \in M).$$

In Definition 1.53, M has the right action (1.7), so $F(tms) = F(m)(s \otimes t^{\mathrm{op}})$. The output is a left S^{en}-module by multiplication on the values, which gives the displayed bimodule action. The

two-sided dual takes values in S^{en}, not in S. On bimodule morphisms it acts by precomposition, as do D, $(-)^*$, and ${}^*(-)$; each is a contravariant functor on the category of S-bimodules. The two-sided dual records both module actions in the target algebra S^{en}.

For a graded bimodule ${}_SM_S$, define its *graded two-sided dual* by

$$M^{(*_{\mathrm{en}})} := \bigoplus_{i\in\mathbb{Z}} (M_{-i})^{*_{\mathrm{en}}}, \qquad (M^{(*_{\mathrm{en}})})_i = (M_{-i})^{*_{\mathrm{en}}}.$$

Definition 1.53 gives a functor

$$(-)^{(*_{\mathrm{en}})} : (\mathrm{GrMod}\, S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \mathrm{GrMod}\, S^{\mathrm{en}}. \tag{1.17}$$

On a degree 0 bimodule map $u : M \to N$, its degree i component is

$$(u^{(*_{\mathrm{en}})})_i : (N_{-i})^{*_{\mathrm{en}}} \longrightarrow (M_{-i})^{*_{\mathrm{en}}}, \qquad F \longmapsto F \circ u|_{M_{-i}} \quad (i \in \mathbb{Z}).$$

The functor restricts to locally finite-dimensional bimodules by (1.14) with $R = S^{\mathrm{en}}$.

The four duals above follow the distinction in Grant–Iyama [GI20, §2.1, pp. 2592–2593], with the following notation changes:

construction	Grant–Iyama	these notes
$\Bbbk$-linear dual	M^*	$D(M)$
left S-linear dual	M^{*_ℓ}	${}^*M$
right S-linear dual	M^{*_r}	M^*
two-sided dual	$M^\vee$	$M^{*_{\mathrm{en}}}$

The last row also changes the convention for enveloping modules. Their $F : M \to S^{\mathrm{en}}$ is left S^{en}-linear. Define

$$\sigma : S^{\mathrm{en}} \longrightarrow S^{\mathrm{en}}, \qquad a \otimes b^{\mathrm{op}} \longmapsto b \otimes a^{\mathrm{op}}.$$

Since $\sigma(xy) = \sigma(y)\sigma(x)$ for $x, y \in S^{\mathrm{en}}$, the map $\sigma \circ F : M \to S^{\mathrm{en}}$ satisfies

$$(\sigma \circ F)(tms) = (\sigma \circ F)(m)(s \otimes t^{\mathrm{op}}) \quad (s, t \in S,\ m \in M),$$

and belongs to our $M^{*_{\mathrm{en}}}$. Thus $M^\vee \to M^{*_{\mathrm{en}}}$, $F \mapsto \sigma \circ F$, translates their two-sided dual into our convention. In particular, their star denotes the linear dual; our star denotes the right S-dual.

1.12 Comparing the duals

The one-sided S-duals and the $\Bbbk$-linear dual are naturally isomorphic after choosing a symmetrising form on S.

Definition 1.54. *A symmetrising form on a finite-dimensional $\Bbbk$-algebra S is a linear map $\tau : S \to \Bbbk$ satisfying*

$$\tau(st) = \tau(ts) \quad \text{for all } s, t \in S, \qquad \big(\tau(st) = 0 \text{ for all } t \in S\big) \implies s = 0.$$

An algebra admitting such a form is symmetric.

The second condition in Definition 1.54 says that the bilinear form $(s, t) \mapsto \tau(st)$ is *non-degenerate*: every non-zero $s \in S$ pairs non-trivially with some $t \in S$.

Every finite-dimensional semisimple $\Bbbk$-algebra admits a symmetrising form. For products of matrix algebras over $\Bbbk$, the forms are explicit:

Example 1.55. *The following maps are symmetrising forms:*

$$\tau : \Bbbk^{\oplus n} \longrightarrow \Bbbk, \quad (a_1, \dots, a_n) \longmapsto \sum_{j=1}^{n} a_j,$$

$$\tau : \mathrm{Mat}_n(\Bbbk) \longrightarrow \Bbbk, \qquad a \longmapsto \mathrm{Tr}(a) := \sum_{i=1}^{n} a_{ii},$$

and more generally, $\tau : \prod_{j=1}^{n} \mathrm{Mat}_{n_j}(\Bbbk) \longrightarrow \Bbbk, \quad (a_1, \dots, a_n) \longmapsto \sum_{j=1}^{n} \mathrm{Tr}(a_j).$

Generally, any finite-dimensional semisimple algebra has a symmetrising form, by Eilenberg–Nakayama.[a] For example, if $\lambda_j : \Delta_j \to \Bbbk$ is a symmetrising form for a finite-dimensional division $\Bbbk$-algebra Δ_j with $j = 1, \dots, n$. Then the semisimple algebra $S = \prod_{j=1}^{n} \mathrm{Mat}_{n_j}(\Delta_j)$ has a symmetrising form given by

$$\tau : S \longrightarrow \Bbbk, \qquad (a_1, \dots, a_n) \longmapsto \sum_{j=1}^{n} \sum_{i=1}^{n_j} \lambda_j((a_j)_{ii}).$$

[a] S. Eilenberg and T. Nakayama, *On the dimension of modules and algebras, II. Frobenius algebras and quasi-Frobenius rings*, Nagoya Math. J. **9** (1955), 1–16, §2, Propositions 3–5, pp. 4–5; doi:10.1017/S0027763000023229.

Theorem 1.56 (Comparison of duals). *Choose a symmetrising form $\tau : S \to \Bbbk$. Postcomposition with τ defines natural isomorphisms from the S-dual (on either side) to D, and also from the graded S-dual (on either side) to $\mathbb{D}$.*

We can spell this out more precisely:

Ungraded duals. For $M \in \mathrm{mod}\, S$ and $N \in \mathrm{mod}(S^{\mathrm{op}})$, the maps

$$\alpha_M : M^* \longrightarrow D(M), \quad f \longmapsto \tau \circ f, \qquad \beta_N : {}^*N \longrightarrow D(N), \quad g \longmapsto \tau \circ g \tag{1.18}$$

are S-module isomorphisms and give natural isomorphisms

$$\begin{aligned} \alpha &: (-)^* \xrightarrow{\sim} D \quad \text{of functors } (\mathrm{mod}\, S)^{\mathrm{op}} \longrightarrow \mathrm{mod}(S^{\mathrm{op}}), \\ \beta &: {}^*(-) \xrightarrow{\sim} D \quad \text{of functors } (\mathrm{mod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \mathrm{mod}\, S. \end{aligned} \tag{1.19}$$

Graded duals. Composition with τ in each degree gives natural isomorphisms

$$\begin{aligned} \alpha &: (-)^{(*)} \xrightarrow{\sim} \mathbb{D} \quad \text{of functors } (\mathrm{grmod}\, S)^{\mathrm{op}} \longrightarrow \mathrm{grmod}(S^{\mathrm{op}}), \\ \beta &: {}^{(*)}(-) \xrightarrow{\sim} \mathbb{D} \quad \text{of functors } (\mathrm{grmod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \mathrm{grmod}\, S. \end{aligned} \tag{1.20}$$

For $M \in \mathrm{grmod}\, S$ and $N \in \mathrm{grmod}(S^{\mathrm{op}})$, the degree i components are $\alpha_{M_{-i}}$ and $\beta_{N_{-i}}$, respectively.

Proof We prove the right module statement, i.e. taking $\alpha_M(f) := \tau \circ f$ yields

$$\text{ungraded case } \alpha : (-)^* \xrightarrow{\sim} D, \quad \text{and} \quad \text{graded case } \alpha : (-)^{(*)} \xrightarrow{\sim} \mathbb{D}.$$

First, non-degeneracy of τ gives an isomorphism $S \to D(S)$, $a \mapsto (s \mapsto \tau(as))$. Hence, for $\lambda \in D(M)$, there is a unique linear map $f : M \to S$ satisfying

$$\tau(f(m)s) = \lambda(ms) \quad \text{for all } m \in M,\ s \in S. \tag{1.21}$$

For $m \in M$ and $s, t \in S$,

$$\tau(f(mt)s) = \lambda(mts) = \tau(f(m)ts),$$

so non-degeneracy implies $f(mt) = f(m)t$. Taking $s = 1$ in (1.21) gives $\tau \circ f = \lambda$; conversely, every right S-linear preimage of λ must satisfy (1.21). Thus α_M is bijective.

Both M^* and $D(M)$ are left S-modules, so the component $\alpha_M : M^* \to D(M)$ must preserve the left action:

$$\alpha_M(sf)(m) = \tau(sf(m)) = \tau(f(m)s) = \tau(f(ms)) = (s\alpha_M(f))(m) \quad (s \in S,\ f \in M^*,\ m \in M).$$

For a right module map $u : M \to N$ and $f \in N^*$, associativity of composition gives $\alpha_M(f \circ u) = \alpha_N(f) \circ u$, proving naturality.

For the graded comparisons, apply the ungraded natural isomorphism degree-wise.

Same argument applies dually to S^{op} and β. □

Remark 1.57. The natural isomorphisms in Theorem 1.56 depend on the chosen symmetrising form. For example, every choice $c_1, \dots, c_n \in \Bbbk \setminus \{0\}$ gives a form $\tau(a_1, \dots, a_n) = \sum_{i=1}^n c_i a_i$ on $\Bbbk^{\oplus n}$.

Remark 1.58. Let A be a graded algebra with $A_0 = S$. The space $\mathrm{Hom}_S(M, S)$ do not have a natural left A-module on the spot, but can be after *picking a symmetrising form τ of S*. Indeed, the comparison (1.20) transports the canonical left A-action on $\mathbb{D}(M)$ to the graded S-dual of a locally finite-dimensional module:

$$af = \alpha_M^{-1}(a\alpha_M(f)) \quad \text{for right } M, \qquad ga = \beta_N^{-1}(\beta_N(g)a) \quad \text{for left } N,$$

where $a \in A$, $f \in M^{(*)}$, and $g \in {}^{(*)}N$. Similarly for right action on left dual.

The comparison theorem has a bimodule version:

Theorem 1.59. *With the actions in* (1.15)*, we obtain natural isomorphisms of functors*

$$\begin{aligned} (-)^* &\xrightarrow[\alpha]{\sim} D \xleftarrow[\beta]{\sim} {}^*(-) \quad &&: (\operatorname{mod} S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \operatorname{mod} S^{\mathrm{en}}, \\ (-)^{(*)} &\xrightarrow[\alpha]{\sim} \mathbb{D} \xleftarrow[\beta]{\sim} {}^{(*)}(-) \quad &&: (\operatorname{grmod} S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \operatorname{grmod} S^{\mathrm{en}}. \end{aligned} \tag{1.22}$$

There is also a natural isomorphism $\gamma : (-)^{*_{\mathrm{en}}} \xrightarrow{\sim} (-)^*$ given by

$$\gamma_M : M^{*_{\mathrm{en}}} \to M^*, \qquad F \mapsto \big(m \mapsto \sum_j a_j \tau(b_j)\big) \quad \text{for} \quad F(m) = \sum_j a_j \otimes b_j^{\mathrm{op}}, \tag{1.23}$$

which also depends on τ, but does not require S^{en} to be semisimple.

Consequently, on finite-dimensional S-bimodules and lfd graded S-bimodules we have natural isomorphisms:

$$\begin{aligned} &\textit{Ungraded:} \quad (-)^{*_{\mathrm{en}}} \xrightarrow[\gamma]{\sim} (-)^* \xrightarrow[\alpha]{\sim} D \xleftarrow[\beta]{\sim} {}^*(-) \ \textit{as functors } (\operatorname{mod} S^{\mathrm{en}})^{\mathrm{op}} \to \operatorname{mod} S^{\mathrm{en}}, \\ &\textit{Graded:} \quad (-)^{(*_{\mathrm{en}})} \xrightarrow[\gamma]{\sim} (-)^{(*)} \xrightarrow[\alpha]{\sim} \mathbb{D} \xleftarrow[\beta]{\sim} {}^{(*)}(-) \ \textit{as functors } (\operatorname{grmod} S^{\mathrm{en}})^{\mathrm{op}} \to \operatorname{grmod} S^{\mathrm{en}}. \end{aligned}$$

Proof The form $\tau_{\mathrm{en}}(a \otimes b^{\mathrm{op}}) = \tau(a)\tau(b)$ is symmetrising on S^{en}. Indeed, $\tau_{\mathrm{en}}((a \otimes b^{\mathrm{op}})(c \otimes d^{\mathrm{op}})) = \tau(ac)\tau(db) = \tau(ca)\tau(bd)$, and its pairing is the tensor product of two non-degenerate

pairings. The formula (1.23) satisfies

$$\gamma_M(F)(ms) = \gamma_M(F)(m)s, \qquad \gamma_M(sFt) = s\gamma_M(F)t, \qquad \alpha_M(\gamma_M(F)) = \tau_{\mathrm{en}} \circ F.$$

The argument (1.21), applied to S^{en} and τ_{en}, makes $F \mapsto \tau_{\mathrm{en}} \circ F$ an isomorphism. Thus γ_M is an isomorphism. For a bimodule map $u : M \to N$ and $F \in N^{*_{\mathrm{en}}}$, the formula (1.23) gives $\gamma_M(F \circ u) = \gamma_N(F) \circ u$, proving naturality. Taking the direct sum of $\gamma_{M_{-i}}$ over $i \in \mathbb{Z}$ gives the graded two-sided comparison. Each component is a bimodule isomorphism, and naturality follows in each degree from the ungraded comparison. □

1.13 Exercises

Exercise 1.60 (Matrix calculations). *Let $n \geq 2$ and $A = \Bbbk A_n$. Verify the grading in Example 1.19 using $e_{ij}e_{pq} = \delta_{jp}e_{iq}$. Compute* $\operatorname{rad} A$ *from its definition as the intersection of all maximal right ideals.*

Hint. *The strictly upper triangular ideal J satisfies $J^n = 0$. Show that $LJ = 0$ for every simple right A-module L, and use the diagonal maps $A \to \Bbbk$, $X \mapsto X_{ii}$, for $1 \leq i \leq n$.*

Exercise 1.61 (Shifts and duals). *Let $A = \Bbbk A_3$ and $M = e_{22}A$ with the inherited grading.*

(1) *Find a homogeneous basis of $\mathbb{D}(M)$, its degrees, and all non-zero actions of matrix units on the basis vectors.*

(2) *Show that $\Bbbk e_{13} \subset Ae_{33}$ is a graded left submodule and that*

$$\mathbb{D}(M) \cong (Ae_{33}/\Bbbk e_{13})(1)$$

as graded left A-modules.

(3) *Deduce that $\mathbb{D}(M) \ncong (Ae_{jj})(r)$ for every $j \in \{1,2,3\}$ and $r \in \mathbb{Z}$.*

Hint for (3). *Compare dimensions and the actions of e_{11}, e_{22}, e_{33}.*

Exercise 1.62 (Homogeneous maps). *The finite-generation hypothesis in Lemma 1.31 is needed. Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ with componentwise multiplication, let $N = e_1S$ have degree 0, and put*

$$M = \bigoplus_{i \geq 0} (e_1S)(-i) = \bigoplus_{i \geq 0} \Bbbk m_i, \qquad \deg m_i = i, \quad m_i e_1 = m_i, \quad m_i e_2 = 0.$$

For the S-linear map $f : M \to N$ with $f(m_i) = e_1$ for every $i \geq 0$, determine all homogeneous components and prove that f is not a finite sum of homogeneous maps.

Exercise 1.63 (Dependence on the form). *Let $S = (\Bbbk A_2)_0 = \Bbbk e_{11} \oplus \Bbbk e_{22}$, $V = (\Bbbk A_2)_1 = \Bbbk e_{12}$, and $c_1, c_2 \in \Bbbk \setminus \{0\}$. Set $\tau(ae_{11} + be_{22}) = c_1a + c_2b$ and let $\varepsilon : V \to \Bbbk$ satisfy $\varepsilon(e_{12}) = 1$. Find the right and left S-linear maps corresponding to ε under* (1.18)*, and determine how they depend on c_1, c_2.*

Exercise 1.64 (Dependence of the action on the form). *The transported A-action can depend on the symmetrising form, but its graded isomorphism class does not. Let $A = \Bbbk A_2$, $S = \Bbbk e_{11} \oplus \Bbbk e_{22}$, and $M = e_{11}A$, with the inherited gradings. For $c = (c_1, c_2) \in (\Bbbk \setminus \{0\})^2$, define*

$$\tau_c : S \longrightarrow \Bbbk, \qquad ae_{11} + be_{22} \longmapsto c_1a + c_2b \quad \text{for all } a, b \in \Bbbk. \tag{1.24}$$

Write $\alpha_c : M^{(*)} \to \mathbb{D}(M)$, $f \mapsto \tau_c \circ f$, and let $a \cdot_c f = \alpha_c^{-1}(a\alpha_c(f))$ be the transported action of Remark 1.58. Consider the right S-linear maps

$$\begin{aligned} &\varphi_0 : M \longrightarrow S, \quad ae_{11} + be_{12} \longmapsto ae_{11}, \\ &\varphi_1 : M \longrightarrow S, \quad ae_{11} + be_{12} \longmapsto be_{22} \quad \text{for all } a, b \in \Bbbk. \end{aligned} \tag{1.25}$$

(1) *Verify that τ_c is symmetrising and that φ_0, φ_1 form a homogeneous basis of $M^{(*)}$. Determine their degrees.*

(2) *Calculate the actions of e_{11}, e_{22}, e_{12} on φ_0, φ_1. For $c, c' \in (\Bbbk \setminus \{0\})^2$, prove that $\cdot_c = \cdot_{c'}$ if and only if $c' = \rho c$ for some $\rho \in \Bbbk \setminus \{0\}$.*

(3) *More generally, let A be a locally finite-dimensional graded algebra with semisimple $S = A_0$, and let M be a locally finite-dimensional graded right A-module. Let τ, τ' be symmetrising forms on S, with comparison maps $\alpha_\tau, \alpha_{\tau'} : M^{(*)} \to \mathbb{D}(M)$ given by composition with τ, τ', respectively, and transported actions $\cdot_\tau, \cdot_{\tau'}$. Prove that there is an invertible $z \in S$ satisfying $\tau'(s) = \tau(zs)$ for every $s \in S$, and that*

$$\gamma : (M^{(*)}, \cdot_{\tau'}) \longrightarrow (M^{(*)}, \cdot_\tau), \qquad (\gamma f)(m) = zf(m)$$

is an isomorphism of graded left A-modules. Deduce that the modules in (2) are isomorphic for every c, c', and give the isomorphism on φ_0, φ_1.

Hint for (3). *Compare $\alpha_{\tau'}$ with α_τ.*

2 Tensor and quadratic algebras

2.1 Tensor products over an algebra

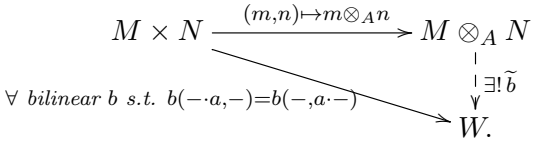
Definition 2.1. *Let A be any $\Bbbk$-algebra, M a right A-module, and N a left A-module. The tensor product of M and N over A is the vector space*

$$M \otimes_A N := M \otimes_{\Bbbk} N / \operatorname{span}_{\Bbbk}\{ma \otimes n - m \otimes an \mid m \in M,\ n \in N,\ a \in A\}.$$

Write $m \otimes_A n$, or simply $m \otimes n$, for the image of $m \otimes n$ in this quotient. Thus $ma \otimes_A n = m \otimes_A an$ for all $m \in M$, $n \in N$, and $a \in A$.

Lemma 2.2 (Basic properties of tensor products). *Let A be any $\Bbbk$-algebra, M a right A-module, and N a left A-module.*

(1) *(Universal property) For a vector space W and a bilinear map $b : M \times N \to W$ satisfying $b(ma, n) = b(m, an)$ for all $m \in M$, $a \in A$, and $n \in N$, there is a unique linear factorisation (commutative diagram)*



(2) *If B, C are $\Bbbk$-algebras and ${}_BM_A$, ${}_AN_C$ are bimodules, then $M \otimes_A N$ is a bimodule ${}_B(M \otimes_A N)_C$ with*

$$b(m \otimes_A n)c = bm \otimes_A nc \qquad (b \in B,\ c \in C,\ m \in M,\ n \in N).$$

For right A-module maps $u : M \to M'$ and left A-module maps $v : N \to N'$, the assignment $(m, n) \mapsto u(m) \otimes v(n)$ induces

$$u \otimes_A v : M \otimes_A N \longrightarrow M' \otimes_A N', \qquad m \otimes n \longmapsto u(m) \otimes v(n).$$

These define functors

$$(-) \otimes_A N : \operatorname{Mod} A \longrightarrow \operatorname{Mod} C, \qquad M \otimes_A (-) : \operatorname{Mod}(A^{\mathrm{op}}) \longrightarrow \operatorname{Mod}(B^{\mathrm{op}}).$$

On bimodule maps, they retain the outer actions:

$$(-) \otimes_A N : B\operatorname{Mod} A \longrightarrow B\operatorname{Mod} C, \qquad M \otimes_A (-) : A\operatorname{Mod} C \longrightarrow B\operatorname{Mod} C.$$

(3) *For $\Bbbk$-algebras A, B, a right A-module U, a bimodule ${}_AV_B$, and a left B-module W, there is a natural linear isomorphism*

$$(U \otimes_A V) \otimes_B W \xrightarrow{\sim} U \otimes_A (V \otimes_B W), \qquad (u \otimes v) \otimes w \longmapsto u \otimes (v \otimes w).$$

There are also natural module isomorphisms

$$M \otimes_A A \xrightarrow{\sim} M, \quad m \otimes a \longmapsto ma, \qquad A \otimes_A N \xrightarrow{\sim} N, \quad a \otimes n \longmapsto an.$$

Moreover, these maps are linear with respect to any additional bimodule action on U, W, M, N.

(4) *Tensor products distribute over direct sums in either variable. For arbitrary families $(M_i)_{i \in I}$ of right A-modules and $(N_j)_{j \in J}$ of left A-modules, indexed by sets I, J,*

$$\Big(\bigoplus_{i \in I} M_i\Big) \otimes_A N \xrightarrow{\sim} \bigoplus_{i \in I} (M_i \otimes_A N), \quad (m_i)_{i \in I} \otimes n \longmapsto (m_i \otimes n)_{i \in I},$$
$$M \otimes_A \Big(\bigoplus_{j \in J} N_j\Big) \xrightarrow{\sim} \bigoplus_{j \in J} (M \otimes_A N_j), \quad m \otimes (n_j)_{j \in J} \longmapsto (m \otimes n_j)_{j \in J}.$$

(5) *For finite sets I, J and right A-modules M_j, N_i, there is a natural isomorphism*

$$\begin{aligned}\operatorname{Hom}_A\Big(\bigoplus_{j\in J} M_j, \bigoplus_{i\in I} N_i\Big) &\xrightarrow{\sim} \bigoplus_{\substack{i\in I\\ j\in J}} \operatorname{Hom}_A(M_j, N_i),\\ f &\longmapsto (\pi_i f \iota_j)_{i\in I,\ j\in J},\end{aligned} \tag{2.1}$$

where $\iota_j : M_j \to \bigoplus_{j'\in J} M_{j'}$ and $\pi_i : \bigoplus_{i'\in I} N_{i'} \to N_i$ are the inclusions and projections.

Proof For (1), Proposition 1.37 gives a unique linear map $M \otimes_{\Bbbk} N \to W$, $m \otimes n \mapsto b(m, n)$. The map vanishes on the defining subspace precisely when $b(ma, n) = b(m, an)$.

For (2), the outer actions preserve that subspace: for $b \in B$, $c \in C$, and $a \in A$,

$$b(ma \otimes n - m \otimes an)c = (bm)a \otimes nc - bm \otimes a(nc).$$

The module axioms follow from those on M and N. For module maps u, v,

$$u(ma) \otimes v(n) = u(m)a \otimes v(n) = u(m) \otimes av(n) = u(m) \otimes v(an),$$

so the displayed formula induces $u \otimes_A v$. The functor identities follow from $(u' \otimes v')(u(m) \otimes v(n)) = u'u(m) \otimes v'v(n)$ and $(\mathrm{id}_M \otimes \mathrm{id}_N)(m \otimes n) = m \otimes n$. If u, v preserve the outer actions, then $u \otimes_A v$ does as well.

For (3), both parenthesisations impose

$$(ua) \otimes v \otimes w = u \otimes (av) \otimes w, \qquad u \otimes (vb) \otimes w = u \otimes v \otimes (bw) \quad (a \in A,\ b \in B).$$

Thus the displayed associativity map and the reverse assignment are well-defined and inverse. The unit inverses are $M \to M \otimes_A A$, $m \mapsto m \otimes 1$, and $N \to A \otimes_A N$, $n \mapsto 1 \otimes n$.

For (4), the bilinear map $((m_i)_{i\in I}, n) \mapsto (m_i \otimes n)_{i\in I}$ satisfies the identity in (1), so it induces the displayed linear map. For each $i \in I$, let $\iota_i : M_i \to \bigoplus_{j\in I} M_j$ be the inclusion. The inverse sends $m \otimes n$ in the ith summand to $\iota_i(m) \otimes n$ and extends over the direct sum. Both composites fix every element $m \otimes n$ from a summand. The other variable is handled in the same way. □

Lemma 2.3 (Hom, direct sums, and direct products). *Let M, N be right A-modules, and let $(M_i)_{i\in I}$ and $(N_i)_{i\in I}$ be arbitrary families of right A-modules, indexed by a set I. Write $\iota_i : M_i \to \bigoplus_{j\in I} M_j$ for the inclusions and $\pi_i : \prod_{j\in I} N_j \to N_i$ for the projections. There are natural linear isomorphisms*

$$\begin{aligned}\operatorname{Hom}_A\Big(\bigoplus_{i\in I} M_i, N\Big) &\xrightarrow{\sim} \prod_{i\in I} \operatorname{Hom}_A(M_i, N), \quad f \longmapsto (f \circ \iota_i)_{i\in I},\\ \operatorname{Hom}_A\Big(M, \prod_{i\in I} N_i\Big) &\xrightarrow{\sim} \prod_{i\in I} \operatorname{Hom}_A(M, N_i), \quad f \longmapsto (\pi_i \circ f)_{i\in I}.\end{aligned}$$

There is also a natural injection

$$\bigoplus_{i\in I} \operatorname{Hom}_A(M, N_i) \longrightarrow \operatorname{Hom}_A\Big(M, \bigoplus_{i\in I} N_i\Big), \qquad (f_i)_{i\in I} \longmapsto \big(m \mapsto (f_i(m))_{i\in I}\big),$$

which is an isomorphism when M is finitely generated.

Proof The inverse of the first map sends $(f_i)_{i\in I}$ to $(m_i)_{i\in I} \mapsto \sum_{i\in I} f_i(m_i)$; the sum is finite because $(m_i)_{i\in I}$ has finite support. The inverse of the second sends $(f_i)_{i\in I}$ to $m \mapsto (f_i(m))_{i\in I}$; a direct product imposes no support condition. For the third map, coordinate projections recover each f_i, proving injectivity. If $m_1, \dots, m_r$ generate M, the images of these generators under $f : M \to \bigoplus_{i\in I} N_i$ are supported on a common finite subset $J \subset I$. Then $f(M) \subset \bigoplus_{i\in J} N_i$, so f comes from a finite family of maps. All three constructions commute with precomposition and postcomposition. □

Proposition 2.4 (Tensor products and direct products). *Let A be a $\Bbbk$-algebra, M, M_i right A-modules, and N, N_i left A-modules, with $i \in I$. The coordinate projections define linear maps*

$$\begin{aligned} \kappa_N &: \Big(\prod_i M_i\Big) \otimes_A N \longrightarrow \prod_i (M_i \otimes_A N), \quad (m_i)_i \otimes n \longmapsto (m_i \otimes n)_i, \\ \lambda_M &: M \otimes_A \Big(\prod_i N_i\Big) \longrightarrow \prod_i (M \otimes_A N_i), \quad m \otimes (n_i)_i \longmapsto (m \otimes n_i)_i. \end{aligned} \tag{2.2}$$

(1) *If I is finite, both maps are isomorphisms.*

(2) *The map κ_N is an isomorphism if N is a direct summand of $A^{\oplus r}$ as a left module for some finite r. The corresponding condition on the right module M makes λ_M an isomorphism.*

(3) *If A is finite-dimensional semisimple, both maps are injective. Moreover, κ_N is an isomorphism when N is finite-dimensional, and λ_M is an isomorphism when M is finite-dimensional.*

Proof The formulas in (2.2) are bilinear. For $s \in A$, the component identities $(m_i s) \otimes n = m_i \otimes sn$ and $(ms) \otimes n_i = m \otimes sn_i$ show that both formulas give well-defined maps from the indicated tensor products. For finite I, products equal sums, and Lemma 2.2(4) proves (1).

For $N = A$, the map κ_A is the identity on $\prod_i M_i$ under $m \otimes s \mapsto ms$. For $N = A^{\oplus r}$, both sides are $(\prod_i M_i)^{\oplus r}$, since a finite direct sum commutes with a direct product. If $N \oplus N' \cong A^{\oplus r}$, the comparison for $N \oplus N'$ is $\kappa_N \oplus \kappa_{N'}$. Its bijectivity implies the bijectivity of κ_N. This proves the assertion about κ_N in (2).

Now let A be finite-dimensional semisimple. For finite-dimensional N, Corollary 1.13 gives (2). For arbitrary N, take $z = \sum_{\ell=1}^{r} m^{(\ell)} \otimes n_\ell$ in $\operatorname{Ker} \kappa_N$ and set $N_0 = \sum_{\ell=1}^{r} An_\ell$. The submodule N_0 is finite-dimensional and has a complement in N by Theorem 1.12. Thus the inclusions

$$\Big(\prod_i M_i\Big) \otimes_A N_0 \longrightarrow \Big(\prod_i M_i\Big) \otimes_A N, \qquad M_i \otimes_A N_0 \longrightarrow M_i \otimes_A N$$

are injective, by the direct sum formula in Lemma 2.2(4). Regard z as an element with second factors in N_0. The equality $\kappa_N(z) = 0$ then gives $\kappa_{N_0}(z) = 0$. Aince κ_{N_0} is an isomorphism, $z = 0$.

For λ_M, apply the same arguments over A^{op}, using $N \otimes_{A^{\mathrm{op}}} M \cong M \otimes_A N$, $n \otimes m \mapsto m \otimes n$. □

The finite-dimensional hypothesis on the fixed factor cannot be dropped from the isomorphism assertion: Example 1.39 gives a non-surjective comparison even for vector spaces. Thus tensor products always distribute over direct sums, but not over arbitrary direct products.

Remark 2.5. For infinite families, Hom need not preserve direct sums in its second variable. For $M = \bigoplus_{i\geq 1} \Bbbk v_i$, the identity $M \to M$ does not come from $\bigoplus_i \operatorname{Hom}_\Bbbk(M, \Bbbk v_i)$: its projection onto every $\Bbbk v_i$ is non-zero. Thus the injection in Lemma 2.3 need not be surjective.

A product in the source of Hom behaves differently from a product in the target. For example, a linear map $\prod_{i\geq 1}\Bbbk \to \Bbbk$ need not be determined by its restrictions to the coordinate copies of $\Bbbk$. Indeed, $(1,1,\ldots) \notin \bigoplus_{i\geq 1}\Bbbk$. Define a functional on $\bigoplus_{i\geq 1}\Bbbk + \Bbbk(1,1,\ldots)$ to vanish on the direct sum and take value 1 at $(1,1,\ldots)$, and extend it to the product by extending a basis. The resulting non-zero functional vanishes on every coordinate copy.

For a bimodule ${}_AV_A$, define

$$V^{\otimes_A 0} := A, \qquad V^{\otimes_A i} := \underbrace{V \otimes_A \cdots \otimes_A V}_{i \text{ factors}} \quad (i \geq 1).$$

Note that there is no parenthesis in the definition of $V^{\otimes_A i}$ since Lemma 2.2(3) says that tenso product is an associatitive operation.

Let A, B, C be $\Bbbk$-algebras. For bimodules ${}_AM_B$, ${}_CX_B$, we have $\mathrm{Hom}_B(M,X) \in C\,\mathrm{Mod}\,A$, with

$$(cfa)(m) = cf(am) \quad (f \in \mathrm{Hom}_B(M,X),\ c \in C,\ a \in A,\ m \in M). \tag{2.3}$$

If $X \in \mathrm{Mod}\,B$ has no specified left action, the same formula defines the right A-action $(fa)(m) = f(am)$. The left-module version follows by passing to opposite algebras.

Proposition 2.6 (Tensor and Hom). *Consider bimodules ${}_AM_B$, ${}_BN_C$, a right C-module X, and a left A-module Y. There are maps, natural in M, N,*

$$\begin{aligned} \theta_r : \mathrm{Hom}_C(N,X) \otimes_B \mathrm{Hom}_B(M,B) &\longrightarrow \mathrm{Hom}_C(M \otimes_B N, X), \\ \theta_r(g \otimes f)(m \otimes n) &= g(f(m)n), \\ \theta_\ell : \mathrm{Hom}_{B^{\mathrm{op}}}(N,B) \otimes_B \mathrm{Hom}_{A^{\mathrm{op}}}(M,Y) &\longrightarrow \mathrm{Hom}_{A^{\mathrm{op}}}(M \otimes_B N, Y), \\ \theta_\ell(g \otimes f)(m \otimes n) &= f(mg(n)). \end{aligned} \tag{2.4}$$

The maps θ_r and θ_ℓ are right A-linear and left C-linear, respectively.

(1) *The map θ_r is an isomorphism if $M \in \mathrm{add}\,B$ as a right B-module.*

(2) *The map θ_ℓ is an isomorphism if ${}_BN \in \mathrm{add}_{\mathrm{mod}\,B^{\mathrm{op}}}(B)$.*

Here $\mathrm{add}\,B = \mathrm{add}_{\mathrm{mod}\,B}(B)$, as in Example 1.5. If ${}_CX_C$ and ${}_AY_A$ are bimodules, the respective maps also preserve the additional actions and are morphisms in $C\,\mathrm{Mod}\,A$.

Proof *Well-definedness of θ_r.* For $f \in \mathrm{Hom}_B(M,B)$, $g \in \mathrm{Hom}_C(N,X)$, and $b \in B$,

$$g(f(mb)n) = g(f(m)bn), \qquad (gb)(f(m)n) = g(bf(m)n) = g((bf)(m)n).$$

These identities respect the defining identifications in both tensor products. Right C-linearity follows from $g(f(m)nc) = g(f(m)n)c$ for $c \in C$. For $a \in A$,

$$\theta_r(g \otimes fa)(m \otimes n) = g(f(am)n) = (\theta_r(g \otimes f)a)(m \otimes n).$$

If X also has a commuting left C-action, replacing g by cg multiplies the value by c, so θ_r preserves that action as well.

Bijectivity of θ_r. Regard M only as a right B-module. For $M = B$, evaluation at 1 identifies $\mathrm{Hom}_B(B,B) \cong B$, while multiplication identifies $B \otimes_B N \cong N$. Under these isomorphisms, θ_r becomes

$$\mathrm{Hom}_C(N,X) \otimes_B B \longrightarrow \mathrm{Hom}_C(N,X), \qquad g \otimes b \longmapsto gb,$$

which is an isomorphism by Lemma 2.2(3). Finite direct sums commute with both constructions by Lemma 2.2(4) and (2.1). Thus θ_r is bijective for $M = B^{\oplus r}$, where $r \geq 0$ is finite. If $M \in \operatorname{add} B$, choose M' with $M \oplus M' \cong B^{\oplus r}$. The comparison for $M \oplus M'$ is the direct sum of the comparisons for M and M', so both summand maps are bijective.

The left comparison. For $f \in \operatorname{Hom}_{A^{\mathrm{op}}}(M, Y)$ and $g \in \operatorname{Hom}_{B^{\mathrm{op}}}(N, B)$, well-definedness follows from

$$f(mb\, g(n)) = f(mg(bn)), \qquad f(m(gb)(n)) = f(mg(n)b) = (bf)(mg(n)) \quad (b \in B).$$

Left A-linearity of f makes the resulting functional left A-linear. For $c \in C$,

$$\theta_\ell(cg \otimes f)(m \otimes n) = f(mg(nc)) = (c\theta_\ell(g \otimes f))(m \otimes n).$$

If Y also has a commuting right A-action, replacing f by fa multiplies the value by a, so θ_ℓ preserves that action as well. For $N = B$ as a left module, evaluation at 1 identifies $\operatorname{Hom}_{B^{\mathrm{op}}}(B, B) \cong B$ and turns θ_ℓ into

$$B \otimes_B \operatorname{Hom}_{A^{\mathrm{op}}}(M, Y) \longrightarrow \operatorname{Hom}_{A^{\mathrm{op}}}(M, Y), \qquad b \otimes f \longmapsto bf.$$

The same finite-sum and direct-summand argument proves (2).

Naturality. Let $p : M \to M'$ and $q : N \to N'$ be bimodule maps. For $f \in \operatorname{Hom}_B(M', B)$ and $g \in \operatorname{Hom}_C(N', X)$, applying θ_r before or after precomposition gives $g(f(p(m))q(n))$ at $m \otimes n$. For $f \in \operatorname{Hom}_{A^{\mathrm{op}}}(M', Y)$ and $g \in \operatorname{Hom}_{B^{\mathrm{op}}}(N', B)$, the two orders give $f(p(m)g(q(n)))$ for θ_ℓ. Thus both comparisons are natural. □

Let S be finite-dimensional semisimple, and consider bimodules ${}_SV_S$, ${}_SW_S$ finitely generated on the right and left, respectively. Corollary 1.13 gives $V_S \in \operatorname{add} S$ and ${}_SW \in \operatorname{add}_{\operatorname{mod} S^{\mathrm{op}}}(S)$. Proposition 2.6 therefore gives natural bimodule isomorphisms

$$\theta_r : W^* \otimes_S V^* \xrightarrow{\sim} (V \otimes_S W)^*, \qquad \theta_\ell : {}^*W \otimes_S {}^*V \xrightarrow{\sim} {}^*(V \otimes_S W), \tag{2.5}$$

$$\theta_r(g \otimes f)(v \otimes w) = g(f(v)w), \qquad \theta_\ell(g \otimes f)(v \otimes w) = f(vg(w)). \tag{2.6}$$

We refer to (2.5) as *anti-distributivity of the S-dual over $\otimes_S$*[2]: dualising reverses the order of the factors.

Example 2.7. *Finite generation alone does not suffice in Proposition 2.6 over a general algebra. Take $A = C = \Bbbk$, $B = \Bbbk[x]$, and $X = Y = \Bbbk$. Consider bimodules ${}_\Bbbk M_B$, ${}_B N_\Bbbk$ with $M = N = B/(x) \cong \Bbbk$ and the outer $\Bbbk$-actions given by scalar multiplication. Every $f \in \operatorname{Hom}_B(M, B)$ satisfies $f(\overline{1})x = f(\overline{1}x) = 0$, hence $f = 0$. Consequently*

$$\operatorname{Hom}_\Bbbk(N, \Bbbk) \otimes_B \operatorname{Hom}_B(M, B) = 0, \qquad \operatorname{Hom}_\Bbbk(M \otimes_B N, \Bbbk) \cong \Bbbk.$$

Both M_B and ${}_B N$ are generated by $\overline{1}$, so neither direct-summand hypothesis can be replaced by finite generation alone. For the left comparison, $\operatorname{Hom}_{B^{\mathrm{op}}}(N, B) = 0$, whereas $\operatorname{Hom}_\Bbbk(M \otimes_B N, \Bbbk) \cong \Bbbk$.

For S, V, W as in (2.5), take $A = B = S$, $C = \Bbbk$, $M = V$, $N = W$, and $X = \Bbbk$ in Proposition 2.6(1). Since $\operatorname{Hom}_\Bbbk(W, \Bbbk) = D(W)$, we obtain the natural bimodule isomorphism [GI20,

[2] this is not an common phrase in the literature, I only use it as a shorthand for clarifications here-and-there.

§2.1]

$$\theta_D : D(W) \otimes_S V^* \xrightarrow{\sim} D(V \otimes_S W),$$
$$\theta_D(\lambda \otimes f)(v \otimes w) = \lambda(f(v)w).$$

The actions on $D(W)$ are those in (1.15). For $s \in S$, naturality with respect to $W \to W$, $w \mapsto ws$, gives left S-linearity of θ_D.

Suppose now that S is equipped with a symmetrising form $\tau : S \to \Bbbk$, as in Definition 1.54. The two-sided dual of Definition 1.53 satisfies

$$\theta_{\mathrm{en}} : W^* \otimes_S V^{*\mathrm{en}} \xrightarrow{\sim} (V \otimes_S W)^{*\mathrm{en}},$$
$$\theta_{\mathrm{en}}(g \otimes F)(v \otimes w) = \sum_j g(a_j w) \otimes b_j^{\mathrm{op}}, \qquad F(v) = \sum_j a_j \otimes b_j^{\mathrm{op}}. \tag{2.7}$$

We check that the formula defines a map on the two indicated tensor products and preserves both outer actions. For example, $F(vs) = F(v)(s \otimes 1)$ replaces a_j by $a_j s$, while $(Fs)(v) = (1 \otimes s^{\mathrm{op}})F(v)$ replaces b_j by $b_j s$. These give, respectively, compatibility with $vs \otimes w = v \otimes sw$ and with the right outer action. Compatibility with $(gs) \otimes F = g \otimes (sF)$ follows from $(gs)(a_j w) = g(sa_j w)$. The other actions follow from right S-linearity of g and $F(tv) = F(v)(1 \otimes t^{\mathrm{op}})$.

The comparison maps γ_M in (1.23) give

$$\gamma_{V \otimes_S W} \circ \theta_{\mathrm{en}} = \theta_r \circ (\mathrm{id}_{W^*} \otimes \gamma_V),$$

since both sides send $g \otimes F$, evaluated at $v \otimes w$, to $\sum_j g(a_j w)\tau(b_j)$. Thus θ_{en} is an isomorphism by Theorem 1.56. Precomposition shows that θ_D and θ_{en} are natural. No assumption that S^{en} is semisimple is used; compare Remark 1.43.

For the maps $\alpha_M(f) = \tau \circ f$ in (1.18),

$$\alpha_{V \otimes_S W} \circ \theta_r = \theta_D \circ (\alpha_W \otimes \mathrm{id}_{V^*}),$$

because both sides send $g \otimes f$ to the functional $v \otimes w \mapsto \tau(g(f(v)w))$.

2.2 Generated algebras

Definition 2.8. *Let A be graded with $S = A_0$. Multiplication makes A_1 an S-bimodule and gives*

$$\mu_i : A_1^{\otimes_S i} \longrightarrow A_i, \qquad a_1 \otimes \cdots \otimes a_i \longmapsto a_1 \cdots a_i \quad (i \geq 1), \qquad \mu_0 = \mathrm{id}_S\,. \tag{2.8}$$

The algebra A is generated in degree 1 (over S) if every μ_i in (2.8) is surjective.

Equivalently, S and A_1 generate A as an algebra.

Example 2.9. *The algebra $\Bbbk A_n$ of Example 1.19 is generated in degree 1 over $S = \bigoplus_i \Bbbk e_{ii}$.*

Example 2.10. *The polynomial algebra $\Bbbk[x_1, \ldots, x_n]$, with each $\deg x_i = 1$ for all i, is generated in degree 1 over $\Bbbk$. Indeed, every monomial of total degree d is a product of d variables, and these monomials span the degree d component.*

Example 2.11. *Let*

$$A = \begin{pmatrix} \Bbbk[x] & \Bbbk[x] \\ 0 & \Bbbk[x] \end{pmatrix}, \qquad \deg(x^m e_{ij}) = m + j - i \quad (m \geq 0,\ 1 \leq i \leq j \leq 2).$$

Then

$$A_0 = \Bbbk e_{11} \oplus \Bbbk e_{22}, \qquad A_d = \Bbbk x^d e_{11} \oplus \Bbbk x^d e_{22} \oplus \Bbbk x^{d-1} e_{12} \quad (d \geq 1).$$

It is generated in degree 1 *over* A_0*, since*

$$x^d e_{11} = (xe_{11})^d, \qquad x^d e_{22} = (xe_{22})^d, \qquad x^{d-1} e_{12} = (xe_{11})^{d-1} e_{12} \quad (d \geq 1).$$

Here A_0 *has two simple summands,* $A_d \neq 0$ *for every* $d \geq 0$*, and* A *is non-commutative:* $e_{11}e_{12} = e_{12}$ *whereas* $e_{12}e_{11} = 0$*.*

2.3 Tensor algebras and the universal property

Definition 2.12. *Let* S *be a* $\Bbbk$*-algebra and consider a bimodule* ${}_S V_S$*. The tensor algebra of* V *over* S *is the graded* $\Bbbk$*-algebra*

$$T_S(V) := \bigoplus_{i \in \mathbb{Z}} T_S(V)_i, \quad \text{where} \quad T_S(V)_i = \begin{cases} V^{\otimes_S i}, & i \geq 1, \\ S, & i = 0, \\ 0, & i < 0, \end{cases}.$$

For $v_1, \ldots, v_{p+q} \in V$*, multiplication is determined* $\Bbbk$*-bilinearly by*

$$(v_1 \otimes_S \cdots \otimes_S v_p)(v_{p+1} \otimes_S \cdots \otimes_S v_{p+q}) = v_1 \otimes_S \cdots \otimes_S v_{p+q} \qquad \text{for all } p, q \geq 1,$$

and by the left and right S*-actions when a factor has degree* 0*.*

The displayed decomposition is the *tensor grading*: $S = T_S(V)_0$ and $V = T_S(V)_1$. Multiplication satisfies $T_S(V)_p T_S(V)_q \subset T_S(V)_{p+q}$ for all $p, q \in \mathbb{Z}$. We also call $T_S(V)$ the tensor algebra generated by V over S. Every element has finitely many non-zero homogeneous components. The unit is 1_S, and associativity follows from Lemma 2.2(3). Write $v_1 \cdots v_d$ for $v_1 \otimes_S \cdots \otimes_S v_d$ in $T_S(V)$. In a quotient algebra (including $T_S(V)$ itself) of $T_S(V)$, the notation $v_1 \cdots v_d$ denotes the image of $v_1 \otimes_S \cdots \otimes_S v_d$.

Example 2.13 (Polynomial and exterior algebras). *Let* V, W *be finite-dimensional vector spaces with bases* $\{x_i \mid 1 \leq i \leq n\}$ *and* $\{y_j \mid 1 \leq j \leq m\}$*, respectively. The symmetric algebra of* V *and the exterior algebra of* W *are*

$$\begin{aligned} \mathsf{S}(V) &= T_\Bbbk(V)/(x_i x_j - x_j x_i \mid 1 \leq i < j \leq n) \cong \Bbbk[x_1, \ldots, x_n], \\ \bigwedge W &= T_\Bbbk(W)/(w^2 \mid w \in W) = T_\Bbbk(W)/(y_j^2, y_i y_j + y_j y_i \mid 1 \leq i < j \leq m). \end{aligned} \tag{2.9}$$

They are generated in degree 1*, with*

$$\mathsf{S}(V)_0 = \Bbbk = (\textstyle\bigwedge W)_0, \qquad \mathsf{S}(V)_1 = V, \qquad (\textstyle\bigwedge W)_1 = W.$$

Their respective bases are

$$\{x_1^{a_1} \cdots x_n^{a_n} \mid a_1, \ldots, a_n \in \mathbb{Z},\ a_1, \ldots, a_n \geq 0\},$$
$$\{y_{j_1} \cdots y_{j_d} \mid 0 \leq d \leq m,\ 1 \leq j_1 < \cdots < j_d \leq m\}.$$

The bases in Example 2.13 can be checked directly. For $\mathsf{S}(V)$, commuting adjacent generators gives the ordered monomials, which are independent under the map to the polynomial ring. For $\bigwedge W$, the relations reduce every product of basis elements to a multiple of a strictly increasing product or to zero. For $1 \le d \le m$, fix $j_1 < \cdots < j_d$ and consider the linear functional $W^{\otimes_{\Bbbk} d} \to \Bbbk$ sending $w_1 \otimes \cdots \otimes w_d$ to the determinant of their coordinates in rows $j_1, \dots, j_d$ relative to the basis $\{y_j \mid 1 \le j \le m\}$. The determinant vanishes whenever two adjacent factors agree, hence on the degree d component of $(w^2 \mid w \in W)$. On increasing products it is 1 on $y_{j_1} \cdots y_{j_d}$ and 0 on the others, proving independence in every characteristic.

Lemma 2.14. *Let $S = \bigoplus_{i=1}^{n} \Bbbk e_i$ with coordinate idempotents, and consider bimodules ${}_S V_S$, ${}_S W_S$. There is a bimodule isomorphism*

$$V \otimes_S W \cong \bigoplus_{i,j,k=1}^{n} e_i V e_j \otimes_{\Bbbk} e_j W e_k. \tag{2.10}$$

Consequently, after choosing a basis of each $e_i V e_j$, a basis of $V^{\otimes_S d}$ for $d \ge 1$ consists of products of d chosen basis elements whose adjacent idempotents match.

Proof The orthogonal idempotents sum to 1, so

$$V = \bigoplus_{i,j=1}^{n} e_i V e_j, \qquad (e_i V e_j) \otimes_S (e_p W e_k) = 0 \quad (1 \le i,j,p,k \le n,\ j \ne p). \tag{2.11}$$

Indeed, $v \otimes_S w = v e_j \otimes_S w = v \otimes_S e_j w = 0$ for $v \in e_i V e_j$, $w \in e_p W e_k$, and $j \ne p$. The map from right to left in (2.10) sends $v \otimes_{\Bbbk} w$ to $v \otimes_S w$ on each summand. Its inverse is

$$V \otimes_S W \longrightarrow \bigoplus_{i,j,k=1}^{n} e_i V e_j \otimes_{\Bbbk} e_j W e_k, \qquad v \otimes_S w \longmapsto \sum_{i,j,k=1}^{n} e_i v e_j \otimes_{\Bbbk} e_j w e_k.$$

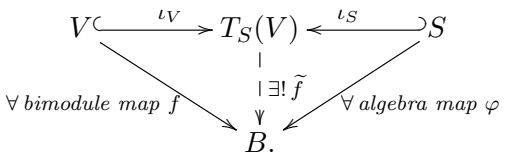
Checking that these two assignments induce well-defined, mutually inverse maps is left to the reader. Iterating (2.10) and using Definition 1.36 gives the stated basis. □

Example 2.15. *We apply Lemma 2.14 to realise $\Bbbk A_3$ as a tensor algebra. Let $S = \Bbbk^{\oplus 3}$ with coordinate idempotents e_1, e_2, e_3, and take the bimodule ${}_S V_S$ with $V = \Bbbk a \oplus \Bbbk b$, $e_1 a e_2 = a$, and $e_2 b e_3 = b$; all other idempotent actions are zero. By Lemma 2.14,*

$$V \otimes_S V = \Bbbk(a \otimes b), \qquad V^{\otimes_S 3} = 0, \qquad T_S(V) = S \oplus \Bbbk a \oplus \Bbbk b \oplus \Bbbk ab.$$

Here $ab = a \otimes b$. The linear map $T_S(V) \to \Bbbk A_3$ given by $e_i \mapsto e_{ii}$, $a \mapsto e_{12}$, $b \mapsto e_{23}$, and $ab \mapsto e_{13}$ preserves multiplication and sends a basis to a basis, hence is an algebra isomorphism.

Proposition 2.16 (Universal property of the tensor algebra). *Let S be a $\Bbbk$-algebra, consider a bimodule ${}_S V_S$, and let $\iota_S : S \to T_S(V)$ and $\iota_V : V \to T_S(V)$ be the canonical inclusions. Given a unital $\Bbbk$-algebra homomorphism $\varphi : S \to B$ and a bimodule map $f : {}_S V_S \to {}_S B_S$, where $sbt = \varphi(s) b \varphi(t)$ on B, there is a unique $\Bbbk$-algebra homomorphism $\widetilde{f} : T_S(V) \to B$ making the diagram commute:*



(2.12)

Proof Define $\widetilde{f}$ on S and each tensor power by

$$\widetilde{f}(s) = \varphi(s), \qquad \widetilde{f}(v_1 \otimes_S \cdots \otimes_S v_i) = f(v_1)\cdots f(v_i) \quad (s \in S,\ i \geq 1). \tag{2.13}$$

The identity $f(vs)f(w) = f(v)\varphi(s)f(w) = f(v)f(sw)$ shows that replacing adjacent factors $vs \otimes w$ by $v \otimes sw$ does not change the value in (2.13). The formulas therefore induce linear maps on the tensor powers, which extend to their direct sum. Multiplication is preserved by (2.13), including the degree 0 factors because $f(sv) = \varphi(s)f(v)$ and $f(vs) = f(v)\varphi(s)$. Also, $\widetilde{f}(1_S) = 1_B$. Every algebra homomorphism making (2.12) commute must have the values in (2.13), proving uniqueness. □

If B is graded, $\varphi(S) \subset B_0$, and $f(V) \subset B_1$, then (2.13) gives $\widetilde{f}(V^{\otimes_S i}) \subset B_i$ for every $i \geq 0$. Thus $\widetilde{f}$ is a graded algebra homomorphism.

Corollary 2.17. *If A is generated in degree 1 over $S = A_0$, then the multiplication map defines a surjective graded algebra homomorphism*

$$\mu : T_S(A_1) \twoheadrightarrow A, \qquad \mu|_S = \mathrm{id}_S, \qquad a_1 \otimes_S \cdots \otimes_S a_i \longmapsto a_1 \cdots a_i \quad \forall i \geq 1.$$

Its degree i component is μ_i from (2.8)*. The kernel $I := \operatorname{Ker}\mu$ is homogeneous, $I_0 = I_1 = 0$, and μ induces a graded algebra isomorphism*

$$T_S(A_1)/I \xrightarrow{\sim} A, \qquad z + I \longmapsto \mu(z).$$

Proof In Proposition 2.16, take $V = A_1$, $B = A$, and the inclusions

$$\varphi : S \hookrightarrow A, \quad s \longmapsto s, \qquad f : A_1 \hookrightarrow A, \quad a \longmapsto a.$$

Then (2.13) gives $\widetilde{f} = \mu$. The map μ preserves degrees, and each component μ_i is surjective by the assumption that A is generated in degree 1. Thus $\operatorname{Ker}\mu$ is homogeneous, and $\mu_0 = \mathrm{id}_S$, $\mu_1 = \mathrm{id}_{A_1}$ give $I_0 = I_1 = 0$. The map $z + I \mapsto \mu(z)$ is therefore a degree-preserving bijective algebra homomorphism. □

Definition 2.18. *For the map μ in Corollary 2.17, the homogeneous two-sided ideal $I := \operatorname{Ker}\mu$ is the ideal of relations.*

The bimodule A_1 specifies the generators and their left and right S-actions; I specifies the relations between their products. Homogeneity is essential for the quotient grading: the relation $x - x^2$ in Example 1.34 does not satisfy this condition.

2.4 Quadratic algebras

We now consider algebras whose ideal of relations is generated in degree 2.

Definition 2.19. *A graded algebra A is quadratic if there exist a finite-dimensional semisimple $\Bbbk$-algebra S, a finite-dimensional S-bimodule V, and an S-sub-bimodule $R \subset V \otimes_S V$ such that $A \cong T_S(V)/(R)$ as graded algebras, where (R) is the two-sided ideal of $T_S(V)$ generated by R.*

An isomorphism $A \cong T_S(V)/(R)$ as in Definition 2.19 is called a quadratic presentation of A.

Proposition 2.20. *A quadratic algebra $A = T_S(V)/(R)$ has $A_0 = S$, $A_1 = V$, and $A_2 = (V \otimes_S V)/R$, is locally finite-dimensional, and is generated in degree 1 over S.*

Proof The ideal (R) has no components in degrees 0 and 1. Its degree 2 component is exactly R. Each A_i is a quotient of the finite-dimensional space $V^{\otimes_S i}$, and $\mu_i : V^{\otimes_S i} \twoheadrightarrow A_i$. □

Thus a quadratic presentation recovers $S \cong A_0$, $V \cong A_1$, and $R \cong \operatorname{Ker}(A_1 \otimes_{A_0} A_1 \to A_2)$. In particular, the kernel of the canonical map $T_{A_0}(A_1) \twoheadrightarrow A$ is generated by its degree 2 component. Quadraticity depends on the grading: for $n \geq 2$, the grading $\deg e_{ij} = 2(j-i)$ on $\Bbbk A_n$ is not quadratic, since it has no degree 1 generators.

Example 2.21. *Let $S = \bigoplus_{i=1}^{n} \Bbbk e_{ii}$ and $V = \bigoplus_{i=1}^{n-1} \Bbbk e_{i,i+1}$ inside $\Bbbk A_n$. Then we have $T_S(V) \cong \Bbbk A_n$ given by*

$$e_{i,i+1} \otimes_S e_{i+1,i+2} \otimes_S \cdots \otimes_S e_{j-1,j} \longmapsto e_{ij} \quad (1 \leq i < j \leq n).$$

Example 2.22. *For S, V, a, b from Example 2.15, the quotient*

$$T_S(V)/(ab) \cong \Bbbk A_3/(e_{13})$$

is quadratic with $R = \Bbbk(a \otimes b)$. Its basis is e_1, e_2, e_3, a, b, in degrees $0, 0, 0, 1, 1$, as in Example 1.33.

Example 2.23. *Let $B = \Bbbk A_4$ and $A = B/(e_{14})$, with the grading of Example 1.19. The ideal $I = (e_{14}) = \Bbbk e_{14}$ has $I_2 = 0$ but $I_3 \neq 0$. Since $e_{14} = e_{12}e_{23}e_{34}$, the algebra A is generated in degree 1 with a relation of degree 3 and no relations of degree 2. By Proposition 2.20, any quadratic presentation would have the same generators and zero quadratic relation space, hence would give $\Bbbk A_4$ itself. Thus A is not quadratic.*

Example 2.24. *The symmetric and exterior algebras of Example 2.13 are quadratic: the defining relations in (2.9) belong to $V \otimes_{\Bbbk} V$ and $W \otimes_{\Bbbk} W$, respectively.*

2.5 Hilbert series

Let A be a locally finite-dimensional graded algebra with $S = A_0$ finite-dimensional semisimple. For a graded right A-module M, regard M_d as an A-module via $A \to A/A_+ = S$. The simple direct summands of the semisimple module M_d are its *composition factors*.

Notation. *By Theorem 1.12, choose a complete set of primitive orthogonal idempotents $e_1, \ldots, e_n \in S$, i.e. $e_1S, \ldots, e_nS$ represent the full set of simple right S-modules up to isomorphism. Then $L_j := e_jS \cong e_jA/e_jA_+$ is a graded simple A-module concentrated in degree 0 for every $1 \leq j \leq n$.*

Definition 2.25. *For a locally finite-dimensional graded right A-module M, write $[M_d : L_j]$ for the multiplicity of the composition factor L_j in M_d. Its degree d dimension vector is the row vector*[3]

$$\mathbf{dim}\, M_d := \big([M_d : L_j]\big)_{1 \leq j \leq n}.$$

For a graded module M, its Hilbert series[4] *is*

$$h_M(t) := \sum_{d \in \mathbb{Z}} (\mathbf{dim}\, M_d) t^d = \sum_{d \in \mathbb{Z}} \big([M_d : L_j]\big)_{1 \leq j \leq n} t^d.$$

The Hilbert series of the algebra A collects the right-module series $h_{e_iA}(t)$ as rows:

$$h_A(t)_{ij} := \big(h_{e_iA}(t)\big)_j = \sum_{d \geq 0} [e_iA_d : L_j] t^d \quad \text{for all } 1 \leq i, j \leq n.$$

[3] In most representation theory literature, it is common to regard dimension vector as a column vector. But in the context of Hilbert series, we follow Beilinson–Ginzburg–Soergel convention to use row vector here.

[4] sometimes it may also be referred as quantum dimension vector; the variable t is the quantum parameter and specialising $t = 1$ yields the usual dimension vector

Row i of $h_A(t)$ records the composition factors of e_iA degree by degree. The entries of the Hilbert series are formal Laurent series; for algebras they are formal power series. Finite support gives Laurent polynomials or polynomials, respectively. Since $e_iA_0 = L_i$, we have $h_A(0) = I_n$, the $n \times n$ identity array. For a finite-dimensional right S-module N, use the same multiplicity notation and write $\mathbf{dim}\, N = ([N : L_j])_{1\le j\le n}$.

The multiplicities can be computed using Hom spaces:

$$[M_d : L_j] = \dim_{\mathrm{End}_A(L_j)} \mathrm{Hom}_A(e_jA, M_d) \quad \text{for all } d \in \mathbb{Z},\ 1 \le j \le n. \tag{2.14}$$

Indeed, $M_dA_+ = 0$, so every map $e_jA \to M_d$ factors uniquely through $e_jA/e_jA_+ \cong L_j$. This identifies $\mathrm{Hom}_A(e_jA, M_d)$ with $\mathrm{Hom}_A(L_j, M_d)$; precomposition gives its right vector-space structure over the division algebra $\mathrm{End}_A(L_j)$. By the Schur's Lemma, A non-zero map between simple modules is an isomorphism, so each copy of L_j contributes one copy of $\mathrm{End}_A(L_j)$, and the other simples contribute zero. This also shows that the multiplicities are independent of the chosen decomposition.

Example 2.26. *If $S = \bigoplus_{j=1}^n \Bbbk e_j$ with coordinate idempotents, then $\mathrm{End}_A(L_j) = \Bbbk$. The Yoneda lemma 1.17 says that*

$$[M_d : L_j] = \dim_{\Bbbk}(M_de_j), \qquad \mathbf{dim}\, M_d = (\dim_{\Bbbk} M_de_j)_{1\le j\le n}.$$

Lemma 2.27. *For locally finite-dimensional graded right A-modules L, M, N, we have the following.*

(1) *A graded isomorphism $M \cong N$ implies $h_M(t) = h_N(t)$.*

(2) *A short exact sequence $0 \to L \to M \to N \to 0$ gives $h_M(t) = h_L(t) + h_N(t)$.*

(3) *For every $a \in \mathbb{Z}$, $h_{M(a)}(t) = t^{-a}h_M(t)$.*

Proof A graded isomorphism preserves each degree and its simple-module multiplicities. In each degree, a graded short exact sequence gives $0 \to L_d \to M_d \to N_d \to 0$, so $[M_d : L_j] = [L_d : L_j] + [N_d : L_j]$ for all $1 \le j \le n$. Finally, $M(a)_d = M_{d+a}$ gives

$$h_{M(a)}(t) = \sum_d (\mathbf{dim}\, M_{d+a})t^d = t^{-a} \sum_j (\mathbf{dim}\, M_j)t^j. \qquad \square$$

Definition 2.28. *For graded right and left S-modules M, N, their tensor product has total grading*

$$(M \otimes_S N)_d = \bigoplus_{p+q=d} M_p \otimes_S N_q.$$

The same definition applies to bimodules, and to vector spaces by taking $S = \Bbbk$.

The decomposition follows by distributing tensor products over direct sums, as in Lemma 2.2(4). When M, N are locally finite-dimensional and bounded below, only finitely many pairs (p, q) contribute in a fixed degree, so the tensor product is locally finite-dimensional and bounded below.

Lemma 2.29. *Let S be finite-dimensional semisimple and ${}_SV_S$ a finite-dimensional bimodule. There are bimodule isomorphisms*

$$\begin{aligned} V \otimes_S V^* &\xrightarrow{\sim} \mathrm{End}_S(V), \qquad v \otimes f \longmapsto \big(u \mapsto vf(u)\big), \\ {}^*V \otimes_S V &\xrightarrow{\sim} \mathrm{End}_{S^{\mathrm{op}}}(V), \quad g \otimes v \longmapsto \big(u \mapsto g(u)v\big). \end{aligned} \tag{2.15}$$

The source actions are those of Lemma 2.2(2). The target actions are defined, for $s, t \in S$ and

$u \in V$, *by*

$$\begin{aligned} (sFt)(u) &= sF(tu) \quad \textit{for } F \in \mathrm{End}_S(V), \\ (sGt)(u) &= G(us)t \quad \textit{for } G \in \mathrm{End}_{S^{\mathrm{op}}}(V). \end{aligned} \tag{2.16}$$

Proof For the first comparison, $(vs)f(u) = v(sf)(u)$ and $vf(us) = vf(u)s$ give a well-defined map to $\mathrm{End}_S(V)$. By Corollary 1.13, identify V_S with a direct summand of $S^{\oplus m}$ for some integer $m \geq 0$, and let $p : S^{\oplus m} \to V$ be the right S-linear projection. For $1 \leq j \leq m$, let $\varepsilon_j \in S^{\oplus m}$ be the vector with 1 in coordinate j and zero elsewhere, and let $q_j : S^{\oplus m} \to S$ be the coordinate projection. Put $v_j = p(\varepsilon_j)$ and $f_j = q_j|_V$. Then $u = \sum_{j=1}^m v_j f_j(u)$ for every $u \in V$. The inverse is

$$\mathrm{End}_S(V) \longrightarrow V \otimes_S V^*, \qquad F \longmapsto \sum_{j=1}^m F(v_j) \otimes f_j.$$

Indeed, one composite sends F to $u \mapsto \sum_{j=1}^m F(v_j) f_j(u) = F(u)$, and the other sends

$$v \otimes f \longmapsto \sum_{j=1}^m v f(v_j) \otimes f_j = v \otimes \sum_{j=1}^m f(v_j) f_j = v \otimes f.$$

Applying this construction over S^{op} proves the second isomorphism, using $V \otimes_{S^{\mathrm{op}}} {}^*V \cong {}^*V \otimes_S V$, $v \otimes g \mapsto g \otimes v$.

The commuting left and right actions on V make (2.16) bimodule actions. For example, $(sFt)(ur) = sF(tur) = sF(tu)r$ for $r \in S$, so sFt is right S-linear. The images of $sv \otimes ft$ and $sg \otimes vt$, evaluated at u, are $svf(tu)$ and $g(us)vt$, respectively. These are exactly the actions in (2.16), proving bimodule linearity of both comparisons. □

Evaluation applies a functional to an element. Coevaluation expresses the identity endomorphism as an element of a tensor product.

Proposition 2.30. *For a bimodule ${}_SV_S$, where S is finite-dimensional semisimple, evaluation gives bimodule maps*

$$\begin{aligned} V^* \otimes_S V &\longrightarrow S, \quad f \otimes v \longmapsto f(v), \\ V \otimes_S {}^*V &\longrightarrow S, \quad v \otimes g \longmapsto g(v). \end{aligned} \tag{2.17}$$

If V is finite-dimensional, the coevaluation maps are

$$\begin{aligned} S &\longrightarrow V \otimes_S V^* \xrightarrow{\sim} \mathrm{End}_S(V), \qquad s \longmapsto \big(v \mapsto sv\big), \\ S &\longrightarrow {}^*V \otimes_S V \xrightarrow{\sim} \mathrm{End}_{S^{\mathrm{op}}}(V), \quad s \longmapsto \big(v \mapsto vs\big). \end{aligned} \tag{2.18}$$

The isomorphisms are those in (2.15)*. Both coevaluation maps are bimodule maps, and each image of* 1 *corresponds to* id_V*.*

Proof The identities $(fs)(v) = f(sv)$ and $(sg)(v) = g(vs)$ from (1.15) make evaluation well-defined on the two tensor products; the other two actions give its bimodule linearity. For coevaluation, left multiplication is right S-linear and right multiplication is left S-linear because the two actions on V commute. Under the endomorphism actions in (2.16), both multiplication assignments preserve the two S-actions. The isomorphisms in (2.15) therefore give the stated bimodule maps. □

For a finite-dimensional graded bimodule ${}_SV_S$, use $V^{(*)}$ and ${}^{(*)}V$ in Proposition 2.30. With the total grading, evaluation and coevaluation preserve degree: evaluation pairs degrees i and $-i$, and id_V has degree 0.

To multiply two arrays $C = (c_{ij})$ and $D = (d_{ij})$, or a row vector $m = (m_i)$ and an array C, use

$$(CD)_{i\ell} = \sum_{j=1}^{n} c_{ij} d_{j\ell}, \qquad (mC)_j = \sum_{i=1}^{n} m_i c_{ij}.$$

Thus multiplication sums over the matching middle index, rather than multiplying individual entries independently. For series, use the same products on coefficients and collect terms with the same total power of t.

Lemma 2.31. *Let N be a finite-dimensional right S-module and ${}_SU_S$ a finite-dimensional bimodule. Then*

$$[N \otimes_S U : L_j] = \sum_{i=1}^{n} [N : L_i][e_i U : L_j] \quad \text{for all } 1 \le j \le n.$$

For a locally finite-dimensional graded right A-module M bounded below,

$$h_{M \otimes_S A}(t) = h_M(t) h_A(t).$$

Proof Decompose $N \cong \bigoplus_{i=1}^{n} (e_i S)^{\oplus [N:L_i]}$. Multiplication $e_i S \otimes_S U \to e_i U$, $e_i s \otimes u \mapsto e_i s u$, is an isomorphism with inverse $u \mapsto e_i \otimes u$. Distribution over direct sums therefore gives

$$N \otimes_S U \cong \bigoplus_{i=1}^{n} (e_i U)^{\oplus [N:L_i]}.$$

Counting copies of L_j proves the first formula. Apply the first formula to $M_p \otimes_S A_q$ and sum over all $p, q \in \mathbb{Z}$ with $p + q = d$. By the definition of $h_A(t)$, the resulting sum is the coefficient of t^d in $h_M(t) h_A(t)$. □

The evaluation and coevaluation maps in Proposition 2.30 are the constructions in the last two paragraphs of [BGS96, §2.7, pp. 484–485], translated to our notation.

For a locally finite-dimensional graded bimodule ${}_SV_S$, (2.18) applies to each V_d; a coevaluation for the whole module need not exist. For example, take $S \neq 0$ and $V = \bigoplus_{d \ge 0} S(-d)$. Every element $\sum_{i=1}^{n} v_i \otimes f_i$ of $V \otimes_S V^{(*)}$ defines an endomorphism with image contained in the finite-dimensional submodule $\sum_{i=1}^{n} v_i S$, so it cannot represent id_V.

Example 2.32 (Hilbert series of a tensor algebra). *Consider the tensor algebra $A = T_S(V)$ and put*

$$C := ([e_i A_1 : L_j])_{1 \le i,j \le n} = ([e_i V : L_j])_{1 \le i,j \le n}.$$

Since $e_i A_0 = L_i$ and $e_i A_{d+1} \cong (e_i A_d) \otimes_S A_1$, Lemma 2.31 gives, by induction,

$$[e_i A_d : L_j] = (C^d)_{ij}, \qquad C^0 = I_n \quad \text{for all } d \ge 0,\ 1 \le i, j \le n.$$

Consequently,

$$\boxed{h_{T_S(V)}(t) = \sum_{d \ge 0} t^d C^d = (I_n - tC)^{-1}.} \tag{2.19}$$

Indeed, multiplying the series on either side by $I_n - tC$ gives constant coefficient I_n; the coefficient of t^d for $d \ge 1$ is $C^d - C^d = 0$. The inverse is therefore an identity of formal power series with the coefficient multiplication defined above.

For $S = \mathrm{Mat}_d(\Bbbk)$, take the bimodule $V = S^{\oplus m}$ and $e = e_{11}$. There is one simple right S-module eS, and $[eV : eS] = m$, whereas $\dim_{\Bbbk} V = md^2$. Hence $h_{T_S(V)}(t) = 1/(1-mt)$.

Example 2.33 (Upper triangular algebras and a quotient). *For $A = \Bbbk A_n$, order the coordinate idempotents as $e_{11}, \dots, e_{nn}$. Let C have $C_{i,i+1} = 1$ for $1 \le i < n$ and all other entries zero. The tensor presentation in Example 2.21 and (2.19) give*

$$h_A(t) = I_n + tC + \cdots + t^{n-1}C^{n-1}, \qquad h_A(t)_{ij} = \begin{cases} t^{j-i}, & i \le j, \\ 0, & i > j. \end{cases}$$

For $n = 3$, killing e_{13} removes the degree 2 contribution in entry $(1,3)$:

$$h_{\Bbbk A_3}(t) = \begin{pmatrix} 1 & t & t^2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}, \qquad h_{\Bbbk A_3/(e_{13})}(t) = \begin{pmatrix} 1 & t & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

The quotient removes $\Bbbk e_{13}$ from degree 2, and hence removes t^2 from entry $(1,3)$.

Example 2.34. *Let $n = \dim_{\Bbbk} V$ and $m = \dim_{\Bbbk} W$ in Example 2.13. Choosing an exponent independently for each x_i, and choosing whether each y_j occurs, gives*

$$h_{\mathsf{S}(V)}(t) = \sum_{a_1,a_2,\cdots a_n \ge 0} t^{a_1+a_2+\cdots a_n} = \prod_{i=1}^{n} \sum_{a_i \ge 0} t^{a_i} = \frac{1}{(1-t)^n} = \sum_{d \ge 0} \binom{d+n-1}{n-1} t^d,$$
$$h_{\bigwedge W}(t) = (1+t)^m.$$

Lemma 2.35. (1) *For locally finite-dimensional graded vector spaces U, W, we have*

Let A, B be locally finite-dimensional graded algebras with $A_0 = \bigoplus_{i=1}^n \Bbbk e_i$ and $B_0 = \bigoplus_{j=1}^m \Bbbk f_j$ where e_i, f_j's are the coordinate idempotents. Give $A \otimes_{\Bbbk} B$ the total grading $(A \otimes_{\Bbbk} B)_d = \bigoplus_{i+j=d} A_i \otimes_{\Bbbk} B_j$ and multiplication $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$. Then

$$h_{A \otimes_{\Bbbk} B}(t)_{(i,p),(j,q)} = h_A(t)_{ij} h_B(t)_{pq} \quad \text{for all } 1 \le i, j \le n,\ 1 \le p, q \le m.$$

Proof For vector spaces, take dimensions in each summand of the total grading. For algebras,

$$(e_i \otimes f_p)(A_a \otimes_{\Bbbk} B_b)(e_j \otimes f_q) = (e_i A_a e_j) \otimes_{\Bbbk} (f_p B_b f_q).$$

Take vector space dimensions and sum over $a + b = d$ to obtain the coefficient of t^d. □

2.6 The degree 2 relations of a quotient

Let S be finite-dimensional semisimple, V a finite-dimensional S-bimodule, and $I \subset T_S(V)$ a homogeneous two-sided ideal with $I_0 = I_1 = 0$. Set $A = T_S(V)/I$. In degree 2,

$$A_2 \cong (V \otimes_S V)/I_2, \quad \text{where} \quad I_2 = I \cap (V \otimes_S V).$$

Definition 2.36. *The quadratic part of A is $qA := T_S(V)/(I_2)$.*

Since $(I_2) \subset I$, there is a surjective graded algebra homomorphism

$$\pi_A : qA \longrightarrow A, \qquad z + (I_2) \longmapsto z + I, \qquad \operatorname{Ker} \pi_A = I/(I_2).$$

Consequently,

$$\pi_A \text{ is an isomorphism} \iff I = (I_2) \iff A \text{ is quadratic.}$$

For the last equivalence, (I_2) gives a quadratic presentation, while Proposition 2.20 identifies the generators and degree 2 relations of any quadratic presentation with A_1 and I_2.

Example 2.37. *For $A = \Bbbk A_4/(e_{14})$ from Example 2.23, we have $I_2 = 0$, so*

$$qA = \Bbbk A_4, \qquad \pi_A : \Bbbk A_4 \twoheadrightarrow \Bbbk A_4/(e_{14}), \qquad \operatorname{Ker} \pi_A = \Bbbk e_{14}.$$

Thus qA does not impose the cubic relation $e_{12}e_{23}e_{34} = 0$.

2.7 Orthogonal complements and quadratic duality

Let S be finite-dimensional semisimple.

Definition 2.38. *For an S-sub-bimodule U of a finite-dimensional S-bimodule M, its right orthogonal complement and left orthogonal complement are*

$$U^{\perp} := \{f \in M^* \mid f(u) = 0 \text{ for all } u \in U\},$$
$$^{\perp}U := \{g \in {}^*M \mid g(u) = 0 \text{ for all } u \in U\}.$$

Both orthogonal complements are S-sub-bimodules. Indeed, for $s, t \in S$ and $u \in U$, the action on S-dual (1.15) reads

$$(sft)(u) = sf(tu) = 0 \quad (f \in U^{\perp}), \qquad (sgt)(u) = g(us)t = 0 \quad (g \in {}^{\perp}U),$$

since $tu, us \in U$. The orthogonal complements are defined without choosing a symmetrising form, but it is possible to realise, through the natural isomorphism associated to a chosen symmeterising form, the orthogonal complements as a sub-bimodule of $\mathbb{D}(M)$.

If S is equipped with a symmetrising form $\tau : S \to \Bbbk$, the comparison maps α_M, β_M of (1.18) satisfy

$$\alpha_M(U^{\perp}) = \{\lambda \in D(M) \mid \lambda(u) = 0 \text{ for all } u \in U\} = \beta_M({}^{\perp}U). \tag{2.20}$$

The middle space sits inside the two outer space, since $\tau \circ \lambda$ satisfies the definition of orthogonal complement. Conversely, $\alpha_M(f)(U) = 0$ gives

$$\tau(f(u)s) = \tau(f(us)) = 0 \quad \text{for all } s \in S,\ u \in U.$$

Non-degeneracy of τ forces $f(u) = 0$ for every $u \in U$. For the left dual, $\tau(sg(u)) = \tau(g(su)) = 0$ gives $g(u) = 0$ instead. Bijectivity of α_M, β_M in Theorem 1.56 proves (2.20).

Example 2.39. *We compute the orthogonal complements of the relation spaces of $\mathsf{S}(V)$ and $\bigwedge W$ for $\dim_{\Bbbk} V = 2 = \dim_{\Bbbk} W$. Take $V = \Bbbk x_1 \oplus \Bbbk x_2$ and $W = \Bbbk y_1 \oplus \Bbbk y_2$ in Example 2.13. The relation spaces in $V \otimes_{\Bbbk} V$ and $W \otimes_{\Bbbk} W$, respectively, are*

$$R_{\mathsf{S}(V)} = \Bbbk(x_1 \otimes x_2 - x_2 \otimes x_1),$$
$$R_{\bigwedge W} = \Bbbk(y_1 \otimes y_1) \oplus \Bbbk(y_2 \otimes y_2) \oplus \Bbbk(y_1 \otimes y_2 + y_2 \otimes y_1).$$

Let x_1^*, x_2^* and y_1^*, y_2^* be the dual bases of $D(V)$ and $D(W)$, respectively. By anti-distributivity of the S-dual (2.5), with $S = \Bbbk$ and the formulas in (2.6),

$$R_{\mathsf{S}(V)}^{\perp} = \operatorname{span}_{\Bbbk}\{x_1^* \otimes x_1^*,\ x_2^* \otimes x_2^*,\ x_1^* \otimes x_2^* + x_2^* \otimes x_1^*\},$$
$$R_{\bigwedge W}^{\perp} = \Bbbk(y_1^* \otimes y_2^* - y_2^* \otimes y_1^*).$$

The first space consists of the exterior relations on $D(V)$; the second consists of the symmetric relations on $D(W)$.

2.8 Exercises

Exercise 2.40 (Anti-distributivity of the S-dual). *Let S be a finite-dimensional semisimple algebra, V a finite-dimensional right S-module, and ${}_SW_S$ a finite-dimensional bimodule. Consider the formula*

$$\theta_r^{V,W} : W^* \otimes_S V^* \longrightarrow (V \otimes_S W)^*, \qquad \theta_r^{V,W}(g \otimes f)(v \otimes w) = g(f(v)w).$$

Write $\theta_r = \theta_r^{V,W}$ when V, W are fixed.

(1) *Check that the formula is bilinear in (g, f) and in (v, w), is unchanged by*

$$vs \otimes w \longleftrightarrow v \otimes sw, \qquad gs \otimes f \longleftrightarrow g \otimes sf \quad \text{for all } s \in S,$$

and gives a right S-linear map $V \otimes_S W \to S$ for each $g \otimes f$. Prove that θ_r is left S-linear. If V is also an S-bimodule, verify $\theta_r(sg \otimes ft) = s\theta_r(g \otimes f)t$ for all $s, t \in S$.

(2) *Prove that θ_r is bijective.*

(3) *Let V' be a finite-dimensional right S-module and ${}_SW'_S$ a finite-dimensional bimodule. For a right S-linear map $p : V \to V'$ and a bimodule map $q : W \to W'$, prove*

$$(p \otimes q)^* \circ \theta_r^{V',W'} = \theta_r^{V,W} \circ (q^* \otimes p^*).$$

Here $p^* : (V')^* \to V^*$, $q^* : (W')^* \to W^*$, and $(p \otimes q)^* : (V' \otimes_S W')^* \to (V \otimes_S W)^*$ are given by precomposition with p, q, and $p \otimes q$, respectively.

Hint for (2). *First take $V = S$, then use Corollary 1.13.*

Remark. *For a finite-dimensional bimodule ${}_SV_S$ and a finite-dimensional left S-module W, the corresponding natural isomorphism of right S-modules is*

$$\theta_\ell : {}^*W \otimes_S {}^*V \longrightarrow {}^*(V \otimes_S W), \qquad \theta_\ell(g \otimes f)(v \otimes w) = f(vg(w)).$$

Exercise 2.41. *For $n \geq 3$ and $Z \subset \{1, \ldots, n-2\}$, let*

$$A_Z = \Bbbk A_n / (e_{k,k+2} \mid k \in Z)$$

with the quotient grading, and write $\bar{e}_{ij}$ for the image of e_{ij} in A_Z.

(1) *Show that a homogeneous basis consists of $\{\bar{e}_{ii} \mid 1 \leq i \leq n\}$ together with all $\bar{e}_{ij}$ for which $1 \leq i < j \leq n$ and no $k \in Z$ satisfies $i \leq k \leq j-2$.*

(2) *Compute $h_{A_Z}(t)$, and give an S-preserving isomorphism of graded algebras $A_Z \cong T_S(V)/(R)$ with explicit S, V, and $R \subset V \otimes_S V$.*

Hint. *For $i \leq k \leq j-2$, use $e_{ij} = e_{ik}e_{k,k+2}e_{k+2,j}$ to determine which matrix units belong to the ideal.*

Exercise 2.42. *Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ with coordinate idempotents, and let the bimodule $V = \Bbbk a \oplus \Bbbk b$ in degree 1 satisfy $e_1 a e_2 = a$, $e_2 b e_1 = b$, with all other idempotent actions zero.*

(1) *Find a basis of $T_S(V)_d$ for every $d \geq 0$ and show that*

$$h_{T_S(V)}(t) = \frac{1}{1-t^2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

(2) *Find a homogeneous basis and the Hilbert series of $A = T_S(V)/(ab)$, and prove that A is finite-dimensional.*

Hint for (1). *Count alternating words separately in each degree, including degree 0.*

Exercise 2.43. *For the algebras in Example 2.13, prove*

$$\dim_{\Bbbk} \mathsf{S}(V)_2 = \frac{n(n+1)}{2}, \qquad \dim_{\Bbbk}(\bigwedge W)_2 = \frac{m(m-1)}{2} \quad \text{where } n = \dim_{\Bbbk} V,\ m = \dim_{\Bbbk} W.$$

Hint. *Count the ordered pairs allowed in the two bases of Example 2.13.*

3 The quadratic dual

3.1 Generators and orthogonal relations

Fix a quadratic presentation $A = T_S(V)/(R)$ as in Definition 2.19.

By anti-distributivity of the S-dual (2.5) we have isomorphisms of bimodules $\theta_r : V^* \otimes_S V^* \xrightarrow{\sim} (V \otimes_S V)^*$ and $\theta_\ell : {}^*V \otimes_S {}^*V \xrightarrow{\sim} {}^*(V \otimes_S V)$. We will identify $V^* \otimes_S V^*$ with $(V \otimes_S V)^*$ via θ_r, and ${}^*V \otimes_S {}^*V$ with ${}^*(V \otimes_S V)$ via θ_ℓ. In particular, we can regard the orthogonal complements of the relation spaces as sub-bimodules

$$R^\perp \subset (V \otimes_S V)^* \cong V^* \otimes_S V^* \quad \text{and} \quad {}^\perp R \subset {}^*(V \otimes_S V) \cong {}^*V \otimes_S {}^*V.$$

Definition 3.1. *The right quadratic dual and left quadratic dual of A are*

$$\begin{aligned} A^! &:= T_S(V^*)/(R^\perp), \quad R^\perp = \{F \in V^* \otimes_S V^* \mid \theta_r(F)(R) = 0\}, \\ {}^!A &:= T_S({}^*V)/({}^\perp R), \quad {}^\perp R = \{F \in {}^*V \otimes_S {}^*V \mid \theta_\ell(F)(R) = 0\}. \end{aligned} \tag{3.1}$$

Note that we only take the ordinary S-duals V^* and ${}^*V$, and they are placed in degree 1 in (3.1) – in contrast to the graded S-dual $V^{(*)}$ which is placed in degree -1.

> Our notations are different from [BGS96, §§2.7–2.8]: their V^* uses left S-linear maps and their ${}^*V$ uses right S-linear maps. Thus our $A^!$ is their ${}^!A$, and our ${}^!A$ is their $A^!$.

By Proposition 2.20, a quadratic presentation identifies S with A_0, V with A_1, and R with $\operatorname{Ker}\mu_2$, where $\mu_2 : A_1 \otimes_{A_0} A_1 \to A_2$ is (2.8). Under $S \cong A_0$ and $V \cong A_1$, anti-distributivity (2.5) gives

$$\begin{aligned} V^* &\cong A_1^*, & R^\perp &\cong (\operatorname{Ker}\mu_2)^\perp \subset A_1^* \otimes_{A_0} A_1^*, \\ {}^*V &\cong {}^*(A_1), & {}^\perp R &\cong {}^\perp(\operatorname{Ker}\mu_2) \subset {}^*(A_1) \otimes_{A_0} {}^*(A_1). \end{aligned}$$

By the universal property of tensor algebra (Proposition 2.16), we can extend $V^* \cong A_1^*$ and ${}^*V \cong {}^*(A_1)$ to tensor algebra isomorphisms, which descend to the quotients by $(R^\perp)$ and $({}^\perp R)$, respectively. Thus both quadratic duals are independent of the chosen presentation up to A_0-preserving isomorphisms of graded algebras.

Proposition 3.2. *The algebra $A^!$ is quadratic and*

$$A^!_0 = S, \qquad A^!_1 = V^*, \qquad A^!_2 \cong R^*.$$

In particular, $\dim_{\Bbbk} A^!_2 = \dim_{\Bbbk} R$.

Proof Restriction to R gives an exact sequence of S-bimodules

$$0 \longrightarrow R^\perp \longrightarrow V^* \otimes_S V^* \xrightarrow{F \mapsto \theta_r(F)|_R} R^* \longrightarrow 0. \tag{3.2}$$

Surjectivity follows from Theorem 1.12: R has a complement as a right S-module, so a right S-linear map $R \to S$ extends by zero on that complement. Taking the quotient in (3.2) gives $A^!_2 \cong R^*$; Proposition 2.20 gives the remaining assertions about $A^!$.

For the dimension assertion, write

$$S \cong \prod_{j=1}^{n} \operatorname{Mat}_{d_j}(\Delta_j), \qquad R_S \cong \bigoplus_{j=1}^{n} (\Delta_j^{1\times d_j})^{\oplus m_j},$$

where each Δ_j is a finite-dimensional division $\Bbbk$-algebra and m_j is the multiplicity of the corresponding simple right module $\Delta_j^{1\times d_j}$) (the row space). For each j, there is a left $\mathrm{Mat}_{d_j}(\Delta_j)$-module isomorphism

$$\Delta_j^{d_j\times 1} \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Mat}_{d_j}(\Delta_j)}(\Delta_j^{1\times d_j}, \mathrm{Mat}_{d_j}(\Delta_j)), \qquad c \longmapsto (x \mapsto cx),$$

where $\Delta_j^{d_j\times 1}$ is the left simple module given by the column space. Thus, as left S-modules,

$${}_SR^* \cong \bigoplus_{j=1}^{n} (\Delta_j^{d_j\times 1})^{\oplus m_j}, \qquad \dim_{\Bbbk} R^* = \sum_{j=1}^{n} m_j d_j \dim_{\Bbbk} \Delta_j = \dim_{\Bbbk} R.$$

□

3.2 Recovering the original algebra

Lemma 3.3. *Let $U \subset M$ be an S-sub-bimodule of a finite-dimensional bimodule ${}_SM_S$. Under $M \cong {}^*(M^*)$ and $M \cong ({}^*M)^*$ from the right and left S-dualities in Lemma 1.52,*

$${}^{\perp}(U^{\perp}) = U, \qquad ({}^{\perp}U)^{\perp} = U.$$

Proof For $u \in U$ and $f \in U^{\perp}$, evaluation state what map this is, from where to where gives $\mathrm{ev}_M(u)(f) = f(u) = 0$, so $U \subset {}^{\perp}(U^{\perp})$ under Lemma 1.52.

Conversely, suppose that there is some $m \notin U$, semisimplicity of S implies (by Corollary 1.13) that $M/U \hookrightarrow S^{\oplus n}$ as a right S-module for some n. Consider the composition $M \twoheadrightarrow M/U \hookrightarrow S^{\oplus n} \twoheadrightarrow S$ which sends m to a non-zero element in a direct summand of $S^{\oplus n}$. Then this is a right S-linear map in $\mathrm{Hom}_S(M,S) = M^*$ satisfying $f(U) = 0$ and $f(m) \neq 0$. Thus $f \in U^{\perp}$ and $m \notin {}^{\perp}(U^{\perp})$. The proof of $({}^{\perp}U)^{\perp} = U$ is similar. □

Theorem 3.4. *Left and right quadratic duality are canonically inverse:*

$${}^!(A^!) \cong A \cong ({}^!A)^! \quad \textit{by } S\textit{-preserving isomorphisms of graded algebras.}$$

Proof *Generators.* The right S-duality in Lemma 1.52 gives $\mathrm{ev}_V : V \xrightarrow{\sim} {}^*(V^*)$, $v \mapsto (f \mapsto f(v))$, identifying the generators of A and ${}^!(A^!)$. It preserves both S-actions by (**??**).

Relations. The right S-action on V^* is $(gs)(w) = g(sw)$ for all $g \in V^*$, $s \in S$, and $w \in V$. Applying (2.6) first to $V^* \otimes_S V^*$ and then to $V \otimes_S V$ gives, for all $v, w \in V$ and $f, g \in V^*$,

$$\begin{aligned} \theta_\ell(\mathrm{ev}_V(v) \otimes \mathrm{ev}_V(w))(g \otimes f) &= \mathrm{ev}_V(w)(gf(v)) = g(f(v)w) \\ &= \mathrm{ev}_{V\otimes_S V}(v \otimes w)(\theta_r(g \otimes f)). \end{aligned} \tag{3.3}$$

Thus $\mathrm{ev}_V \otimes \mathrm{ev}_V$ is the bidual identification for $V \otimes_S V$ under anti-distributivity of the S-dual (2.5). Lemma 3.3, applied to $R \subset V \otimes_S V$, identifies the relations ${}^{\perp}(R^{\perp})$ of ${}^!(A^!)$ with R. The presentations (3.1) therefore give an S-preserving isomorphism of graded algebras ${}^!(A^!) \cong A$. Interchanging left and right gives $({}^!A)^! \cong A$. □

Proposition 3.5. *A choice of symmetrising form $\tau : S \to \Bbbk$ gives S-preserving isomorphisms of graded algebras $A^! \cong {}^!A$ and $(A^!)^! \cong A$.*

Proof The maps $\alpha_V(f) = \tau \circ f$ and $\beta_V(g) = \tau \circ g$ in (1.18) give a bimodule isomorphism $\beta_V^{-1}\alpha_V : V^* \to {}^*V$. For $f, g \in V^*$, write $f' = \beta_V^{-1}\alpha_V(f)$ and $g' = \beta_V^{-1}\alpha_V(g)$. Then

$$\begin{aligned} \tau\big(f'(vg'(w))\big) = \tau\big(f(v)g'(w)\big) &= \tau\big(g'(f(v)w)\big) \\ = \tau\big(g(f(v)w)\big) \qquad (v, w \in V). \end{aligned}$$

By (2.20), $(\beta_V^{-1}\alpha_V)^{\otimes_S 2}$ therefore identifies $R^\perp$ with ${}^\perp R$. The bimodule map $\beta_V^{-1}\circ\alpha_V$ extends by Proposition 2.16 to an isomorphism $T_S(V^*) \to T_S({}^*V)$ and hence descends to $A^! \cong {}^!A$. Apply the same comparison to $A^!$ and then Theorem 3.4 to obtain $(A^!)^! \cong {}^!(A^!) \cong A$. □

Theorem 3.4 needs no choice of form. Proposition 3.5 permits the usual statement that quadratic duality is an involution, but the identification of left and right duals uses τ.

3.3 Tensor algebras and monomial relations

Example 3.6. *Quadratic duality exchanges having no relations with setting every product of two generators equal to zero. That is, taking $R = 0$ or $R = V \otimes_S V$ in* (3.1) *gives*

$$\big(T_S(V)\big)^! = T_S(V^*)/(V^* \otimes_S V^*) \cong S \oplus V^*,$$
$$\big(T_S(V)/(V \otimes_S V)\big)^! = T_S(V^*).$$

In $S \oplus V^*$, $(V^*)^2 = 0$ and $(s, f)(t, g) = (st, sg + ft)$; this construction is also known as trivial extension of S by V^*.

For the rest of this subsection, take semisimple $S = \bigoplus_{i=1}^n \Bbbk e_i$ with coordinate idempotents e_i. Choose a basis $\mathcal{B}_{ij}$ of each $e_i V e_j$ and let $\mathcal{B} = \bigcup_{i,j} \mathcal{B}_{ij}$. A non-empty word $b_1 \cdots b_d$ is *composable* if $b_t \in \mathcal{B}_{i_{t-1},i_t}$ for $1 \le t \le d$ and some $i_0, \ldots, i_d \in \{1, \ldots, n\}$. (Thus adjacent factors have matching local units.) By Lemma 2.14, the composable words form a basis of the positive degree part of $T_S(V)$.

Definition 3.7. *The presentation $A = T_S(V)/(R)$ is monomial with respect to $\mathcal{B}$ if R is spanned by a subset $\mathcal{R}$ of the composable words in $\mathcal{B}$.*

For $b \in \mathcal{B}_{ij}$, define the right S-linear map $b^* : V \to S$ by its values on $\mathcal{B}$:

$$b^*(c) = \delta_{bc} e_j \quad (c \in \mathcal{B}), \qquad e_j b^* e_i = b^*. \tag{3.4}$$

The elements b^* form a $\Bbbk$-basis of V^*; the left and right local units of each basis element are reversed.

Proposition 3.8. *For a monomial quadratic presentation:*

(1) *A basis of A consists of the e_i and the non-empty composable words containing no adjacent pair from $\mathcal{R}$.*

(2) *The quadratic dual is monomial, with*

$$R^\perp = \operatorname{span}_\Bbbk\{c^* \otimes_S b^* \mid bc \text{ is composable and } bc \notin \mathcal{R}\}. \tag{3.5}$$

Proof The ideal generated by $\mathcal{R}$ is spanned by the composable words containing an element of $\mathcal{R}$ as an adjacent pair. Removing those basis words proves (1). For $b \in \mathcal{B}_{ij}$ and $c \in \mathcal{B}_{j\ell}$, (2.6) and (3.4) give

$$\theta_r(c^* \otimes b^*)(b \otimes c) = c^*(e_j c) = e_\ell.$$

The functional vanishes on every other basis tensor. By Lemma 2.14, the composable $c^* \otimes b^*$ form a basis of $V^* \otimes_S V^*$. Evaluating a linear combination on each $b \otimes c \in \mathcal{R}$ forces precisely the coefficient of $c^* \otimes b^*$ to be zero, since $e_\ell \neq 0$. Thus the annihilator is the span of the complementary basis tensors, proving (2). □

Example 3.9. *We apply Example 3.6 to the tensor algebra description of $\Bbbk A_n$ in Example 2.21. Take $A = \Bbbk A_n$, $S = \bigoplus_{i=1}^n \Bbbk e_{ii}$, and $V = \bigoplus_{i=1}^{n-1} \Bbbk e_{i,i+1}$. There are no quadratic relations, so $A^!$ has the*

reversed generators $e^*_{i,i+1} \in e_{i+1,i+1}V^*e_{ii}$ and every composable product of two generators is zero. With C as in Example 2.33, this gives

$$h_{A^!}(t) = I_n + tC^{\mathsf{T}}.$$

Here ^{T} denotes transpose: $(C^{\mathsf{T}})_{ij} = C_{ji}$. For $n = 3$,

$$(\Bbbk A_3)^! \cong \big(\Bbbk A_3/(e_{13})\big)^{\mathrm{op}}, \qquad \big(\Bbbk A_3/(e_{13})\big)^! \cong (\Bbbk A_3)^{\mathrm{op}}.$$

These are graded isomorphisms fixing the diagonal idempotents.

Example 3.10. *For $A = \Bbbk A_2$, Examples 2.33 and 3.9 give*

$$h_A(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad h_{A^!}(-t)^{\mathsf{T}} = \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix}, \qquad h_A(t)h_{A^!}(-t)^{\mathsf{T}} = I_2.$$

The transpose aligns the indices because an S-dual reverses the two idempotents in (3.4).

Example 3.11. *Proposition 3.8 exchanges the forbidden and surviving composable pairs. Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ with coordinate idempotents, and let the bimodule $V = \Bbbk a \oplus \Bbbk b$ satisfy $e_1ae_2 = a$, $e_2be_1 = b$, with all other idempotent actions zero. Take $A = T_S(V)/(ab)$. The only composable words of length 2 are ab and ba. Thus*

$$A^! = T_S(V^*)/(a^*b^*), \qquad A^! = S \oplus \Bbbk a^* \oplus \Bbbk b^* \oplus \Bbbk b^*a^*.$$

The original relation $ab = 0$ corresponds to the surviving word b^*a^*; the surviving word ba corresponds to the dual relation $a^*b^* = 0$. In the order e_1, e_2, their Hilbert series are

$$h_A(t) = \begin{pmatrix} 1 & t \\ t & 1+t^2 \end{pmatrix}, \qquad h_{A^!}(t) = \begin{pmatrix} 1+t^2 & t \\ t & 1 \end{pmatrix}.$$

The degree 2 terms occur in different entries: $ba \in e_2Ae_2$, whereas $b^*a^* \in e_1A^!e_1$.

3.4 Symmetric and exterior algebras

Proposition 3.12. *For finite-dimensional $\Bbbk$-vector spaces V, W,*

$$\mathsf{S}(V)^! \cong \bigwedge D(V), \qquad \big(\bigwedge W\big)^! \cong \mathsf{S}(D(W)).$$

Proof For $\mathsf{S}(V)$, choose a basis $x_1, \ldots, x_n$ of V, with dual basis $x_1^*, \ldots, x_n^*$ of $D(V)$. Here $S = \Bbbk$, and the evaluation (2.6) reads

$$\theta_r(x_j^* \otimes x_i^*)(x_p \otimes x_q) = \delta_{ip}\delta_{jq}.$$

A tensor $\sum_{i,j=1}^n c_{ij}x_i^* \otimes x_j^*$ vanishes on each $x_p \otimes x_q - x_q \otimes x_p$ precisely when $c_{pq} = c_{qp}$ for $1 \le p < q \le n$. Thus the orthogonal complement of the symmetric relations is spanned by $(x_i^*)^2$ for $1 \le i \le n$ and $x_i^*x_j^* + x_j^*x_i^*$ for $1 \le i < j \le n$, which are the exterior relations on $D(V)$.

For $\bigwedge W$, choose a basis $y_1, \ldots, y_m$ of W, with dual basis $y_1^*, \ldots, y_m^*$ of $D(W)$. A tensor $\sum_{i,j=1}^m c_{ij}y_i^* \otimes y_j^*$ annihilates the exterior relations precisely when $c_{ii} = 0$ for $1 \le i \le m$ and $c_{pq} + c_{qp} = 0$ for $1 \le p < q \le m$. The orthogonal complement is therefore spanned by $y_i^*y_j^* - y_j^*y_i^*$ for $1 \le i < j \le m$, which are the symmetric relations on $D(W)$. The presentations in (2.9) give both isomorphisms. □

Consequently, to pair $\mathsf{S}(V)$ with $\bigwedge W$ by quadratic duality, take $W = D(V)$ and $y_i = x_i^*$, so $m = n$. The $\Bbbk$-linear duality $D(D(V)) \cong V$ from Example 1.10 then identifies $(\bigwedge W)^!$ with $\mathsf{S}(V)$.

The proof works in characteristic 2: the relations $(x_i^*)^2 = 0$ remain part of the exterior presentation even when the signs $+$ and $-$ coincide. Example 2.39 is the calculation for two generators. Quadratic duality need not preserve Hilbert series: if $\dim_{\Bbbk} V > 0$, Example 2.34 gives different series for $\mathsf{S}(V)$ and $\mathsf{S}(V)^! \cong \bigwedge D(V)$.

The *Koszul condition*, to be defined in Lecture 5, implies the identity

$$h_A(t)h_{A^!}(-t)^{\mathsf{T}} = I_n \qquad \text{when } A_0 = \bigoplus_{i=1}^{n} \Bbbk e_i. \tag{3.6}$$

This is [BGS96, Lemma 2.11.1]; the proof will be given in Lecture 8. Equation (3.6) is not asserted for arbitrary quadratic algebras. For $\mathsf{S}(V)$ and its quadratic dual $\bigwedge D(V)$, with $n = \dim_{\Bbbk} V$, the case $S = \Bbbk$ gives $(1-t)^{-n}(1-t)^n = 1$.

3.5 Tensor products and signs

Definition 3.13. *For graded $\Bbbk$-algebras A, B, the ordinary tensor product $A \otimes_{\Bbbk} B$ and the graded tensor product $A \otimes_{\Bbbk}^{\mathrm{gr}} B$ have underlying vector space $A \otimes_{\Bbbk} B$ and the total grading of Definition 2.28, with products*

$$\begin{aligned} (a \otimes b)(a' \otimes b') &= aa' \otimes bb' && \text{in } A \otimes_{\Bbbk} B, \\ (a \otimes b)(a' \otimes b') &= (-1)^{\deg b \deg a'} aa' \otimes bb' && \text{in } A \otimes_{\Bbbk}^{\mathrm{gr}} B \end{aligned} \tag{3.7}$$

for homogeneous $a, a' \in A$ and $b, b' \in B$, extended bilinearly.

Both products have identity $1_A \otimes 1_B$. For the graded product, associativity follows from $\deg b \deg a' + (\deg b + \deg b') \deg a'' = \deg b' \deg a'' + \deg b(\deg a' + \deg a'')$ for homogeneous $a', a'' \in A$ and $b, b' \in B$. For homogeneous $a \in A$ and $b \in B$, the elements $a \otimes 1$ and $1 \otimes b$ commute in the ordinary tensor product; in the graded tensor product they satisfy

$$(1 \otimes b)(a \otimes 1) = (-1)^{\deg a \deg b}(a \otimes 1)(1 \otimes b).$$

Suppose $A_0 = S$ and $B_0 = T$ are finite-dimensional semisimple and $S \otimes_{\Bbbk} T$ is semisimple. The sign does not change the underlying graded $(S \otimes_{\Bbbk} T)$-bimodule, so the two tensor products have the same Hilbert series whenever A, B are locally finite-dimensional. For $S = \Bbbk^{\oplus n}$ and $T = \Bbbk^{\oplus m}$, Lemma 2.35 gives its entries:

$$h_{A \otimes_{\Bbbk} B}(t)_{(i,p),(j,q)} = h_A(t)_{ij} h_B(t)_{pq}.$$

In particular, when $S = T = \Bbbk$, this is $h_A(t)h_B(t)$.

Proposition 3.14. *Let $A = T_S(V)/(R)$ and $B = T_T(W)/(Q)$ be quadratic algebras, and assume that $S \otimes_{\Bbbk} T$ is semisimple. Then both tensor products are quadratic over $S \otimes_{\Bbbk} T$, and*

$$(A \otimes_{\Bbbk} B)^! \cong A^! \otimes_{\Bbbk}^{\mathrm{gr}} B^!, \qquad (A \otimes_{\Bbbk}^{\mathrm{gr}} B)^! \cong A^! \otimes_{\Bbbk} B^!. \tag{3.8}$$

The mixed commuting relations become anticommuting relations under the reversed tensor pairing.

The assumption on $S \otimes_{\Bbbk} T$ holds when S, T are products of matrix algebras over $\Bbbk$, since $\mathrm{Mat}_a(\Bbbk) \otimes_{\Bbbk} \mathrm{Mat}_b(\Bbbk) \cong \mathrm{Mat}_{ab}(\Bbbk)$. It does not follow from semisimplicity alone: take $S = T$ to be the field in Example 1.44.

Proof The degree 1 part of $A \otimes_{\Bbbk} B$ is $(V \otimes_{\Bbbk} T) \oplus (S \otimes_{\Bbbk} W)$. By Lemma 2.2, its tensor square over $S \otimes_{\Bbbk} T$ decomposes into the four summands

$$(V \otimes_S V) \otimes_{\Bbbk} T \;\oplus\; \underbrace{V \otimes_{\Bbbk} W}_{\text{order } VW} \;\oplus\; \underbrace{V \otimes_{\Bbbk} W}_{\text{order } WV} \;\oplus\; S \otimes_{\Bbbk} (W \otimes_T W).$$

Its quadratic relations consist of $R \otimes_{\Bbbk} T$, $S \otimes_{\Bbbk} Q$, and the mixed relations

$$(v \otimes 1)(1 \otimes w) - (1 \otimes w)(v \otimes 1) \quad (v \in V,\ w \in W). \tag{3.9}$$

Here the products are taken in the tensor algebra on the degree 1 part, and their span is an $(S \otimes_{\Bbbk} T)$-sub-bimodule. Let C be the quotient of that tensor algebra by the three specified relation spaces. The generator maps induce algebra homomorphisms $A \to C$, $a \mapsto \overline{a}$, and $B \to C$, $b \mapsto \overline{b}$. Their images commute: the degree 0 actions already commute across the two factors, and (3.9) makes the degree 1 generators commute. Consequently $A \otimes_{\Bbbk} B \to C$, $a \otimes b \mapsto \overline{a}\,\overline{b}$, is an algebra homomorphism. It is inverse to the canonical homomorphism $C \to A \otimes_{\Bbbk} B$, since both composites fix the generating copies of S, T, V, W. Thus the stated relations give exactly $A \otimes_{\Bbbk} B$. For the graded tensor product, the same construction uses $\overline{b}\,\overline{a} = (-1)^{\deg a \deg b} \overline{a}\,\overline{b}$ for homogeneous a, b.

The right $(S \otimes_{\Bbbk} T)$-duals of the two generator summands are $V^* \otimes_{\Bbbk} T$ and $S \otimes_{\Bbbk} W^*$, with evaluations

$$(f \otimes t)(v \otimes t') = f(v) \otimes tt', \qquad (s \otimes g)(s' \otimes w) = ss' \otimes g(w).$$

These identifications hold for finite free right modules and hence for their direct summands by Corollary 1.13. The reversed tensor evaluation pairs each pure summand with its corresponding dual summand and exchanges the two mixed summands; all other pairings are zero. The pure relation spaces have orthogonal complements $R^{\perp} \otimes_{\Bbbk} T$ and $S \otimes_{\Bbbk} Q^{\perp}$ in the corresponding pure summands. The mixed relation space is the set of pairs $(z, -z)$ in the two copies of $V \otimes_{\Bbbk} W$. For mixed tensors, the two reversed evaluations of $f \in V^*$ and $g \in W^*$ both give $f(v) \otimes g(w)$ on the corresponding copy. Thus an annihilating pair of functionals must have equal components. As these simple tensors span the full dual, the mixed annihilator is spanned by

$$(f \otimes 1)(1 \otimes g) + (1 \otimes g)(f \otimes 1) \quad (f \in V^*,\ g \in W^*).$$

These are the mixed relations of $A^! \otimes^{\mathrm{gr}}_{\Bbbk} B^!$, proving the first isomorphism in (3.8). For $A \otimes^{\mathrm{gr}}_{\Bbbk} B$, the minus sign in (3.9) is a plus sign. The mixed annihilator then has a minus sign, giving the ordinary tensor product on the dual side. □

Example 3.15. *We compute how Proposition 3.14 changes a commutation relation in a matrix example. Take $A = \Bbbk A_2 \otimes_{\Bbbk} \Bbbk A_2$. Its degree 0 algebra has basis $e_{ii} \otimes e_{jj}$ for $i, j \in \{1, 2\}$, and its degree 1 part has basis*

$$a = e_{12} \otimes e_{11}, \quad b = e_{22} \otimes e_{12}, \quad c = e_{11} \otimes e_{12}, \quad d = e_{12} \otimes e_{22}.$$

By Lemma 2.14, the tensor square of the degree 1 part has basis ab, cd, and both map to $e_{12} \otimes e_{12}$. Thus the relation space is $\Bbbk(ab - cd)$. Using the reversed generators from (3.4), its orthogonal complement is $\Bbbk(b^*a^* + d^*c^*)$. Hence the dual presentation has an anticommuting relation:

$$ab = cd \text{ in } A, \qquad b^*a^* = -d^*c^* \text{ in } A^!.$$

Order the degree 0 idempotents as $e_{11}\otimes e_{11}, e_{22}\otimes e_{11}, e_{11}\otimes e_{22}, e_{22}\otimes e_{22}$. The homogeneous bases give

$$h_A(t) = \begin{pmatrix} 1 & t & t & t^2 \\ 0 & 1 & 0 & t \\ 0 & 0 & 1 & t \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad h_{A^!}(t) = h_A(t)^{\mathsf{T}}.$$

The signs agree with Proposition 3.14; in characteristic 2, the same presentations remain valid with $-1 = 1$. Here replacing d^* by $-d^*$ removes the minus sign. In this example the sign distinguishes the specified presentations, not the isomorphism classes of algebras.

Example 3.16. *The symmetric/exterior pair shows that the sign in a graded tensor product cannot always be omitted. Let $V = \Bbbk x_1 \oplus \Bbbk x_2$ and let y_1, y_2 be the dual basis of $W = D(V)$. Proposition 3.14 gives*

$$\mathsf{S}(V)^! \cong \big(\Bbbk[x_1] \otimes_{\Bbbk} \Bbbk[x_2]\big)^! \cong \frac{\Bbbk[y_1]}{(y_1^2)} \otimes_{\Bbbk}^{\mathrm{gr}} \frac{\Bbbk[y_2]}{(y_2^2)} \cong \bigwedge W.$$

If $\operatorname{char}\Bbbk \neq 2$, the ordinary tensor product of the two square-zero algebras is commutative, whereas the graded tensor product is not: $y_1y_2 = -y_2y_1 \neq 0$. They are therefore not isomorphic. This sign cannot be removed by a change of generators.

3.6 Exercises

Exercise 3.17. *For $A = \Bbbk A_4/(e_{13})$, write a presentation of $A^!$ using reversed generators and find homogeneous bases of A and $A^!$. Check the double dual using Proposition 3.8.* Hint. *Among $e_{12}e_{23}$ and $e_{23}e_{34}$, only the first is a relation. Complement the forbidden pairs and reverse their order.*

Exercise 3.18. *Let $S = \bigoplus_{i=1}^n \Bbbk e_i$ with coordinate idempotents, and let ${}_SV_S$ be a finite-dimensional bimodule in degree 1. Set $C_{ij} = [e_iV : e_jS] = \dim_{\Bbbk}(e_iVe_j)$ for $1 \le i, j \le n$, and put*

$$A = T_S(V)/(V \otimes_S V) = S \oplus V.$$

(1) *Prove that*

$$A^! = T_S(V^*), \qquad h_A(t) = I_n + tC, \qquad h_{A^!}(t) = (I_n - tC^{\mathsf{T}})^{-1}.$$

(2) *Deduce that $h_A(t)h_{A^!}(-t)^{\mathsf{T}} = I_n$.*

Hint for (1). *Find $(V \otimes_S V)^{\perp}$ and compute the multiplicities $[e_iV^* : e_jS]$.*

Exercise 3.19. *Let $S = \bigoplus_{i=1}^4 \Bbbk e_i$ with coordinate idempotents, and let ${}_SV_S$ have basis $\{a, b, c, d\}$ with*

$$a \in e_1Ve_2, \qquad b \in e_2Ve_4, \qquad c \in e_1Ve_3, \qquad d \in e_3Ve_4,$$

and $e_iVe_j = 0$ for all other pairs. For $\lambda \in \Bbbk$, put $R_\lambda = \Bbbk(ab - \lambda cd) \subset V \otimes_S V$ and $A_\lambda = T_S(V)/(R_\lambda)$.

(1) *Using θ_r, compute $R_\lambda^{\perp}$ and a presentation of $A_\lambda^!$ for every λ, including $\lambda = 0$.*

(2) *Compute ${}^!(A_\lambda^!)$ and recover the relation $ab - \lambda cd$ under the canonical identification ${}^*(V^*) \cong V$.*

(3) *Prove that $A_\lambda \cong A_\mu$ as graded algebras if and only if λ, μ are both zero or both non-zero.*

Hint for (1). *Evaluate $\alpha b^*a^* + \beta d^*c^*$ on $ab - \lambda cd$ for $\alpha, \beta \in \Bbbk$.*

Hint for (3). *A graded isomorphism permutes the coordinate idempotents of S. The degree 1 components force it to fix e_1, e_4 and either fix or interchange e_2, e_3. Compare which of the multiplication maps*

$$e_1(A_\lambda)_1 e_i \otimes_{\Bbbk} e_i(A_\lambda)_1 e_4 \longrightarrow e_1(A_\lambda)_2 e_4 \quad \text{for } i \in \{2,3\}$$

are zero. For non-zero λ, μ, rescale d.

4 Homological algebra

4.1 Free and projective modules

Until Subsection 4.5, A is an ordinary $\Bbbk$-algebra and all modules are right A-modules.

Definition 4.1. *A module F is free with basis $\{u_i \mid i \in I\}$ if every $v \in F$ has a unique expression $v = \sum_{i\in I} u_i a_i$, where $a_i \in A$ and only finitely many a_i are non-zero. Equivalently, $F \cong \bigoplus_{i\in I} A$, with u_i corresponding to 1 in the ith summand.*

A map from a free module is determined by arbitrary images of its basis elements. In particular, a generating family $\{m_i \mid i \in I\}$ of M gives a surjection

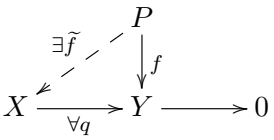
$$\bigoplus_{i\in I} A \longrightarrow M, \qquad (a_i)_{i\in I} \longmapsto \sum_{i\in I} m_i a_i.$$

Every module is therefore a quotient of a free module. If M is finitely generated, the free module can be chosen to be $A^{\oplus m}$ for some integer $m \geq 0$.

Definition 4.2. *A short exact sequence $0 \to L \xrightarrow{i} X \xrightarrow{q} Y \to 0$ splits if there is an A-linear map $s : Y \to X$ with $qs = \mathrm{id}_Y$. The map s is called a section of q.*

Such a section gives $X = i(L) \oplus s(Y)$: write $x = (x - sq(x)) + sq(x)$, and use $qs = \mathrm{id}_Y$ to check that $i(L) \cap s(Y) = 0$. Thus $(l, y) \mapsto i(l) + s(y)$ is an isomorphism $L \oplus Y \to X$ compatible with the inclusion and quotient maps.

Definition 4.3. *A module P is projective if, for every surjective A-linear map $q : X \to Y$ and every A-linear map $f : P \to Y$, there is an A-linear map $\widetilde{f} : P \to X$ with $q\widetilde{f} = f$ (diagram commutes):*



exact.

Free modules have this lifting property: for each basis element u_i, choose $x_i \in X$ with $q(x_i) = f(u_i)$, and extend $u_i \mapsto x_i$ A-linearly. Projective modules retain this property without requiring a basis.

For fixed P, postcomposition defines the functor $\mathrm{Hom}_A(P, -) : \mathrm{Mod}\, A \to \mathrm{Mod}\, \Bbbk$, acting on $u : X \to Y$ by

$$\mathrm{Hom}_A(P, u) : \mathrm{Hom}_A(P, X) \longrightarrow \mathrm{Hom}_A(P, Y), \qquad f \longmapsto u \circ f.$$

Postcomposition preserves identities and composition.

Lemma 4.4. *The functor $\mathrm{Hom}_A(P, -)$ is left exact, that is, every short exact sequence $0 \to L \xrightarrow{i} X \xrightarrow{q} Y \to 0$ induces an exact sequence*

$$0 \longrightarrow \mathrm{Hom}_A(P, L) \xrightarrow{i\circ -} \mathrm{Hom}_A(P, X) \xrightarrow{q\circ -} \mathrm{Hom}_A(P, Y).$$

Proof Since i is injective, $i \circ f = 0$ implies $f = 0$. For $g : P \to X$, the equality $qg = 0$ is equivalent to $g(P) \subseteq \mathrm{Ker}\, q = i(L)$. It is therefore equivalent to a unique factorisation $g = i\widetilde{g}$ with $\widetilde{g} : P \to L$ A-linear. □

Surjectivity of q does not in general imply surjectivity of $q \circ -$. Having a projective module is precisely the additional condition needed.

Proposition 4.5. *For a right A-module P, the following are equivalent.*

(1) *P is projective.*

(2) *The functor $\mathrm{Hom}_A(P,-)$ is exact.*

(3) *Every surjective A-linear map $q : X \to P$ has an A-linear section.*

(4) *P is isomorphic to a direct summand of a free module.*

If P is finitely generated, condition (4) can be replaced by $P \in \operatorname{add} A$.

Proof We prove (1) $\Leftrightarrow$ (2) and (1) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (1).

(1) $\Leftrightarrow$ (2). By Lemma 4.4, exactness is equivalent to surjectivity of $q\circ - : \mathrm{Hom}_A(P,X) \to \mathrm{Hom}_A(P,Y)$ for every surjection $q : X \to Y$. This says exactly that every $f : P \to Y$ lifts to X.

(1) $\Rightarrow$ (3). Apply the lifting property to $f = \mathrm{id}_P$.

(3) $\Rightarrow$ (4). Choose a free module F and a surjection $q : F \to P$. A section $s : P \to F$ gives $F = \operatorname{Ker} q \oplus s(P)$. If P is finitely generated, choose $F = A^{\oplus m}$.

(4) $\Rightarrow$ (1). Identify P with a summand of a free module F, with inclusion $j : P \to F$ and projection $r : F \to P$ satisfying $rj = \mathrm{id}_P$. For $q : X \to Y$ surjective and $f : P \to Y$, freeness gives $h : F \to X$ with $qh = fr$. Then $\widetilde{f} = hj$ satisfies $q\widetilde{f} = frj = f$. □

Corollary 4.6. *Direct summands of projective modules are projective. For an arbitrary family $(P_i)_{i\in I}$ of projective modules indexed by a set I, the module $\bigoplus_{i\in I} P_i$ is projective.*

Proof For a direct summand, use the inclusion and projection as in (4) $\Rightarrow$ (1) above. A map from $\bigoplus_{i\in I} P_i$ is determined by its restrictions to the P_i; lift each restriction and combine the lifts. □

Example 4.7. *For an idempotent $e \in A$, the decomposition $A = eA \oplus (1-e)A$ makes eA projective. Such modules need not be free: for $A = \Bbbk A_2$, the module $e_{22}A$ has dimension 1, whereas every non-zero free module has dimension at least $\dim_\Bbbk A = 3$.*

Example 4.8. *A simple module need not be projective, even over $\Bbbk A_2$. Set $L_1 = e_{11}A/\Bbbk e_{12}$ and let $q : e_{11}A \to L_1$ be the quotient map. If q had an A-linear section s, then $s(e_{11} + \Bbbk e_{12}) = e_{11} + ce_{12}$ for some $c \in \Bbbk$. But*

$$s((e_{11} + \Bbbk e_{12})e_{12}) = s(0) = 0, \qquad (e_{11} + ce_{12})e_{12} = e_{12} \neq 0,$$

contradicting A-linearity. Thus L_1 is not projective; equivalently, id_{L_1} is not in the image of $q \circ - : \mathrm{Hom}_A(L_1, e_{11}A) \to \mathrm{End}_A(L_1)$.

If A is finite-dimensional semisimple, every surjection splits by Theorem 1.12; hence every A-module is projective.

4.2 Tops and Nakayama's lemma

The ideal $\operatorname{rad} A$ is two-sided: it consists precisely of the elements that act by zero on every simple right A-module.

> For a simple module L and $0 \neq l \in L$, the map $A \to L$, $a \mapsto la$, is surjective with maximal right ideal as kernel. Thus $l \operatorname{rad} A = 0$. Conversely, if $a \in A$ acts by zero on every simple module, apply this to A/U for each maximal right ideal U to obtain $a \in U$, hence $a \in \operatorname{rad} A$. The

identities $l(ba) = (lb)a = 0$ and $l(ab) = (la)b = 0$ for $a \in \operatorname{rad} A$, $b \in A$, and $l \in L$ show that $ba, ab \in \operatorname{rad} A$.

For a finitely generated right A-module M, a set of elements generates M if its images generate $M/M \operatorname{rad} A$. This is the content of Nakayama's lemma.

Lemma 4.9 (Nakayama's lemma). *Let M be a finitely generated right A-module. If $M = M \operatorname{rad} A$, then $M = 0$. More generally, if $N \subseteq M$ is a submodule and $N + M \operatorname{rad} A = M$, then $N = M$.*

Proof For $a \in \operatorname{rad} A$, the right ideal $(1-a)A$ equals A. Otherwise it is contained in a maximal right ideal U; then $a, 1-a \in U$ gives $1 \in U$, a contradiction. Thus $(1-a)b = 1$ for some $b \in A$.

Suppose $M = M \operatorname{rad} A \neq 0$ and choose generators $m_1, \dots, m_r$ with r minimal. Since $\operatorname{rad} A$ is a two-sided ideal, $M \operatorname{rad} A = \sum_{j=1}^{r} m_j \operatorname{rad} A$. Write $m_r = \sum_{j=1}^{r} m_j a_j$ with $a_j \in \operatorname{rad} A$ and choose $b \in A$ with $(1 - a_r)b = 1$. Then

$$m_r(1 - a_r) = \sum_{j=1}^{r-1} m_j a_j, \qquad m_r = \sum_{j=1}^{r-1} m_j a_j b,$$

contradicting the minimality of r. For the last assertion, apply the first to the finitely generated quotient M/N, which satisfies $(M/N) \operatorname{rad} A = M/N$. □

Remark 4.10. If A is finite-dimensional, then $\operatorname{rad} A$ is *nilpotent*: $(\operatorname{rad} A)^r = 0$ for some integer $r \geq 1$. Indeed, the powers stabilise at $(\operatorname{rad} A)^r = (\operatorname{rad} A)^{r+1}$, and Lemma 4.9 applied to the finite-dimensional module $(\operatorname{rad} A)^r$ gives $(\operatorname{rad} A)^r = 0$. Thus finite generation is unnecessary in Nakayama's lemma: $M = \operatorname{rad} M$ implies $M = M(\operatorname{rad} A)^r = 0$. Applying this to M/N also gives $N + \operatorname{rad} M = M \Longrightarrow N = M$ for every submodule $N \subseteq M$.

For the rest of this subsection, assume that A is finite-dimensional. Every maximal right ideal of A contains $\operatorname{rad} A$, so $\operatorname{rad}(A/\operatorname{rad} A) = 0$; hence $A/\operatorname{rad} A$ is semisimple by Theorem 1.12.

Definition 4.11. *Let A be a finite-dimensional algebra. The top of a right A-module M is*

$$\operatorname{top} M := M/\operatorname{rad} M.$$

The module $\operatorname{top} M$ is semisimple because $\operatorname{rad} A$ acts by zero and $A/\operatorname{rad} A$ is semisimple. Moreover, every map from M to a semisimple module T vanishes on $\operatorname{rad} M$, since $\operatorname{rad} T = 0$, and therefore factors uniquely through $M \to \operatorname{top} M$. Thus $\operatorname{top} M$ is the maximal semisimple quotient of M.

For an A-linear map $f : M \to N$, the inclusion $f(\operatorname{rad} M) \subseteq \operatorname{rad} N$ gives

$$\operatorname{top} f : \operatorname{top} M \longrightarrow \operatorname{top} N, \qquad m + \operatorname{rad} M \longmapsto f(m) + \operatorname{rad} N.$$

If M is finitely generated, Nakayama's lemma says that lifts to M of generators of $\operatorname{top} M$ generate M. In particular, if $f : X \to M$ induces a surjection on tops, then $f(X) + \operatorname{rad} M = M$, so f is surjective.

Example 4.12. *For $A = \Bbbk A_2$, we have $\operatorname{rad} A = \Bbbk e_{12}$ and*

$$\operatorname{top}(e_{11}A) = e_{11}A/\Bbbk e_{12} = L_1, \qquad \operatorname{top}(e_{22}A) = e_{22}A = L_2.$$

4.3 Projective covers

A free surjection onto M may use unnecessary generators. A projective cover is a surjection from a projective module whose domain cannot be replaced by a proper submodule still mapping onto M.

Definition 4.13. *Let M be an A-module. A projective cover is a surjective map $p : P \to M$ with P projective such that*

$$U \subseteq P \text{ a submodule and } p(U) = M \quad \Longrightarrow \quad U = P.$$

Proposition 4.14. *Let A be finite-dimensional, let P be a projective right A-module, and let $p : P \to M$ be surjective. The following are equivalent.*

(1) *p is a projective cover.*

(2) $\operatorname{Ker} p \subseteq \operatorname{rad} P$.

(3) $\operatorname{top} p : \operatorname{top} P \to \operatorname{top} M$ *is an isomorphism.*

Proof We prove (1) $\Rightarrow$ (2) $\Rightarrow$ (1) and (2) $\Leftrightarrow$ (3).

(1) $\Rightarrow$ (2). If $\operatorname{Ker} p \not\subseteq \operatorname{rad} P$, there is a maximal submodule $U \subset P$ not containing $\operatorname{Ker} p$. Then $U + \operatorname{Ker} p = P$, so $p(U) = M$, contradicting (1).

(2) $\Rightarrow$ (1). If $p(U) = M$, then $U + \operatorname{Ker} p = P$. Condition (2) gives $U + \operatorname{rad} P = P$, hence $\operatorname{rad}(P/U) = P/U$. Remark 4.10 gives $P/U = 0$, so $U = P$.

(2) $\Leftrightarrow$ (3). The equality $p(\operatorname{rad} P) = \operatorname{rad} M$ makes $\operatorname{top} p$ surjective, with kernel $(\operatorname{Ker} p + \operatorname{rad} P)/\operatorname{rad} P$. □

Theorem 4.15. *Let A be a finite-dimensional algebra. Every right A-module M has a projective cover; if M is finite-dimensional, then so is its cover. Over any algebra, for any two projective covers $p : P \to M$ and $q : Q \to M$, there is an isomorphism $f : P \to Q$ with $qf = p$.*

Proof Uniqueness: Projectivity of P gives $f : P \to Q$ with $qf = p$. Since $q(f(P)) = M$ and q is a cover, $f(P) = Q$. Projectivity of Q now gives a section $s : Q \to P$ of f. The equality $p(s(Q)) = q(Q) = M$ and the cover property of p give $s(Q) = P$. Thus f and s are inverse isomorphisms.

Existence for finite-dimensional M: Choose a surjection $p : P \to M$ from a finite-dimensional projective module with $\dim_{\Bbbk} P$ as small as possible. Such a surjection exists because M is a quotient of some $A^{\oplus m}$. Suppose $U \subsetneq P$ and $p(U) = M$. Projectivity gives $h : P \to U$ with $(p|_U)h = p$. Composing h with the inclusion $U \hookrightarrow P$ gives $u : P \to P$ with $pu = p$ and $u(P) \subseteq U$.

Choose an integer $r \geq 1$ such that the kernels and images of the powers of u have stabilised. Then

$$P = \operatorname{Ker} u^r \oplus \operatorname{Im} u^r.$$

Indeed, if $u^r(x) \in \operatorname{Ker} u^r$, then $x \in \operatorname{Ker} u^{2r} = \operatorname{Ker} u^r$, so $u^r(x) = 0$; equality follows by the dimension formula. Thus $\operatorname{Im} u^r$ is projective, and $pu^r = p$ shows that $p(\operatorname{Im} u^r) = M$. But $\operatorname{Im} u^r \subseteq U \subsetneq P$ contradicts the minimal choice of $\dim_{\Bbbk} P$. Therefore p is a projective cover.

Existence for arbitrary M: By Theorem 1.12, choose a decomposition $\operatorname{top} M \cong \bigoplus_{\lambda \in I} L_\lambda$ into simple right $A/\operatorname{rad} A$-modules, indexed by a set I. Each L_λ is finite-dimensional, so the preceding argument gives a finite-dimensional projective cover $q_\lambda : Q_\lambda \to L_\lambda$. Projectivity lifts each

composite $Q_\lambda \to L_\lambda \hookrightarrow \operatorname{top} M$ through the quotient $M \to \operatorname{top} M$ to a map $f_\lambda : Q_\lambda \to M$. Set

$$P := \bigoplus_{\lambda \in I} Q_\lambda, \qquad p : P \longrightarrow M, \qquad (x_\lambda)_{\lambda \in I} \longmapsto \sum_{\lambda \in I} f_\lambda(x_\lambda).$$

The sum defining p is finite for each element of P. Corollary 4.6 makes P projective. The equality $\operatorname{rad} P = \bigoplus_{\lambda \in I} \operatorname{rad} Q_\lambda$ gives $\operatorname{top} P \cong \bigoplus_{\lambda \in I} \operatorname{top} Q_\lambda$. Thus the isomorphisms $\operatorname{top} q_\lambda$ identify $\operatorname{top} p$ with $\bigoplus_{\lambda \in I} L_\lambda \xrightarrow{\sim} \operatorname{top} M$. Thus $M = p(P) + \operatorname{rad} M$, and Remark 4.10 applied to $M/p(P)$ shows that p is surjective. Proposition 4.14 now makes p a projective cover.

□

Remark 4.16. For finite-dimensional M, finite generation gives a surjection $A^{\oplus m} \to M$, and finite-dimensionality of A makes its domain finite-dimensional. Finite-dimensionality of P ensures that the kernels and images of powers of u stabilise. For arbitrary M, the argument instead uses semisimplicity of $A/\operatorname{rad} A$ and nilpotence of $\operatorname{rad} A$. Uniqueness of projective covers requires no finiteness assumption.

Example 4.17. *For $A = \Bbbk A_2$, the maps $e_{11}A \to L_1$ and $\mathrm{id} : e_{22}A \to L_2$ are projective covers: their kernels are respectively $\operatorname{rad}(e_{11}A)$ and 0. In contrast, the surjection $A = e_{11}A \oplus e_{22}A \to L_1$ obtained by killing $e_{22}A$ is not a cover, since its restriction to $e_{11}A$ is already surjective.*

4.4 Complexes and resolutions

Starting with a surjection from a projective module onto M, one can repeat the construction for each successive kernel. Composing each surjection with the inclusion of its target kernel gives an exact sequence. We now define such resolutions over an arbitrary $\Bbbk$-algebra A.

Definition 4.18. *A chain complex $C_\bullet$ of right A-modules is a sequence*

$$\cdots \longrightarrow C_{i+1} \xrightarrow{d_{i+1}} C_i \xrightarrow{d_i} C_{i-1} \longrightarrow \cdots, \qquad d_i d_{i+1} = 0 \quad \text{for all } i \in \mathbb{Z}.$$

The maps d_i are A-linear and are called differentials; the index i is the homological degree. Its homology in degree i is

$$H_i(C_\bullet) := \operatorname{Ker} d_i / \operatorname{Im} d_{i+1}.$$

The identity $d_i d_{i+1} = 0$ ensures that $\operatorname{Im} d_{i+1} \subseteq \operatorname{Ker} d_i$, so the quotient is defined. Exactness at C_i means $H_i(C_\bullet) = 0$. A complex concentrated in non-negative homological degrees has the form $\cdots \to C_2 \to C_1 \to C_0 \to 0$; in particular, $H_0(C_\bullet) = C_0/\operatorname{Im} d_1$.

Definition 4.19. *A projective resolution of M is an exact sequence*

$$\cdots \longrightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{p_0} M \longrightarrow 0 \tag{4.1}$$

with every P_i projective. We write $P_\bullet \to M$ for this data; the complex $P_\bullet$ itself ends in $P_0 \to 0$. A resolution is free if every P_i is free.

Thus p_0 is surjective, $\operatorname{Ker} p_0 = \operatorname{Im} d_1$, and $\operatorname{Ker} d_i = \operatorname{Im} d_{i+1}$ for all $i \geq 1$. Equivalently, p_0 induces $H_0(P_\bullet) \cong M$, and $H_i(P_\bullet) = 0$ for all $i \geq 1$. The module M is not an additional term of $P_\bullet$.

Proposition 4.20. *Every right A-module has a free resolution, hence a projective resolution.*

Proof For every right A-module N, the free module $A^{\oplus N} := \bigoplus_{n \in N} A$, indexed by the underlying set of N, surjects onto N by

$$A^{\oplus N} \longrightarrow N, \qquad (a_n)_{n \in N} \longmapsto \sum_{n \in N} n a_n.$$

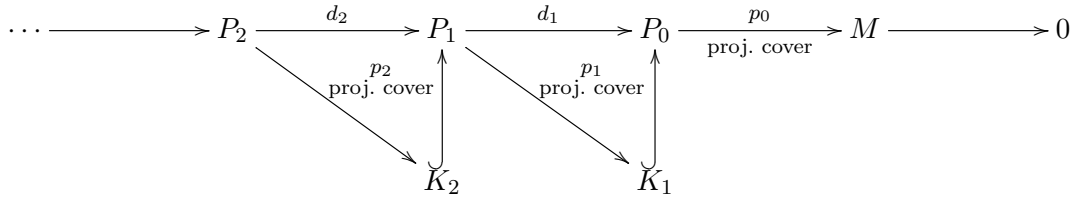
Start with $N = M$ and repeat for each successive kernel. Compose each new surjection with the inclusion of that kernel into the preceding free module; its image is exactly that kernel, so the resulting sequence is exact. □

Definition 4.21. *A projective resolution* (4.1) *is minimal if* $p_0 : P_0 \to M$ *and the surjections*

$$\begin{aligned} p_1 : P_1 &\longrightarrow \operatorname{Ker} p_0, \qquad x \longmapsto d_1(x), \\ p_i : P_i &\longrightarrow \operatorname{Ker} d_{i-1}, \quad x \longmapsto d_i(x) \quad \text{for all } i \geq 2 \end{aligned}$$

are projective covers.

For the diagram, put $K_1 := \operatorname{Ker} p_0$ and $K_i := \operatorname{Ker} d_{i-1}$ for all $i \geq 2$. In a minimal resolution, each differential factors as a projective cover onto its image followed by inclusion:



When A is finite-dimensional, Proposition 4.14 gives the criterion

$$\operatorname{Im} d_i \subseteq \operatorname{rad} P_{i-1} \quad \text{for all } i \geq 1. \tag{4.2}$$

Indeed, exactness gives $\operatorname{Ker} p_0 = \operatorname{Im} d_1$ and $\operatorname{Ker} d_i = \operatorname{Im} d_{i+1}$ for all $i \geq 1$.

Theorem 4.22. *Every right module over a finite-dimensional algebra has a minimal projective resolution. If the module is finite-dimensional, every term of its minimal resolution is finite-dimensional. Over any algebra, any two minimal resolutions* $P_\bullet \to M$ *and* $P'_\bullet \to M$ *are isomorphic: there are isomorphisms* $f_i : P_i \to P'_i$ *satisfying*

$$p'_0 f_0 = p_0, \qquad d'_i f_i = f_{i-1} d_i \quad \text{for all } i \geq 1.$$

More generally, maps f_i satisfying these identities are called a *map of resolutions*; the identities express the commutativity of the diagram:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & P_1 & \xrightarrow{d_1} & P_0 & \xrightarrow{p_0} & M & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_0} & & \downarrow{\scriptstyle \mathrm{id}_M} & & \\ \cdots & \longrightarrow & P'_1 & \xrightarrow{d'_1} & P'_0 & \xrightarrow{p'_0} & M & \longrightarrow & 0. \end{array}$$

Proof Use projective covers in place of free surjections in Proposition 4.20. For finite-dimensional A, Theorem 4.15 applies to every successive kernel. If M is finite-dimensional, choose a finite-dimensional cover at each step; its kernel is again finite-dimensional. For uniqueness, choose an isomorphism $f_0 : P_0 \to P'_0$ between the covers of M. It restricts to an isomorphism $\operatorname{Ker} p_0 \to \operatorname{Ker} p'_0$; uniqueness of covers of these kernels gives $f_1 : P_1 \to P'_1$. Repeating gives f_i with $d'_i f_i = f_{i-1} d_i$ at every step, without finiteness assumptions. □

Remark 4.23. For a minimal projective resolution of M, the kernels $\operatorname{Ker} p_0$ and $\operatorname{Ker} d_i$ for $i \geq 1$ are called the successive *syzygies* of M. Their isomorphism classes are independent of the chosen minimal resolution by Theorem 4.22.

Remark 4.24. Over an arbitrary algebra, Proposition 4.20 still gives a free resolution, but projective covers of the successive kernels need not exist. The obstruction is to existence, not to uniqueness of a minimal resolution.

Example 4.25. *The covers in Example 4.17 give a minimal resolution of L_1:*

$$0 \longrightarrow e_{22}A \xrightarrow{d_1} e_{11}A \xrightarrow{p_0} L_1 \longrightarrow 0, \qquad d_1(e_{22}a) = e_{12}a.$$

Indeed, $\operatorname{Ker} p_0 = \Bbbk e_{12}$*, and* $d_1 : e_{22}A \to \Bbbk e_{12}$ *is an isomorphism. Thus no further projective terms are needed.*

Example 4.26. *A finite-dimensional algebra can have a simple module with an infinite minimal resolution. Let* $E = \Bbbk[\epsilon]/(\epsilon^2)$ *and* $\pi : E \to \Bbbk$*,* $a + b\epsilon \mapsto a$*. Every* $a + b\epsilon$ *with* $a \neq 0$ *has inverse* $a^{-1} - a^{-2}b\epsilon$*, so* $\Bbbk\epsilon$ *is the unique maximal ideal and* $\operatorname{rad} E = \Bbbk\epsilon$*. The sequence*

$$\cdots \xrightarrow{\epsilon\cdot -} E \xrightarrow{\epsilon\cdot -} E \xrightarrow{\epsilon\cdot -} E \xrightarrow{\pi} \Bbbk \longrightarrow 0$$

is a minimal resolution: multiplication by ϵ *has kernel and image both equal to* $\Bbbk\epsilon$*. At every step the kernel is non-zero, so the construction by projective covers never terminates.*

4.5 Graded version

Now let $A = \bigoplus_{d\in\mathbb{Z}} A_d$ be locally finite-dimensional non-negatively graded with $S = A_0$ finite-dimensional semisimple. Recall that $A_+ = \bigoplus_{d>0} A_d$ and $A/A_+ \cong S$ so that we can regard S as a graded right A-module concentrated in degree 0. All maps in this subsection are degree-preserving unless otherwise stated.

Definition 4.27. *A graded free module is a direct sum* $\bigoplus_{\lambda\in I} A(-d_\lambda)$*, where* I *is a set and* $d_\lambda \in \mathbb{Z}$*. A graded module is graded projective if it has the lifting property of Definition 4.3 for degree-preserving maps.*

The generator 1 of $A(-d)$ has degree d. For a graded surjection $q : X \to Y$, an element of Y_d has a preimage in X_d. Thus the basis-lifting proof works for graded free modules. Every graded module is a quotient of a graded free module, by taking homogeneous generators.

Corollary 4.28. *For a graded right A-module P, the following are equivalent.*

(1) *P is graded projective.*

(2) *The functor* $\operatorname{Hom}_A^{\mathbb{Z}}(P, -)$ *is exact on graded modules.*

(3) *Every degree-preserving surjection onto P has a degree-preserving section.*

(4) *P is a graded direct summand of a graded free module.*

Arbitrary direct sums and graded direct summands of graded projective modules are graded projective. In particular, $Q \otimes_S A(-d)$ *is graded projective for every finite-dimensional right S-module Q and every* $d \in \mathbb{Z}$*.*

Proof We follow (1) $\Leftrightarrow$ (2) and (1) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (1) from Proposition 4.5, using degree-preserving maps. (1) $\Leftrightarrow$ (2). The proof of Lemma 4.4 applies to graded Hom; surjectivity on Hom is the graded lifting property. (1) $\Rightarrow$ (3). Lift id_P. (3) $\Rightarrow$ (4). Split a graded free surjection onto P. (4) $\Rightarrow$ (1). Restrict lifts from the graded free module to its summand P. The proof of Corollary 4.6 also applies with degree-preserving maps. For the last assertion, Corollary 1.13 makes Q a direct summand of $S^{\oplus m}$ for some m, so $Q \otimes_S A(-d)$ is a graded direct summand of $A(-d)^{\oplus m}$. □

Definition 4.29. *The graded top of a graded right A-module M is the graded right S-module*

$$\operatorname{top}_{\mathrm{gr}} M := M/MA_+.$$

For a degree-preserving map $f : M \to N$, write $\operatorname{top}_{\mathrm{gr}} f : \operatorname{top}_{\mathrm{gr}} M \to \operatorname{top}_{\mathrm{gr}} N$, $m + MA_+ \mapsto f(m) + NA_+$.

Lemma 4.30. *If A is finite-dimensional, then $A_+ = \operatorname{rad} A$. Consequently, $\operatorname{top}_{\mathrm{gr}} M = \operatorname{top} M$ for every graded right A-module M, with $\operatorname{top} M$ carrying the quotient grading.*

Proof If A is finite-dimensional, then $A_d = 0$ for all sufficiently large d, so $(A_+)^r = 0$ for some integer $r \geq 1$. For any simple right module L, the submodule LA_+ is either 0 or L. The latter gives $L = L(A_+)^r = 0$, a contradiction. Thus A_+ annihilates every simple module and is contained in $\operatorname{rad} A$. Since $A/A_+ \cong S$ is semisimple, every non-zero element of S acts non-trivially on some simple right S-module. Regarding these as simple A-modules shows that $\operatorname{rad} A \subseteq A_+$. □

Example 4.31. *The equality $A_+ = \operatorname{rad} A$ can fail when A is infinite-dimensional. Take $A = \Bbbk[x]$ with $\deg x = 1$. Then $A_+ = (x)$, whereas $\operatorname{rad} A = 0$. Indeed, if $0 \neq f \in \operatorname{rad} A$, then $xf \in \operatorname{rad} A$, and the first paragraph of the proof of Lemma 4.9 gives a right inverse to $1 - xf$. This is impossible because $1 - xf$ has positive degree. In particular, $A/\operatorname{rad} A = A$, whereas $\operatorname{top}_{\mathrm{gr}} A = \Bbbk$.*

Over this algebra, even the ordinary module $\Bbbk = A/(x)$ has no projective cover. Suppose $p : P \to \Bbbk$ were one, and choose $u \in P$ with $p(u) = 1$. Then $p(uA) = \Bbbk$, so $P = uA$. The surjection $A \to P$, $a \mapsto ua$, splits because P is projective. The resulting projection $A \to A$ is multiplication by an idempotent $e \neq 0$, and $P \cong eA$. Since $\Bbbk[x]$ has no idempotents other than 0 and 1, we have $P \cong A$. But every surjection $A \to \Bbbk$ maps the proper submodule $(x-1)A$ onto $\Bbbk$, contradicting the cover property.

A graded module M is *bounded below* if there is an integer b such that $M_d = 0$ for all $d < b$. Every finitely generated graded module is bounded below: replace a finite generating set by its homogeneous components and take the least of their degrees.

Lemma 4.32 (Graded Nakayama lemma). *For a bounded below graded right module M,*

$$\operatorname{top}_{\mathrm{gr}} M = 0 \quad \Longrightarrow \quad M = 0.$$

If $N \subseteq M$ is a graded submodule and $N + MA_+ = M$, then $N = M$.

Proof If $M = MA_+ \neq 0$, choose the least integer d with $M_d \neq 0$. But $(MA_+)_d = \sum_{r>0} M_{d-r}A_r = 0$, a contradiction. The second assertion follows by applying the first to M/N. □

This version uses boundedness below, not finite generation. In particular, homogeneous lifts of generators of $\operatorname{top}_{\mathrm{gr}} M$ generate M.

Definition 4.33. *A graded projective cover of a bounded below locally finite-dimensional graded module M is a surjection $p : P \to M$, with P bounded below, locally finite-dimensional, and graded projective, such that*

$$\operatorname{top}_{\mathrm{gr}} p : \operatorname{top}_{\mathrm{gr}} P \xrightarrow{\sim} \operatorname{top}_{\mathrm{gr}} M.$$

As in Proposition 4.14, this condition is equivalent to $\operatorname{Ker} p \subseteq PA_+$, and to no proper graded submodule of P mapping onto M.

The identity $p(PA_+) = MA_+$ gives $\operatorname{Ker}(\operatorname{top}_{\mathrm{gr}} p) = (\operatorname{Ker} p + PA_+)/PA_+$. If $\operatorname{Ker} p \subseteq PA_+$ and $p(U) = M$ for a graded submodule $U \subseteq P$, then $U + PA_+ = P$; graded Nakayama gives $U = P$. Conversely, if $\operatorname{Ker}(\operatorname{top}_{\mathrm{gr}} p) \neq 0$, choose a graded S-module complement W to this kernel in

$\operatorname{top}_{\mathrm{gr}} P$. The inverse image U of W under $P \to \operatorname{top}_{\mathrm{gr}} P$ is a proper graded submodule of P. Since $\operatorname{top}_{\mathrm{gr}} p(W) = \operatorname{top}_{\mathrm{gr}} M$, we have $p(U) + MA_+ = M$; graded Nakayama gives $p(U) = M$.

Proposition 4.34. *Every bounded below locally finite-dimensional graded right A-module M has a graded projective cover, unique up to a graded isomorphism commuting with the maps to M. Its domain is isomorphic to*

$$P = \bigoplus_{d\in\mathbb{Z}} \big((\operatorname{top}_{\mathrm{gr}} M)_d \otimes_S A\big)(-d), \tag{4.3}$$

where $(\operatorname{top}_{\mathrm{gr}} M)_d$ is placed in degree 0 before shifting.

Proof Since S is semisimple, choose right S-linear sections $s_d : (\operatorname{top}_{\mathrm{gr}} M)_d \to M_d$ of the quotient maps for every $d \in \mathbb{Z}$. Define on each summand of (4.3)

$$p : \big((\operatorname{top}_{\mathrm{gr}} M)_d \otimes_S A\big)(-d) \longrightarrow M, \qquad v \otimes a \longmapsto s_d(v)a.$$

The map is well-defined by S-linearity of s_d, and it preserves degree. Each summand of P is graded projective. Its graded top is $(\operatorname{top}_{\mathrm{gr}} M)_d$ placed in degree d, since multiplication by A_+ generates precisely its components above degree d. Thus $\operatorname{top}_{\mathrm{gr}} p$ identifies $\operatorname{top}_{\mathrm{gr}} P$ with $\operatorname{top}_{\mathrm{gr}} M$. Graded Nakayama, applied to $M/p(P)$, makes p surjective. If $M_d = 0$ for $d < b$, then $P_d = 0$ for $d < b$, and

$$P_q = \bigoplus_{b\le d\le q} (\operatorname{top}_{\mathrm{gr}} M)_d \otimes_S A_{q-d}$$

is finite-dimensional for every $q \in \mathbb{Z}$. Thus p is a graded projective cover. The uniqueness proof of Theorem 4.15 applies with degree-preserving lifts and sections. □

Definition 4.35. *A graded projective resolution is a projective resolution (4.1) in which the modules are graded, the maps preserve degree, and the P_i are graded projective. It is minimal if $p_0 : P_0 \to M$ and the maps $p_i : P_i \to K_i$, $x \mapsto d_i(x)$, with $K_1 = \operatorname{Ker} p_0$ and $K_i = \operatorname{Ker} d_{i-1}$ for $i \ge 2$, are graded projective covers for all $i \ge 1$.*

The homological degree i and the grading degree d are distinct:

$$d_i : (P_i)_d \longrightarrow (P_{i-1})_d.$$

Exactness is checked separately in each grading degree, as in Remark 1.29. For bounded below locally finite-dimensional terms, minimality is equivalent to

$$\operatorname{Im} d_i \subseteq P_{i-1}A_+ \quad \text{for all } i \ge 1. \tag{4.4}$$

Theorem 4.36. *Every bounded below locally finite-dimensional graded right A-module has a minimal graded projective resolution. Any two such resolutions are isomorphic by degree-preserving maps commuting with the differentials and the maps to M.*

Proof Repeat the construction and uniqueness proof of Theorem 4.22, using Proposition 4.34. Every kernel remains bounded below and locally finite-dimensional, so the construction continues at every step. □

For each term, (4.3) gives

$$P_i \cong \bigoplus_{d\in\mathbb{Z}} \big((\operatorname{top}_{\mathrm{gr}} P_i)_d \otimes_S A\big)(-d).$$

Thus the graded S-module $\operatorname{top}_{\mathrm{gr}} P_i$ records both the types and the degrees of the generators needed in homological degree i.

Example 4.37. *The ordinary resolution in Example 4.25 becomes graded after a shift. Let* $A = \Bbbk A_2$ *with* $A_0 = S = \Bbbk e_{11} \oplus \Bbbk e_{22}$ *and* $A_1 = \Bbbk e_{12}$*, and place* L_1, L_2 *in degree* 0*. Then*

$$0 \longrightarrow e_{22}A(-1) \xrightarrow{d_1} e_{11}A \xrightarrow{p_0} L_1 \longrightarrow 0, \qquad d_1(e_{22}a) = e_{12}a$$

is minimal: the generator e_{22} *of the shifted module and its image* e_{12} *both have degree* 1*, and* $\operatorname{Ker} p_0 = \Bbbk e_{12} = (e_{11}A)A_+$*. Its projective terms have tops* L_1 *and* $L_2(-1)$*. The other simple module* $L_2 = e_{22}A$ *is already projective.*

Example 4.38. *Let* $A = \Bbbk[x]$ *with* $\deg x = 1$*. The quotient* $\Bbbk = A/(x)$ *has the minimal graded resolution*

$$0 \longrightarrow A(-1) \xrightarrow{d_1} A \xrightarrow{\pi} \Bbbk \longrightarrow 0, \qquad d_1(a) = xa, \quad \pi(f) = f(0).$$

Multiplication by x *is injective, its image is* $xA = \operatorname{Ker}\pi$*, and the shift makes it degree-preserving. The kernel of* π *lies in* AA_+*, and the next kernel is zero. The two projective terms therefore have one generator each, in degrees* 0 *and* 1*, respectively.*

Forgetting the grading need not preserve projective covers. In Example 4.38, the proper ungraded submodule $(x-1)A \subset A$ still maps onto $\Bbbk$ under π, since $\pi(x-1) = -1$. Thus π is a graded projective cover but not an ordinary projective cover.

Example 4.39. *Give* $E = \Bbbk[\epsilon]/(\epsilon^2)$ *the grading* $\deg \epsilon = 1$*. The ordinary resolution in Example 4.26 becomes*

$$\cdots \longrightarrow E(-3) \xrightarrow{d_3} E(-2) \xrightarrow{d_2} E(-1) \xrightarrow{d_1} E \xrightarrow{\pi} \Bbbk \longrightarrow 0, \qquad d_i(a) = \epsilon a.$$

For every $i \geq 1$*, the generator of* $E(-i)$ *and its image under* d_i *both have degree* i*. Exactness was checked in the ordinary resolution, and* $\operatorname{Im} d_i = \epsilon E(-i+1) \subseteq E(-i+1)E_+$ *proves minimality. The term in homological degree* i *has one generator, in degree* i*.*

Remark 4.40. The multiplicities in the graded tops are sometimes called graded Betti numbers. For representatives $L_1, \ldots, L_n$ of the simple right S-modules, one can write

$$(\operatorname{top}_{\mathrm{gr}} P_i)_d \cong \bigoplus_{j=1}^{n} L_j^{\oplus \beta_{i,d,j}(M)}, \qquad \beta_{i,d}(M) := (\beta_{i,d,j}(M))_{1 \leq j \leq n}.$$

Here $i \geq 0$ is the homological degree and $d \in \mathbb{Z}$ is the grading degree. These multiplicities are independent of the chosen minimal resolution by Theorem 4.36. We will use the modules $\operatorname{top}_{\mathrm{gr}} P_i$ directly, without this additional indexing.

Definition 4.41. *An algebra* A *is* right noetherian *if every submodule of every finitely generated right* A*-module is finitely generated. It is* left noetherian *if the same condition holds for left* A*-modules. It is* noetherian *if it is both right and left noetherian.*

If a graded module M is finitely generated, then $\operatorname{top}_{\mathrm{gr}} M$ is finite-dimensional: the images of a finite homogeneous generating set generate $\operatorname{top}_{\mathrm{gr}} M$ over the finite-dimensional algebra S. Its graded projective cover is therefore finitely generated by (4.3). The kernel need not be finitely generated unless we impose an additional condition.

Corollary 4.42. *Assume that* A *is right noetherian. Every finitely generated graded right* A*-module has a minimal graded projective resolution whose terms and successive kernels are finitely generated. In particular, this applies to* $S = A/A_+$*.*

Proof A finitely generated graded module is bounded below and locally finite-dimensional, since it is a quotient of a finite direct sum of shifts of A. Its graded projective cover is finitely generated by the

preceding paragraph. Right noetherianity makes the kernel finitely generated, so the same argument applies successively to every kernel. □

Right noetherianity ensures finite generation at every step. Without that assumption, boundedness below and local finite-dimensionality still suffice for existence of a minimal graded resolution by Theorem 4.36.

5 Koszul complex and Koszul algebras

5.1 The Koszul complex of a polynomial algebra

Let V be an n-dimensional $\Bbbk$-vector space, choose a basis $x_1, \dots, x_n$, and let $A = \mathsf{S}(V) = \Bbbk[x_1, \dots, x_n]$, with each x_j in degree 1. Let $\pi : A \to \Bbbk$ be projection onto degree 0, so $\Bbbk = A/A_+$. For $0 \le i \le n$, put

$$K_i := \bigwedge^i V \otimes_{\Bbbk} A(-i), \qquad K_0 = A.$$

The right A-action is on the second factor. The factor $\bigwedge^i V$ is placed in grading degree 0, so the generators of K_i have grading degree i.

Definition 5.1. *For $1 \le i \le n$, define the Koszul differential*

$$\begin{aligned} d_i : K_i &\longrightarrow K_{i-1}, \\ (v_1 \wedge \cdots \wedge v_i) \otimes a &\longmapsto \sum_{r=1}^{i} (-1)^{r-1} (v_1 \wedge \cdots \wedge \widehat{v_r} \wedge \cdots \wedge v_i) \otimes v_r a. \end{aligned} \tag{5.1}$$

The formula is multilinear and alternating in $v_1, \dots, v_i$, so it defines a map on $\bigwedge^i V$. The sequence $K_\bullet \to \Bbbk$ consists of these maps followed by $\pi : K_0 = A \to \Bbbk$.

Proposition 5.2. *The sequence*

$$0 \longrightarrow K_n \xrightarrow{d_n} \cdots \xrightarrow{d_2} K_1 \xrightarrow{d_1} K_0 \xrightarrow{\pi} \Bbbk \longrightarrow 0$$

is an exact minimal graded free resolution of $\Bbbk$ over A.

Proof Each d_i preserves degree and is right A-linear. For $i \ge 2$, compute $d_{i-1}d_i$ on a basis wedge. Every term is indexed by two distinct positions $r < s$ deleted from that wedge. Deleting position r and then s gives coefficient $(-1)^{r+s-3} x_{j_s} x_{j_r}$; deleting s and then r gives coefficient $(-1)^{r+s-2} x_{j_r} x_{j_s}$. Since $x_{j_s} x_{j_r} = x_{j_r} x_{j_s}$, these two terms sum to zero. Hence $d_{i-1}d_i = 0$; also $\pi d_1 = 0$ because every term of d_1 is divisible by a variable.

To prove exactness, give A and $\bigwedge V$ the multigrading with $\deg x_j$ equal to the jth standard basis vector. For $\alpha \in \mathbb{Z}^n_{\ge 0}$, let $\operatorname{supp}(\alpha) = \{j \mid \alpha_j > 0\}$. The degree α part of K_i has basis

$$b_J := (x_{j_1} \wedge \cdots \wedge x_{j_i}) \otimes x^{\alpha - \mathbf{1}_J}, \qquad J = \{j_1 < \cdots < j_i\} \subseteq \operatorname{supp}(\alpha),$$

where $\mathbf{1}_J$ has entry 1 at indices in J and 0 elsewhere. The differential is $d_i(b_J) = \sum_{r=1}^{i} (-1)^{r-1} b_{J \setminus \{j_r\}}$.

If $\alpha \ne 0$, let q be the least index in $\operatorname{supp}(\alpha)$. For each $i \ge 0$, define $h_{\alpha,i} : K_{i,\alpha} \to K_{i+1,\alpha}$ on basis vectors by

$$h_{\alpha,i}(b_J) = \begin{cases} b_{\{q\} \cup J}, & q \notin J, \\ 0, & q \in J, \end{cases}$$

inserting q first in the ordered wedge. For $q \notin J$ and $i \ge 1$, direct substitution gives

$$\begin{aligned} d_{i+1} h_{\alpha,i}(b_J) &= b_J - \sum_{r=1}^{i} (-1)^{r-1} b_{\{q\} \cup (J \setminus \{j_r\})}, \\ h_{\alpha,i-1} d_i(b_J) &= \sum_{r=1}^{i} (-1)^{r-1} b_{\{q\} \cup (J \setminus \{j_r\})}. \end{aligned}$$

For $i = 0$, $J = \varnothing$ and $d_1 h_{\alpha,0}(b_\varnothing) = b_\varnothing$. For $q \in J$, the term of $d_i(b_J)$ deleting q is the only term on which $h_{\alpha,i-1}$ is nonzero, and it gives $h_{\alpha,i-1} d_i(b_J) = b_J$ while $d_{i+1} h_{\alpha,i}(b_J) = 0$. Thus $d_{i+1} h_{\alpha,i} + h_{\alpha,i-1} d_i =$

$\mathrm{id}_{K_{i,\alpha}}$ for every $i \geq 1$ in positive multidegree; the case $i = 0$ was checked separately. All positive multidegrees are therefore exact. In multidegree 0, only $K_0 \cong \Bbbk$ is nonzero and π is the identity onto $\Bbbk$. This proves exactness of the displayed sequence.

Finally, each K_i is a finite direct sum of copies of $A(-i)$, and (5.1) has image in $K_{i-1}A_+$. Exactness gives $\operatorname{Ker} d_i = \operatorname{Im} d_{i+1} \subseteq K_i A_+$ for $i < n$, and $\operatorname{Ker} d_n = 0$. Also, $\operatorname{Ker} \pi = AA_+$. Thus each map from K_i onto the preceding kernel has kernel contained in $K_i A_+$, so it is a projective cover by Definition 4.33. The resolution is therefore minimal. □

5.2 Koszul algebras

Definition 5.3. *A minimal graded projective resolution $P_\bullet \to M$ is linear if P_i/P_iA_+ is concentrated in degree i for every $i \geq 0$. This is equivalent to P_i being generated in degree i: generation in degree i implies the stated concentration because A_+ raises degree. Conversely, if the top is concentrated in degree i, boundedness below and Lemma 4.32 show that P_i has no components below degree i; the quotient $P_i/(P_i)_iA$ then has zero graded top, so Lemma 4.32 makes that quotient zero.*

Definition 5.4. *Let $A = T_S(V)/(R)$ be a quadratic algebra with $S = A_0$ finite-dimensional semisimple. The algebra A is Koszul if the degree zero quotient module $S = A/A_+$ has a linear minimal graded projective resolution as a right A-module.*

The generators in homological degree i must then all have grading degree i. Since S can have several simple modules, the definition requires a linear resolution of their direct sum, not just of one chosen simple module.

Proposition 5.5. *For every finite-dimensional vector space V, the symmetric algebra $\mathsf{S}(V)$ is Koszul.*

Proof Proposition 5.2 gives a minimal resolution with $K_i = \bigwedge^i V \otimes_\Bbbk A(-i)$. Thus K_i is generated in degree i for every i. □

Example 5.6. *The definition also applies when S has several simple modules. For $A = \Bbbk A_2$, the resolution in Example 4.37 is linear for L_1, and $L_2 = e_{22}A$ is projective in degree 0. Taking their direct sum gives a linear resolution of $S = L_1 \oplus L_2$; hence $\Bbbk A_2$ is Koszul.*

Example 5.7. *The exterior algebra $E = \Bbbk[\epsilon]/(\epsilon^2)$ is quadratic, and Example 4.39 gives a minimal resolution of its degree zero quotient with $P_i = E(-i)$. Thus E is Koszul, although its minimal resolution is infinite.*

List of Symbols

Page numbers link to the first defining passage. Entries marked advanced refer to optional material.

Symbol	Meaning	Page
Algebras and categories		
$\Bbbk$	Base field.	1
$\mathrm{Mat}_n(\Bbbk)$	Algebra of $n \times n$ matrices over $\Bbbk$.	2
$\Bbbk A_n$	Upper triangular $n \times n$ matrix algebra.	2
e_{ij}	Unit matrix with 1 in position (i, j).	2
$\mathrm{Hom}_{\mathcal{C}}(X, Y),\ \mathrm{id}_X$	Morphisms from X to Y; identity of X.	2
$\mathrm{Mod}\, A$	Category of right A-modules and A-linear maps.	2
$\mathrm{mod}\, A$	Category of finitely generated right A-modules.	2
$\mathrm{Hom}_A(M, N)$	A-linear maps, with no degree restriction.	2
$\mathrm{add}\, M, \mathrm{add}_{\mathrm{mod}\, A}\, M$	Direct summands of finite direct sums of the right module M.	2
$\mathcal{C}^{\mathrm{op}}$	Opposite category: morphisms have their directions reversed.	3
$\mathrm{rad}\, A$	Jacobson radical of the algebra A.	4
$\mathrm{rad}\, M$	Intersection of the maximal submodules of M.	4
$eA,\ eAe$	Right ideal and corner subalgebra associated with an idempotent e.	5
A^{op}	Opposite algebra: $a^{\mathrm{op}} b^{\mathrm{op}} = (ba)^{\mathrm{op}}$.	6
Gradings, sums, and bimodules		
A_g, M_g	Degree g component of a group-graded algebra A or module M	6, 6
$M(d)$	Grading shift: $M(d)_j = M_{j+d}$.	7
$\mathrm{Hom}_A^{\mathbb{Z}}(M, N)$	Space of degree-preserving A-linear maps.	7
$\mathrm{GrMod}\, A$	Category of graded right A-modules and degree-preserving maps.	7
$\mathrm{grmod}\, A$	Category of locally finite-dimensional graded right A-modules.	7
A_+	Positive degree ideal $\bigoplus_{i>0} A_i$.	9
$\bigoplus_{i \in I} M_i$	Direct sum.	10
$M^{\oplus r}$	Direct sum of r copies of M.	2
$\prod_{i \in I} M_i$	Direct product.	10
$\Bbbk[[x]]$	Formal power series ring.	10
${}_B M_C$	Bimodule with left B-action and right C-action.	11
$B\,\mathrm{Mod}\, C \simeq \mathrm{Mod}\, C \otimes_{\Bbbk} B^{\mathrm{op}}$	Category of (B, C)-bimodule.	11
A^{en}	Enveloping algebra $A \otimes_{\Bbbk} A^{\mathrm{op}}$.	12
Duals and comparison maps		
$D(V)$	Ungraded $\Bbbk$-linear dual $\mathrm{Hom}_{\Bbbk}(V, \Bbbk)$.	4
$D(u)$	Dual map, given by precomposition with u.	4
ev_V	Evaluation $V \to D(D(V))$, sending v to $\lambda \mapsto \lambda(v)$.	4
$\mathbb{D}(V)$	Graded $\Bbbk$-linear dual, with $\mathbb{D}(V)_i = D(V_{-i})$.	13
$M^*,\ {}^*N$	Ungraded right and left S-linear duals: $\mathrm{Hom}_S(M, S)$ and $\mathrm{Hom}_{S^{\mathrm{op}}}(N, S)$.	15
$M^{(*)},\ {}^{(*)}N$	Graded right and left S-duals.	15
$M^{*_{\mathrm{en}}}$	Two-sided dual $\mathrm{Hom}_{S^{\mathrm{en}}}(M, S^{\mathrm{en}})$; advanced.	16
$M^{(*_{\mathrm{en}})}$	Graded two-sided dual, with degrees reversed; advanced.	17

List of Symbols (continued)

Symbol	Meaning	Page
$\tau : S \to \Bbbk$	Symmetrising form on S.	17
$\alpha_M,\ \beta_N$	Natural transformation from S-duals to linear duals, by composition with τ.	18
γ_M	Comparison from the two-sided dual to the right S-dual.	19
Tensor products, presentations, and Hilbert series		
$U \otimes_{\Bbbk} V$	Tensor product of vector spaces.	9
$M \otimes_A N$	Tensor product over A, with $ma \otimes n = m \otimes an$.	22
$V^{\otimes_A i}$	ith tensor power over A; $V^{\otimes_A 0} = A$.	25
$\theta_r,\ \theta_\ell$	Tensor-Hom comparison maps; anti-distributivity of the S-dual in the semisimple case.	25
$\mu_i : A_1^{\otimes_S i} \to A_i$	Multiplication of i elements of degree 1.	27
$T_S(V)$	Tensor algebra generated by the bimodule ${}_S V_S$.	28
$\mathsf{S}(V)$	Symmetric algebra of V.	28
$\bigwedge W$	Exterior algebra of W.	28
$I = \operatorname{Ker} \mu$	Ideal of relations for $\mu : T_S(A_1) \to A$.	30
$A = T_S(V)/(R)$	Quadratic presentation: generators V in degree 1, relations R in degree 2.	30
$L_j = e_j S,\ e_j A$	Simple modules and the corresponding right-module summands of A used to index Hilbert series.	31
$[M_d : L_j]$	Composition multiplicity in degree d; computed by (2.14).	31
$\mathbf{dim}\, M_d$	Row vector $([M_d : L_j])_{1 \le j \le n}$.	31
$h_M(t)$	Module Hilbert series $\sum_{d \in \mathbb{Z}} (\mathbf{dim}\, M_d) t^d$.	31
$h_A(t)$	Algebra Hilbert series, with entries recording both simple-module indices.	31
qA	Quadratic part $T_S(V)/(I_2)$ of $A = T_S(V)/I$; advanced.	35
$U^{\perp},\ {}^{\perp}U$	Right and left S-linear functionals vanishing on U.	36
Quadratic duality and resolutions		
$A^!,\ {}^!A$	Right and left quadratic dual algebras.	38
$A \otimes_{\Bbbk} B$	Ordinary algebra tensor product, without signs in multiplication.	42
$A \otimes_{\Bbbk}^{\mathrm{gr}} B$	Graded algebra tensor product, with the sign in (3.7).	42
$\operatorname{top} M = M/\operatorname{rad} M$	Top of an ordinary module over a finite-dimensional algebra.	48
$d_i : C_i \to C_{i-1}$	Chain differential; lowers homological degree by 1.	50
$H_i(C_\bullet)$	Homology $\operatorname{Ker} d_i / \operatorname{Im} d_{i+1}$.	50
$P_\bullet \to M$	Projective resolution of M, with the map $P_0 \to M$ specified.	50
$\operatorname{top}_{\mathrm{gr}} M = M/MA_+$	Graded top of a graded right A-module.	52
$\beta_{i,d,j}(M),\ \beta_{i,d}(M)$	Graded Betti multiplicities and their row vector; advanced.	55
$K_i,\ K_\bullet$	Koszul terms $K_i = \bigwedge^i V \otimes_{\Bbbk} A(-i)$ and their complex, for $A = \mathsf{S}(V)$.	57
$d_i : K_i \to K_{i-1}$	Koszul differential.	57

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